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**FITSUM DERIBA**

# Enhancing the Education of Accessibility with Artificial Intelligence

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Fitsum Deriba

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## ABSTRACT

The global reliance on digital technologies in all aspects of life highlights the fundamental need for universal accessibility. This study addresses these need to effectively integrate accessible design education within Ethiopian Higher Education (EHE), recognizing the importance for digital inclusion and the limited pedagogical strategies for its effective integration to create universally accessible digital environments. It investigates the multifaceted role of Artificial Intelligence (AI) as a pedagogical tool to enhance students' design learning and application of accessible design principles. This dissertation includes four studies—one systematic and three empirical—that explored on the accessibility and inclusive design gaps in digital technologies and EHE computing curricula, focusing on three aspects: i) the role of Generative AI (GenAI) in accessibility and design education, ii) pedagogical efficacy of human and AI feedback modalities, and iii) the lasting impact of AI interventions in the design education domain.

Using a mixed-methods sequential design, the research followed a structured process of data collection, preparation, exploration, and analysis. Participants were recruited from computing schools within EHE institutions. The qualitative analysis, employing thematic analysis, analyzed student experiences regarding GenAI's role in design, challenges associated with its use, and its potential future impact on inclusive design and accessibility. The

quantitative analysis examined students' implementation of accessibility features and self-efficacy, as well as gains in foundational accessibility knowledge, competence, and motivation. Additionally, the heuristic evaluation examined the scores of students' post-intervention on User Interface (UI) designs and the association of high-achiever and low-achiever student patterns on key codes—accessibility, knowledge, design, and improvement—as experienced by students. Through statistical analysis, it further examined the lasting impact of GenAI by analyzing the short-term and long-term effects of instructors versus GenAI feedback, and provided insights into whether their feedback has a lasting impact on students' design and accessibility learning.

The finding provides a comprehensive understanding of the potential and limitations of integrating GenAI into accessible design education. It also underscored the need for pedagogically informed integration of GenAI in HCI and software engineering curricula, emphasizing the complementary roles of AI and human instruction to effectively equip future professionals with the skills to design more inclusive digital experiences. Future research should focus on addressing the identified challenges, enhancing AI integration strategies, and exploring innovative approaches to leverage AI for supporting both practical competence and theoretical understanding in accessible design education.

**Keywords:** AI in Education, Accessible Design Education, Learning Accessibility, Design Skills, Multimodal-LLM, Experimental Study, Design Skills, Student Learning

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## LIST OF ABBREVIATIONS AND ACRONYMS

| | |
|---|---|
| ADA | The Americans with Disabilities Act |
| AI | Artificial Intelligence |
| AIEd | Artificial intelligence in education |
| ATAG | Authoring Tool Accessibility Guidelines |
| EAA | The European Accessibility Act |
| EHE | Ethiopian Higher Education |
| ENA | Epistemic Network Analysis |
| GenAI | Generative Artificial Intelligence |
| HCI | Human Computer Interaction |
| ICT | Information and Communication Technologies |
| ITS | Intelligent Tutoring System |
| LMS | Learning Management System |
| M-LLMs | Multimodal Large Language Models |
| MOOCs | Massive Open Online Courses |
| NLP | Natural Language processing |
| PBL | Problem Based Learning |
| PjBL | Project Based Learning |
| POUR | Perceivable, Operable, Understandable, and Robust |
| SDT | Self-Deterministic Theory |
| UDL | Universal Design for Learning |
| UAAG | User Agent Accessibility Guidelines |
| UI | User Interface |
| UNCRPD | United Nations Convention on the Rights of Persons with Disabilities |
| UX | User Experience |
| V-Labs | Virtual labs |
| W3C | World Wide Web Consortium |
| WAI | Web Accessibility Initiative |
| WCAG | Web Content Accessibility Guidelines |
| WebAIM | Web accessibility in mind |
| WWW | World Wide Web |

## List of Original Publications

The foundation of this dissertation is based on the data from articles presented by roman numerals I – IV.

**I** Deriba, F. G., Saqr, M., & Tukiainen, M. (2024, June). Assessment of accessibility in virtual laboratories: a systematic review. In Frontiers in Education (Vol. 9, p. 1351711). Frontiers Media SA. Doi: https://doi.org/10.3389/feduc.2024.1351711

**II** Deriba, F. G., Saqr, M., & Tukiainen, M. (2024, October). Exploring the Role of AI in Enhancing Accessibility in User Interface Design: A Qualitative Study of Student Experiences. In International conference on technological ecosystems for enhancing multiculturality (pp. 437-447). Singapore: Springer Nature Singapore. Doi: https://doi.org/10.1007/978-981-96-5658-5_4

**III** Deriba FG, Saqr M and Tukiainen M (2025). Enhancing Students' Skill through AI: An Experimental Study in Learning Accessible Design. Manuscript submitted for publication to Interactive Learning Environments.

**IV** Deriba FG, Saqr M and Tukiainen M (2025). Does the Impact of GenAI Persist in Learning? Comparing AI and Instructor Feedback on Accessibility. Manuscript accepted for publication at Universal Access in the Information Society.

These publications have included at the end of this dissertation with their copyright holders' permission.

## Author's contribution

The articles included in this dissertation are original research papers focusing on accessibility and design education with a particular emphasis on Artificial Intelligence (AI) in developing countries. These articles highlight the role of AI in advancing design and accessibility education. The following is a detailed description of the author's contributions to each of the research papers (Papers I to IV).

**Paper I**: This paper presents a systematic review of literatures that assesses the accessibility of learning technology, particularly within V-Labs. The author of this article formulated the research questions, conducted the systematic review, and prepared the initial draft of the manuscript. The co-authors contributed to refining the research focus, provided comments on the manuscript, and addressed with the reviewer feedback from the main author prior to final acceptance to Journal.

**Paper II**: This paper examined the role of AI in enhancing accessibility in User Interface Design. The author developed the initial concept for the article in consultation with the supervisors, who served as co-authors, and designed the research approach in cooperation with them. The author conducted qualitative data collection through semi-structured interviews with students from Ethiopian higher education institutions. The author drafted the paper, while the co-authors supervised, edited, reviewed, and guided the work through to publication. The second author presented this paper at TEEM 2024 Conference.

**Paper III**: This experimental study is an extension of the previous research work (Paper II). The author of this research contributed to the initial draft of the manuscript with Prof. Mohammed, who provided valuable input, editing the final version, and assisting the author until the manuscript's final submission.

**Paper IV**: This paper focuses on examining the GenAI impact on student learning of accessibility, comparing its feedback with instructor feedback, and investigating its lasting effects. The author was the main contributor to this paper, collecting data from an EHEI and preparing the article in consultation with the co-authors, who provided valuable input and editing the final version of the manuscript.

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# 1 INTRODUCTION

## 1.1 Background, Context, and Motivation

Technology has become integral to numerous aspects of daily life, encompassing communication, information retrieval, publishing, entertainment, commerce, and education. While the ubiquity of technological interaction is often taken for granted, improperly designed systems can present barriers, hindering individuals from fully engaging in these activities (Wilson et al., 1997). Recognizing accessibility impediments, legislative measures, such as the Americans with Disabilities Act (ADA), Accessible Rich Internet Applications (ARIA), Section 508, and, recently, the European Accessibility Act (Commission, 2025), have been enacted to mandate equitable access to resources, products, and services, including technologies, that are typically accessible to all individuals. Analogous to physical accessibility requirements for ramps, elevators, and Braille signage, these laws extend to digital realms, necessitating the accessibility of websites, digital publications, software, videos, and other technological platforms (W3C, 2018). Therefore, accessibility has evolved into a fundamental value for both product development and organizational principles. Digital accessibility transcends mere user experience enhancement; it embodies the principle of making technology universally accessible, particularly to marginalized individuals or those facing challenges in its usage. This indicates that accessibility is an inherent right, rather than a privilege. In turn, this underscores the expectation that digital technology should be inclusive and supportive of the diverse needs of the individual, while being usable, easy to understand, and enjoyed by everyone (Guedes, 2024).

The integration of AI into the educational domain has garnered global attention for its role in supporting teaching and learning practices, particularly in computer science and design education (Walter, 2024). AI-based tools, including intelligent tutoring systems (ITS) and multimodal large language models (M-LLMs), are widely used to support personalized learning, automate

feedback (Mustafa et al., 2024; Zawacki-Richter et al., 2019), and assist in tasks such as software development and user interface (UI) design (Belzner et al., 2023). As technology advances, there is a growing imperative to ensure AI technologies serve diverse populations through inclusive and accessible design (Trewin et al., 2019).

The need for digital accessibility for all people, globally, remains a major problem (Kulkarni, 2019), which involves the interplay of the principles of equality, social justice, and sustainable development. According to the World Health Organization 2023 report, approximately 1.3 billion individuals (16% of the global population) experience some form of disability, a figure rising due to aging demographics and environmental factors (WHO, 2023). These include varied barriers for different individuals, particularly in developing regions like Africa. For example, people who are blind, have low vision, or are color blind face barriers due to incompatibility with screen reader technology, such as the inability to increase the size of text on screen, poor color contrast, or the lack of a function that converts text to speech (Hamideh Kerdar et al., 2024). Likewise, people who are hard of hearing face barriers related to the limited access to assistive devices and a lack of or limited accessibility features in technology. People with limited mobility or dexterity encounter barriers due to the height of kiosks, interfaces that require the use of a mouse, and incompatibility with voice recognition software. Individuals with cognitive and neurological disabilities face barriers on websites with complex navigation and page layouts that are difficult to understand, the use of lengthy sentences and difficult vocabulary, and distracting background images or noises.

Despite the ratification of the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD) by 185 nations (Assembly, 2006), barriers to physical, digital, and social inclusion remained across geographic and socio-economic contexts. While these countries officially agreed to be legally bound by the Convention's rule, namely, they promise to work towards protecting and promoting the rights of people with disabilities within their own countries, implementation in developing regions has been limited. Inaccessible infrastructure, technologies, and policies limit opportunities for persons with disabilities in education, employment, healthcare, and civic participation (F. G. Deriba et al., 2023; Grech & Soldatic, 2016). These

disparities are worsened in low- and middle-income countries, where 80% of individuals with disabilities reside (Schulze, 2010), and resource constraints and sociocultural stigma intensify exclusion.

Rapid digitization further complicates accessibility. According to Park (2022), people with varied abilities have less access to information and communication technologies (ICTs) than their non-persons with disabilities peers, corresponding to isolation from technological advancements and socio-economic opportunities (Warschauer, 2004). For this study, we used the definition of accessibility by the World Wide Web Consortium (W3C) as the ability of persons with disabilities to perceive, understand, operate, navigate, and interact with digital technologies in the same way as those without disabilities (W3C, 2018). "Perceivable" refers to the availability of content (e.g., captions for deaf users), "operable" pertains to the users' ability to engage with content (e.g., using a keyboard versus a mouse), "understandable" relates to how users process content (e.g., ease of making necessary changes), and "robust" indicates the extent to which content can function with current and future technologies.

While efforts to address digital accessibility include technical frameworks (e.g., WCAG), legal mandates (e.g., ADA, EAA), and inclusive design principles like Universal Design for Learning (UDL), there are gaps in inconsistent accessibility policy implementation (Lazar, 2019), lack of common ground on objectives and agendas of accessibility, and lack of accessibility and inclusive design experts, particularly in developing regions, which leads to the continued development of limited accessible systems. For example, dualisms across different stakeholders, such as "profit versus human rights, market share versus accessibility, competition versus inclusion," can change digital accessibility (Stienstra et al., 2007). This issue primary arises from a lack of emphasis on accessibility in computing education.

The 2024 Web accessibility in mind (WebAIM) analysis of one million top home pages across the globe revealed unresolved accessibility gaps, with 95.9% containing WCAG 2 failures—a marginal 1.9% improvement over five years. Detected errors rose to 56.8 per page (13.6% increase from 2023), driven by unresolved issues like low-contrast text (81% of pages), missing alt text (54.5%), and unlabeled forms (48.6%). Despite rising ARIA usage (up

15%), pages using ARIA averaged 34% more errors, underscoring ongoing challenges in implementing accessible design and restricting equitable access to digital technologies to wider users. While frameworks like UDL aim to bridge gaps, their limited integration into core curricula is the main challenge (Elglaly et al., 2024).

Equipping students the skills and knowledge of accessibility by providing feedback in teaching design and accessibility instruction is one approach; however, in developing countries like Ethiopia (Jima et al., 2025), where human expertise and resources are scarce, traditional feedback mechanisms are limited in their ability to provide personalized guidance due to different factors, such as workload, and large class size, leaving educators unprepared to teach accessibility effectively (H. Zhang et al., 2025).

In the Ethiopian context, accessibility challenges are characterized by infrastructural, educational, and sociocultural realities. Although national initiatives, such as the Inclusive Education Strategy (Affairs, 2012), the Digital Transformation Strategy for Ethiopia (FEDRE, 2020), digital strategy for inclusive prosperity (Ahmed Ali, 2020)and our youth succeed in the new world. We are witnessing a global transformation driven by new technologies such as Artificial Intelligence, Internet of Things, Nanotechnology, and Big Data, amongst many others that offer new models for production, communication, and lifestyle. Our children require new skills and knowledge, and it is our responsibility to position them better in seizing the future. Ethiopia is yet to realize its potential in the digital space and leverage technology to build a more prosperous society. Urgent, bold, and coordinated action is needed so we can make this transformation successful. We are on a journey of rapid economic and social change. In 2019, we adopted the Homegrown Economic Reform Agenda and the Ten-Year National Development Plan (2020-2030, and Education sector development program (Federal Ministry of Education, 2021), emphasize equity and technology-enabled learning, their implementation in higher education remains uneven, or these strategic goals often conflict with significant on-the-ground realities. Limited broadband connectivity, a shortage of accessibility-trained instructors and ICT personnel, and the absence of standardized digital accessibility frameworks hinder meaningful inclusion across universities (F. Deriba et al., 2025). Moreover, sociocultural

stigma surrounding disability and inadequate assistive technology provision restrict the participation of student with disability. These contextual barriers informed the methodological choices of this dissertation particularly the adoption of scalable, AI-assisted feedback mechanisms and participatory design approaches—as pragmatic responses to resource scarcity, expertise gaps, and infrastructural limitations within Ethiopian higher education institutions.

Different scholars have further highlighted the concerns regarding the exclusion of persons with disabilities' lived experiences from digital technologies (Fitsum Gizachew Deriba et al., 2024), marginalized co-design processes (S. Guedes et al., 2023), insufficient accessibility in digital learning technologies, and the absence of or limited inclusive design practice in computing and design education. Although initiatives like the W3C's accessibility fundamentals have made progress, resource disparities and inadequate global harmonization highlight the need for user-centered strategies that co-design participation and introduce students early to accessibility and inclusive design before joining the tech industry.

In this dissertation, emerging educational technologies like virtual labs (V-Labs) were used to illustrate gaps in the accessibility elements of existing digital learning technologies. While such technologies can enable safe, gamified, and self-directed learning, they often exclude users through the inadequacy of their accessibility features (F. G. Deriba et al., 2024). For instance, learners with visual impairments or linguistic or cultural barriers face obstacles in engaging with virtual experiments. Meanwhile, AI's potential to democratize accessibility education remains underexplored (Paper I). While studies have emphasized AI's role in enhancing motivation and self-efficacy (Jia & Tu, 2024), its application in supporting inclusive design skills, such as accessible UI creation or WCAG compliance evaluation, is nascent (paper IV).

This dissertation addresses two relevant aspects. First, it critiques the exclusionary design of digital technologies; the researcher examined V-Labs, proposing participatory design and how students could include accessibility elements and inclusive design to improve inclusivity in the design process. Second, it investigates how M-LLMs (e.g., Gemini, ChatGPT) can scale accessibility education by providing equitable feedback, enhancing

problem-solving skills, and fostering autonomy in project-based learning (PBL; Paper III). In this dissertation, our motivation is not to specifically examine accessibility among person with disability or to directly assess the accessibility features of the AI tools used. Instead, we aim to equip students (aspiring digital technology designers and developers) with the knowledge, awareness, and competencies to create inclusive technologies by leveraging AI-based interventions into HCI and web development curricula, as well as the skills to design and develop technologies through inclusive practices and advocate for policy reforms in institutions in developing regions like Ethiopian Higher education to prioritize accessibility as a core educational mandate.

## 1.2 Problem Statement

While educational technologies are making progress, the challenges of digital accessibility in providing equitable access and inclusivity for diverse populations of the developing regions have not been adequately addressed (Kulkarni, 2019). Although digital technologies provide across diverse sphares of human life, their inclusive design, development, and implementation retain some exclusionary practices. Four interrelated challenges underpin this issue.

The first is lack of accessible design practices. Current digital educational platforms frequently fail to meet accessibility standards due to insufficient awareness of accessibility among professionals such as digital technology designers and developers regarding legal mandates (e.g., ADA, UNCRPD) and industry obligations for developing accessible design (Abuaddous et al., 2016; Durdu & Yerlikaya, 2020). The lack of developer awareness is a technical oversight and a failure that entrenches exclusion. According to the literature, this exclusion of design practice may be due to the accessibility of knowledge gaps, or a lack of motivation (M. V. R. Leite et al., 2021). Furthermore, the inherent complexity of accessibility standards and guidelines, entails diverse considerations related to visual, auditory, and cognitive design (Kulkarni, 2019). Therefore, educational institutions and industry practices should prioritize accessibility literacy, particularly in underdeveloped areas, to

make digital platforms serve all people equally. Furthermore, despite the availability of WCAG guidelines and UDL frameworks that guide inclusive practice and learning accessibility, many stakeholders (IT professionals, web designers or developers, and educational faculty) lack or have limited familiarity with accessibility and inclusive design-related resources, leading to underutilization in practice (Sanderson et al., 2022). To this end, during and after the design of technologies, accessibility testing is often perceived as time-intensive, reducing motivation among developers to prioritize inclusive design (M. V. R. Leite et al., 2021).

Second, there is a gap in accessibility and inclusive design practice through training and in formal school education (Khribi et al., 2022). A critical bottleneck lies in computing education curricula, which emphasize technical proficiency over accessible design practices. The marginalization of accessibility topics or content within curricula, often considered as elective categories instead of a core component or omitted entirely (Diasse & Kawai, 2025), excludes students from foundational exposure to inclusive design. Consequently, graduates join the workforce inadequately prepared to meet the requirements of 1.3 billion persons with disabilities, thereby sustaining the ongoing problem of limited accessible technology (Persson et al., 2015). This pedagogical oversight increases the "digital divide" in education (Friedman, 2024; Tuikka et al., 2018), as well as raises legal and ethical problems because nations implement digital accessibility legislation (Parthasarathy & Joshi, 2024).

Third, educational interventions to overcome accessibility challenges are limited. Efforts to bridge this gap, such as Universal UDL frameworks, remain constrained by inconsistent integration into core curricula. In regions like the Global South, educators are overburdened to deliver instructions, including accessibility and design education, due to large class sizes, limited instructor expertise in accessibility and inclusive design-related areas, and inadequate feedback mechanisms to students in teaching and learning accessible design, further hindering effective pedagogy (Iraj et al., 2021; Maringe & Sing, 2014). Traditional feedback modalities, reliant on subjective evaluations and instructor-dependent critiques, struggle to scale personalized guidance essential for retaining accessibility competencies (Carless & Winstone, 2023; Selvaraj et al., 2021).

Fourth, the potential of adaptive technologies like AI to fill gap in the accessibility experts is underexplored. Emerging AI-based tools, particularly M-LLMs, offer unprecedented opportunities to deliver scalable, standards-aligned feedback in accessibility education. However, their application in cultivating domain-specific skills, such as inclusive UI development, remains nascent (Kishore et al., 2024; Torres et al., 2023). In existing studies, empirical insights into how AI-assisted feedback compares to human expertise in fostering accessibility principles in design education are scarce (paper IV), particularly for learners in resource-constrained contexts like Ethiopia (Kim et al., 2022).

These challenges underscore a supplemental need to reimagine pedagogical strategies to leverage adaptive technologies, such as AI, in accessibility education. By investigating how AI-based tools could augment traditional teaching practices, particularly in personalizing learning pathways, this study seeks to address critical gaps in education and practice.

## 1.3 Research Objectives

Recognizing the global consensus regarding the importance of accessibility as a fundamental human right and a key factor in creating a digitally inclusive environment for all, different stakeholders, such as educators, designers, developers, and academic institutions, need to prioritize accessible design (Persson et al., 2015). This includes promoting a culture of inclusion in all aspects of digital design and development, such as designing websites and applications that are usable by all. Although schools, designers, and technology industries primarily bear this responsibility, the inaccessibility or partial accessibility of digital technologies and the dearth of accessibility knowledge are apparent in formal education, which requires an urgent need to promote awareness and provide interventions to tackle these issues. Specifically, this research identified two specific problems, which led to the development of these research objectives and research questions that will be addressed throughout this dissertation. First, the lack of accessibility in digital learning environments, specifically V-labs (F. G. Deriba et al., 2024),

indicates a gap in the accessibility of existing learning technologies and environments, raising critical concerns that need to be answered. Second, according to Campoverde-Molina (2023), there is a gap in the accessibility of educational resources, such as websites, particularly in under-developed regions. Our study also confirmed the existence of these problem and the need to overcome them (paper IV).

The overarching goal of this dissertation is to advance accessibility and inclusivity in learning environments for digital educational technology design and development education, thereby providing a more equitable and accessible digital future for all learners and educational stakeholders. Furthermore, this study contributes to the theoretical and practical understanding of how AI could be effectively leveraged to support inclusive design practices in software development. Specifically, the accessibility of V-Labs and Ethiopian higher educational technology websites is evaluated. Thereafter, student learning of UI design and development was assessed by utilizing support from M-LLMs and assessing their effectiveness in integrating LLM feedback into their design. Additionally, the study aims to understand how AI-based feedback enhances software development students' technical accessibility awareness and practical knowledge of inclusive design practices. Thus, this dissertation aimed to address accessibility challenges such as gap in design practices through three interconnected objectives.

First, it assesses the status of accessibility in educational technologies, particularly by conducting a comprehensive analysis of existing accessibility features in V-labs and understanding how these labs were usable and accessible to all users (Paper I). Additionally, the researcher examined Ethiopian higher educational Institutions (EHEI) educational website accessibility through student-led auditing using PjBL (paper IV). This elucidated existing gap in learning technologies and professional accessibility experts, which created a foundation for our second study.

Second, building on the first paper, the second objective was to explore the potential of AI-based solutions, particularly M-LLM, in addressing accessibility challenges in existing educational technologies - specifically, in the teaching and learning of accessibility and inclusive design practices. Therefore, the researcher identified the need for interventions with AI to tackle accessibility

and inclusivity challenges in learning settings, particularly by offering software development students early support. This support would equip learners with the technical skills and practical knowledge of accessibility and inclusive design practice in design and software development education, as computing education produces the software developer, designer, and tester. Therefore, we investigated how AI-assisted interventions, particularly M-LLMs (e.g., Gemini, ChatGPT), support and enhance students' understanding of accessibility concepts, principles, frameworks, and inclusive design practices through PjBL. Further, the study measured the change in students' accessibility knowledge and skills after using M-LLMs and evaluated the impact of AI-based feedback on students' ability to design accessible UIs in PjBL (paper IV).

The third objective is to understand the lasting effectiveness of AI in students' accessibility learning. Specifically, the researcher compared M-LLM-supported learning with traditional instructional methods for teaching accessibility concepts. A key aspect of this comparison was evaluating the impact of AI-based feedback versus instructor feedback on students' accessibility proficiency. Furthermore, the researcher assessed the efficacy of the different feedback modalities in the context of higher education web accessibility tasks, examining both the short- and long-term effects on students' accessibility knowledge and problem-solving skills.

## 1.4 Research Questions

The research objectives of the dissertation, as stated in section 1.3, are operationalized through the following research questions:

> **RQ1:** What accessibility elements are implemented in learning technologies, specifically in V-Labs, and how do they accommodate users with diverse needs?

This RQ was answered by paper I. As such, the main goal of Paper I is to ascertain the effectiveness of educational technologies in terms of accessibility and inclusivity. The study aims to identify the gap in current knowledge by

synthesizing empirical studies to evaluate accessibility from four different perspectives including, instructional design features and content availability, interaction support features, persons with disabilities, and diverse linguistic and cultural backgrounds. Furthermore, the study examines the assessment methods used in evaluating V-labs and highlights areas that require further research to improve accessibility.

> **RQ2:** How do Multimodal LLMs support software development students' awareness of accessibility concepts and their ability to integrate inclusive design practices into UI design?

This question was answered by Papers II and III. Paper II answers RQ2 by examining the effectiveness of AI tools, particularly M-LLM, in enhancing student learning of UI design within a Human-Computer Interaction (HCI) course. It explores how these tools support accessibility in UI design. Additionally, the study investigates students' perceptions of M-LLM in developing accessible UI designs, identifying both benefits and challenges to inform future AI-assisted design practices. Furthermore, RQ2 was answered in Paper III by investigating how M-LLMs can enhance software development students' understanding and implementation of accessible UI design, thus addressing gaps in accessibility education. By conducting a 12-week teaching experiment with undergraduate computer science students, the research aims to evaluate AI's pedagogical role in improving accessibility knowledge, designing applications, and cultivating self-efficacy while identifying challenges of using M-LLMs in the design process.

> **RQ3:** To what extent does AI-based feedback improve students' proficiency in identifying and resolving accessibility issues in design education compared to instructor feedback in the long and short term?

RQ3 was answered by paper IV. The primary goal of this question is to investigate the effectiveness of AI-generated compared to instructor-delivered feedback in teaching accessibility and inclusive design practice to undergraduate computer science students. The RQ3 will be further answered

by assessing how different feedback sources impact students' short-term and long-term performance of accessibility concepts in a web development course. Specifically, it examines whether AI-generated feedback can match or surpass human feedback in improving students' understanding and application of accessibility principles, as well as whether any gender-based differences exist in students' learning outcomes.

To answer the above-stated three RQs, section 1.4, particularly in Table 1, outlines what kind of research design and methodology the study followed.

**Table 1:** Relationships of the research questions (RQs) with methods and publications

| RQs | Research method and strategies | Publication | Research Goal |
|---|---|---|---|
| RQ1 | Systematic review | Paper I | To assess the status of accessibility in digital educational technologies and identify existing gaps. |
| RQ2 | Qualitative method, including interviews | Paper II | (i) To explore the role of AI tools in enhancing UI design accessibility, focusing on student experiences and perceptions. |
| | Mixed method including artifact collection, Quasi-experimental design including pre-post test | Paper III | (ii) To investigate the effectiveness of M-LLMs in improving software development students' comprehension and implementation of accessible UI design. |
| RQ3 | Quantitative method, Quasi-experimental design includes a written exam, pre-post questionnaires | Paper IV | To provide empirical evidence on AI-generated feedback versus instructor-delivered feedback on students' learning and understanding of accessibility principles and inclusive design skills in web development. |

## 1.5 Contributions of the Dissertation articles to the study

While this research makes several contributions to computing education, accessibility, and design learning, each article contributed uniquely to the study. First, practically, the study provides insights into how AI-assisted accessibility education by linking SDT to inclusive design outcomes, demonstrating how autonomy and competence mediate students' engagement with AI, particularly M-LLMs. Furthermore, the dissertation provides actionable insights on V-Lab accessibility by outlining design accessible cultural, linguistic, and inclusive elements and provides insights on how AI feedback can support students in learning inclusive design using M-LLMs in accessibility education.

Methodologically, the study advocates hybrid evaluation metrics, combining accessibility audit scores with SDT-aligned psychosocial scales to assess AI's pedagogical impact. Finally, societally, the work promotes policy reforms to prioritize accessibility as a core competency in computing education, reducing the digital divide by equipping students (the future technology developers) with inclusive design skills. Specifically, each article makes unique contributions to the dissertation as follows:

**Paper I**: it contributes conceptually to the dissertation by identifying diverse virtual learning environments across disciplines, by assessing V-Labs' accessibility in terms of usability, user experience, and engagement, and analyzing key accessibility features for all students, diverse backgrounds, instructional design, and interaction support. It also highlights existing gaps and calls for future research to enhance inclusivity and improve accessibility assessment methods in learning design.

**Paper II**: This article practically contributes to the dissertation by analyzing students' experiences in integrating AI tools into UI design education, providing insights into how AI tools enhance accessibility awareness and usability in HCI projects. The study identifies the benefits of AI in developing inclusive UI development. Additionally, it highlights challenges such as AI response variability, integration difficulties, and accessibility knowledge gaps. The study further emphasizes the need for AI training in design education and suggests future research directions to refine AI-assisted accessibility improvements in UI development.

**Paper III**: contributed practically and methodologically to the dissertation by experimentally demonstrating AI's role in accessibility education. The article highlights AI's role as a collaborative design partner and accessibility information retrieval tool. The research identifies distinct AI engagement patterns among high- and low-achieving students, revealing differences in strategic learning versus reliance on AI corrections. It also underscores the need for AI training in accessibility education, advocating for pedagogical frameworks that integrate AI tools into HCI curricula to foster inclusive design practices.

**Paper IV**: Finally, this study contributed practically and methodologically to the dissertation by providing empirical evidence on the comparative

effectiveness of AI-generated and instructor-delivered feedback in design education. While the study finds that AI feedback is similar to human feedback in immediate post-intervention performance, it highlights the advantage of human feedback in lasting support for students' accessibility concepts. Additionally, it sheds light on the limitations of AI-generated feedback in supporting cognitive processes critical for learning about accessibility and inclusive design education. The study also explores gender-based trends, revealing that while AI feedback may initially benefit female learners slightly, it does not sustain long-term engagement and understanding. These findings reinforce the importance of integrating AI as a complementary tool rather than a replacement for human instruction in accessibility education as well as the value of recommended hybrid feedback model.

Therefore, by AI, such as Multimodal LLM, in pedagogy, particularly accessibility in design education, this dissertation provides insights on how educational technologies can support a more inclusive learning.

## 1.6 Structure of the Dissertation

This dissertation is structured into seven chapters. *Chapter One, Introduction,* provides the general context of the study. This includes an overview, a statement the problem the study aims to address, the research objectives and questions, highlights of the research methodology used, and finally, the study's contribution. *Chapter Two, Literature Review,* lays the background for the dissertation. This chapter covers the context, concepts, and definitions of accessibility, explores the foundations of accessibility in digital education, and discusses strategies and standards for assessing and evaluating design accessibility. *Chapter Three, Background on AI in Education,* offers a brief overview and background of how AI facilitates teaching and learning practices in education. It provides an overview of the paradigm shift AI brings to education and discusses the challenges and issues observed in AI education. Chapter four, *Reviews the previous literature on AI, Accessibility and Education*, examines existing studies that are related to this dissertation, identifies gaps in the previous literature, and provides an overview of how

this study addresses these gaps. Specifically, it first explores the role of AI in accessibility education and closes with how AI supports, in the long-short term, overcoming accessibility barriers in design and accessibility education and its lasting impact.

Following these foundational chapters, *Chapter Five, Research Methodology,* presents the research approach and specific methodology used in the dissertation. This section details the methodological procedures, including an overview of the research design, the research context and participants involved in the study, the research procedures followed in the empirical studies, the data collection methods required to investigate the studies, the techniques used for data analysis and preprocessing, and the research ethics adhered to throughout this dissertation. *Chapter Six, Results of the Study,* then presents the findings. It begins with an overview of the findings, followed by those related to problem identification and needs analysis obtained from the findings of both secondary and primary studies. Subsequently, the study presents findings related to AI in accessible and design education, structured through quantitative insights into student experiences in UI design and the role of M-LLMs in their design learning process. This includes findings on accessibility awareness, self-efficacy, and the impact of AI on the design process; insights from epistemic network analysis (ENA) of student experiences with Generative AI; and findings from a heuristic evaluation of student-developed UI in project-based learning. Finally, this chapter presents findings related to feedback in accessibility and inclusive design education__ specifically, a comparative analysis of AI and instructor feedback's impact on accessibility learning.

The *final chapter, Chapter Seven, Discussion and Conclusion,* presents a discussion guided by research questions, concludes the study, and suggests future research directions. The discussion begins with a reflection on each research question posed in this dissertation. This is followed by a reflection on the overall study's findings and their implications for various stakeholders, including educators and academic institutions, AI developers and researchers, UI or UX designers, developers, and future practitioners, as well as the broader technology industry and society. This chapter also details the dissertation's

contributions, acknowledges its limitations, and, finally, outlines future directions for further study relevant to different parties.

# 2 LITERATURE REVIEW

This chapter presents a comprehensive review of the existing body of literature relevant to this study, focusing on digital technology accessibility. First, it provides an overview of the basic accessibility context, concepts, and definitions used in this dissertation. Following this, the chapter describes a basic outlay of the landscape of digital technologies by exploring how the accessibility of digital technologies has evolved. Subsequently, it examines theories, guidelines, methodologies and pedagogies relevant to this dissertation, particularly those that guide the design and development of accessible digital technologies and support the teaching and learning of accessibility and inclusive design practice in computing education. The chapter then explores the strategies and standards used for assessing and evaluating design accessibility. Finally, it investigates how AI has been used to overcome accessibility barriers, especially the potential of Generative AI (GenAI) in providing personalized user support and offering intelligent assistance to students during the inclusive design of systems or software.

## 2.1 Accessibility: Context, Concepts, and Definitions

Given the pervasive and increasing reliance on digital technologies across all facets of modern society, it is important to understand why accessibility matters (Kulkarni, 2019). First, accessibility is essential for creating digital products for a wider audience. This practice promotes inclusivity and equity, removes barriers, and improves the overall user experience, enabling full participation in digital society (Gilbert, 2019). Second, it is necessary to fulfill the ethical, social, and legal imperatives of digital inclusion and accessibility for all users. Several domains, including websites, software, the built environment, and education, require accessibility, and various standards and legislations have been amended to mandate accessibility in each domain. Although this study focuses on digital accessibility and inclusive design education to

establish a robust foundation, it is essential to acknowledge the theoretical and conceptual underpinnings of accessibility as defined by various actors.

The term “accessibility” is multifaceted, and understanding its diverse interpretations across different bodies is essential for a comprehensive analysis. Numerous stakeholders, including architects, engineers, researchers, healthcare professionals, policymakers, regulatory bodies, standard development organizations, advocacy groups, non-governmental organizations, educators, and academic institutions, have defined accessibility based on their specific context (Iwarsson & Ståhl, 2003). For example, the World Wide Web Consortium (W3C), which focuses on the accessibility of web content, applications, and authoring tools, outlines its POUR principles ((WWC), n.d.), whereas the UN emphasizes equal rights and barrier removal across all life domains. In contrast, regulatory bodies like the U.S. Access Board often focus on technical standards, ISO underscores broad usability, and organizations of persons with disabilities highlight the importance of dignity and full participation (Gilbert, 2019). Researchers, in turn, tend to adopt a more analytical, pedagogical, and theoretical approach to understanding and defining accessibility. Thus, while the core concept of accessibility remains consistent, which is removing barriers, different bodies emphasize distinct aspects based on their specific mandates and perspectives.

For example, W3C defines web accessibility, guided by POUR principles (Perceivable, Operable, Understandable, and Robust), stating that websites, tools, and technologies must be designed and developed for universal usability. Specifically, they state that users must be able to perceive information and UI elements through their senses, ensuring all content is perceivable. Users should also be able to operate UI components and navigation without encountering impossible interactions. Furthermore, users should understand both the information presented and how to operate the interface. Lastly, content has to be robust enough, for example, to be interpreted by various user agents, such as assistive technologies, and remain accessible as technology evolves.

Similarly, the United Nations (UN), particularly through the Convention on the Rights of Persons with Disabilities (CRPD) (Assembly, 2007), emphasizes accessibility as equal access and the elimination of barriers across all

domains, framing it as a fundamental human right applicable to all aspects of life. Specifically, Article 9 of the CRPD ensures equal access for persons with disabilities to various aspects of life. This includes the physical environment and public facilities or services such as transportation, information, or communication technologies in both urban and rural areas. These measures require identifying and eliminating obstacles and barriers to accessibility across different regions, especially where there is a gap in implementing accessibility.

Considering these diverse definitions and facets of accessibility, this dissertation adopts the WCAG definition as its primary framework due to its specific relevance to inclusive design in digital technologies.

Furthermore, while expressions such as *"differently-abled," "disabled people,"* or *"individuals with special needs"* appear in public and academic discourse, this dissertation adopts the term *"persons with disabilities"* in accordance with the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD, 2006). This terminology aligns with a person-first and rights-based perspective, emphasizing dignity, autonomy, and full participation of individuals in society rather than focusing on their impairments.

## 2.2 Foundations of Accessibility in Digital Education

Understanding the evolution of accessibility in education provides important context for practices in the design and development of digital technologies as well as future directions. Specifically, understanding their evolution helps developers, institutions, and educators stay compliant with legal mandates and uphold ethical standards of inclusivity (Rutter et al., 2007); therefore, it also allows educators to adopt more effective teaching strategies grounded in universal design principles (Rao & Meo, 2016). This inspires further innovation in creating more user-friendly and adaptable digital learning tools and promotes a culture of inclusion within the educational community (Bong & Chen, 2024). While there have been several debates regarding the precise trajectory of digital accessibility, its current state has been shaped by historical developments and key influencing factors (Hanna, 2018; Keates,

2005). Although a definitive starting point for digital accessibility in education remains a subject of scholarly discussion, researchers have highlighted its roots in the pre-web era with initiatives aimed at democratizing learning opportunities for marginalized groups, including adult learners and individuals facing physical or socioeconomic barriers. We have framed these developments into six stages or eras.

First, the early stage spanned from the pre-web to the early internet age. During this period (Vanderheiden et al., 2022), the initial focus was twofold: ensuring physical access to learning environments and providing assistive technologies for all users. The disability rights movement championed the understanding that societal barriers, stemming from prejudicial attitudes and exclusionary practices, posed far greater obstacles to participation (Scotch, 1989). Initially, the concept of accessibility in education largely centered on physical access to learning environments and the provision of assistive technologies, such as Braille, large print materials, and specialized equipment (Segal & Marciano, 1988). Second, the global disability rights movements of the 20th century (Scotch, 1989) advocated for equal opportunities in all aspects of life, including education. These efforts laid the foundation for addressing barriers beyond the physical. Subsequently, the early adoption of computers and assistive software, such as screen readers and alternative keyboards, highlighted the potential of technology to both offer solutions and create new barriers (F. K. Ahmad, 2015; Botelho, 2021).

The second era, commencing in the late 20th century, was marked by the rise of the internet and the increasing need for digital accessibility (DiMaggio & Hargittai, 2001). First, the advent of the World Wide Web (WWW) introduced new barriers. While the evolution of accessibility considerations in digital education predates the widespread adoption of the internet, the emergence of the WWW in the 1990s, offering unprecedented educational resources, simultaneously revealed that websites and digital content often presented new accessibility challenges for diverse users (Shneiderman, 1997). Following this, early advocacy for web accessibility by persons with disabilities, user advocates, and researchers began to call attention to the inaccessibility of the web and the urgent need for guidelines and standards (Badre, 2002; Cunliffe, 2000; Schmidt, 1997). This advocacy was instrumental in the development

of the WCAG. The Web Accessibility Initiative (WAI) of the W3C was formed in 1997 and released the first version of the Web Content Accessibility Guidelines (WCAG 1.0) in 1999 (Consortium, 1999). This was a foundational moment, providing the first internationally recognized set of guidelines for making web content accessible. WCAG 1.0 focused on technical aspects and provided checkpoints for developers.

The third stage involved the development of educational theories and frameworks related to accessibility, inclusive design, and learning. More details of these are presented in section 2.3. For instance, Universal Design, originating in or closely linked to the fields of architecture, building environment, ergonomics, and product design, has gained relevance across a much broader set of domains, including education (Story et al., 1998). In the educational context, the core idea of UDL, also known as Universal Design for Education, was to design and develop flexible learning materials and activities. Subsequently, the development of UDL, based on research in learning sciences, was formalized in the early 2000s (Rose, 2000). UDL provided a pedagogical framework for creating flexible and inclusive learning experiences by addressing the what, how, and why of learning.

The fourth stage encompassed legislative and policy advancements. This stage commenced with the strengthening of disability rights laws. Various stakeholders, for example, disability organizations and legal scholars, found that existing disability rights legislation across countries began to be interpreted and updated to include digital accessibility (Lazar et al., 2015). This further contributed to the development of new accessibility legislation due to the innovations of advanced digital technology. Following this, different countries and regions started enacting specific legislation related to digital accessibility in various sectors, including education: for instance, Section 508 in the US (Mueller, 2008), and accessibility laws within the European Union (Ferri & Favalli, 2018). This pushed forward the adoption of accessibility policies by educational institutions. Subsequently, educational institutions across regions began to examine existing gaps in this matter (Beyene et al., 2023) and implement the developed accessibility policies (even though the implementation varies across regions due to resource, infrastructure, and

expert gaps) to ensure their digital resources and platforms were accessible to all students.

The fifth stage involved the refinement of accessibility-related and inclusive design guidelines and standards. For instance, the refinement of WCAG has been an ongoing process. While the initial WCAG 1.0 has undergone several revisions, for example, WCAG 2.0 in 2008 (Caldwell et al., 2008) WCAG 2.1 in 2018 (W3C, 2018), and the ongoing development of WCAG 3.0 (WWC, 2024), updates reflect the dynamic nature of accessibility challenges and solutions. The detailed design principles, guidelines, success criteria, and conformance levels of WCAG versions are presented in section 2.4. These guidelines have been updated continuously by considering the features of new technologies to address the challenges of a wider range of users and provide more testable success criteria. Furthermore, the development of the Authoring Tool Accessibility Guidelines (ATAG) and the User Agent Accessibility Guidelines (UAAG) represented another development by WAI to make the tools used to create and access web content more accessible (Gunderson, 2004).

The sixth stage was marked by an increased focus on inclusive design and user experience (UX) in the technology industry and academic sectors. This involved a shift from mere compliance with technical accessibility standards to a deeper understanding of all users through inclusive design (Gilbert, 2019). Thus, the increasing emphasis was placed on inclusive design practices, advocating for the incorporation of accessibility considerations early in the design process, rather than as an afterthought (Gilbert, 2019; Worsley & Bar-El, 2022), with recognition of the value of involving individuals with disabilities in the design, development, and testing of digital educational resources. Finally, the recent state is characterized by the exploration of pedagogical approaches for accessibility (Lewthwaite & Sloan, 2016; Putnam et al., 2016); the need for providing awareness of accessibility principles and concepts to students, educators, instructional designers, and technology developers; and the importance of digital accessibility (Gilligan, 2020; Lewthwaite et al., 2023). Thus, the ongoing consideration of technological advancements, with constantly emerging new technologies and features, requires the continuous adaptation of guidelines and practices. This shows how the foundations of accessibility in digital education have evolved from an initial focus on physical

access and assistive technologies to a more comprehensive understanding including technical standards, pedagogical frameworks, legal mandates, inclusive design principles, and a user-centered approach.

## 2.3 Theories, Guidelines, and Pedagogies for Accessibility and Inclusive Design

Understanding the theoretical foundations, practical principles, guidelines, and pedagogical approaches that support accessibility education and inclusive design skills is necessary for social inclusion and individual rights. Although inclusive design has a strong history, dating back to the 1970s with Ron Mace in the product design profession (King-Sears, 2009), the theoretical aspects of its practice, as well as the opportunities and challenges of inclusivity, are poorly understood, particularly in academic settings (Clarkson et al., 2013). While foundational theories such as ”universal design” in the US and its European counterpart, ”design for all,” have emerged, initiatives to implement inclusive design practices are needed (Persson et al., 2015). These theoretical perspectives, including universal design and inclusive design, provide the essential groundwork for understanding why accessibility and inclusive design education are paramount.

Building upon these theoretical foundations, various guidelines, such as WCAG and Universal Design Principles, have been developed to address accessibility barriers across diverse user groups in accessing digital technologies. For instance, while WCAG are technical standards that help make the digital world accessible to all people, the universal design principles guide the design of products and environments (Burgstahler, 2009). As Burgstahler (2009) explained, the seven universal design principles (see Figure 1) can inform both the overall structure of instruction and specific elements such as materials, facilities, and teaching strategies, including lectures, classroom discussions, group activities, web-based learning, laboratory work, fieldwork, and demonstrations. Building upon this, a universally designed curriculum ensures that students with a wide spectrum of disabilities , ethnic origins, language skills, and learning styles are provided

multiple means of representation, expression, and engagement (King-Sears, 2009). While the uniform implementation of these guidelines and principles remains a challenge across regions due to different factors like a lack of experience among educators, inadequate resources, and a shortage of trained professionals, the principles and guidelines offer specific recommendations for creating more accessible digital content and learning materials. Therefore, such principles serve as practical tools that translate the broader theoretical concepts into practical steps for developers and designers, outlining what needs to be done to improve accessibility.

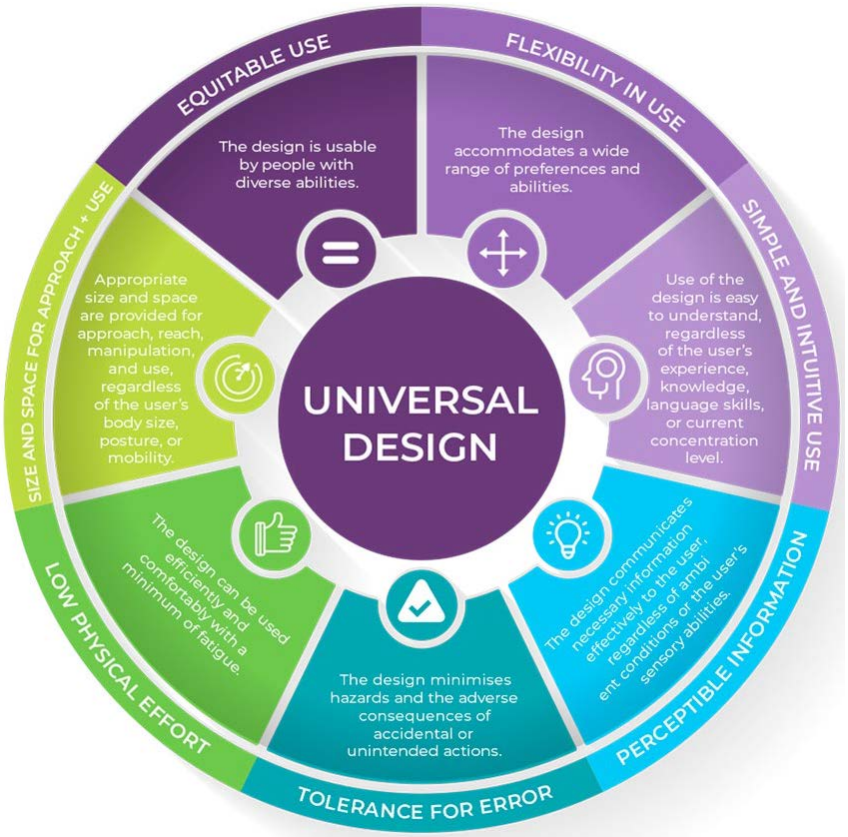



**Figure 1:** Seven universal design principles (Burgstahler, 2009)

Besides these guidelines, some pedagogies are used in the teaching and learning of accessibility and inclusive design in computing education. Widely used pedagogies include the UDL framework (CAST, 2024), culturally sustaining pedagogy (Paris, 2012), project or problem-based learning

(D'Elia et al., 2025), and collaborative or participatory approaches. Educating students about accessible design that aligns with pedagogical frameworks can enhance their technical accessibility skills and provide knowledge and awareness of standards with principles of inclusive design. For example, Rehman et al. (2024) underscored that teaching students with experimental design is essential to assessing collaborative skill development, in our case inclusive design skill, in students engaged in PjBL activities.

A range of pedagogical approaches foster this connection; for example, UDL underscores the proactive integration of accessibility considerations throughout the entire design process, encouraging students to address the diverse needs of all potential users from the initial concept to the final implementation (Burgstahler & Cauce, 2020; Hyatt & Owenz, 2024). Similarly, PjBL, collaborative, and participatory approaches facilitate iterative design cycles where students develop design prototype solutions and evaluate them against established accessibility guidelines (Xu et al., 2020).

Therefore, instructing students aligning with the accessibility principle and inclusive design in design education helps them understand accessibility design competence and problem-solving skills (Diasse & Kawai, 2025; Kulkarni, 2019). The areas where accessibility concepts should be promoted are in assessing the accessibility of existing products, websites, or environments and designing products or software using recognized guidelines and principles (Putnam et al., 2016), which are all supported by relevant pedagogical approaches or theories (e.g., cognitive load theory and self-deterministic theory were used in our dissertations-Paper IV).

## 2.4 Strategies and Standards for Assessing and Evaluating Design Accessibility

After the design and development of learning technologies, whether the designed products are accessible to learners or users should be evaluated (Seixas Pereira & Duarte, 2025). There are several methods and approaches to examine the accessibility of digital platforms, including websites and mobile applications, both during and after their development. These methods assess

whether the platform or device complies with established standards and guidelines. Hence, educating students about accessible design necessitates that they internalize various evaluation strategies as ongoing tools for refining their creative work. For example, adherence to WCAG is often presented as a final verification step rather than as a lens through which to critically examine design decisions throughout the prototyping process (Campoverde-Molina et al., 2020). According to Kumar et al. (2021), assessing compliance with the four WCAG principles requires understanding several guidelines under each principle, and for each guideline, there is a need to recognize testable success criteria at three levels: A (the most basic), AA (widely accepted), and AAA (highest level).

Initially, while WCAG 1.0 did not possess formal design principles (although the guidelines addressed similar concepts), it comprised 16 guidelines and utilized 65 checkpoints, which later served a similar function to the "success criteria" in WCAG 2.0 (see Figure *2*). The subsequent iteration, WCAG 2.0, represented an improvement, incorporating 4 principles, 12 guidelines, 61 success criteria, and 3 conformance levels (A with 21 criteria, AA with 13, and AAA with 23). Building upon this, WCAG 2.1 version maintained the 4 principles and 13 guidelines but expanded to 78 success criteria across the same three conformance levels (A with 30 criteria, AA with 20, and AAA with 28). This version introduced 2.5 input modalities and 17 new success criteria. The most recent guideline, WCAG 2.2, similarly retains the four principles and 13 guidelines of its predecessor but includes 86 success criteria within the three conformance levels (A with 34 criteria, AA with 23, and AAA with 29). Further, eight new success criteria are added in this version. Looking ahead, WCAG 3.0 is currently in draft, representing the next generation of web accessibility guidelines designed to further enhance digital content accessibility for all individuals and to address evolving technologies and user needs.

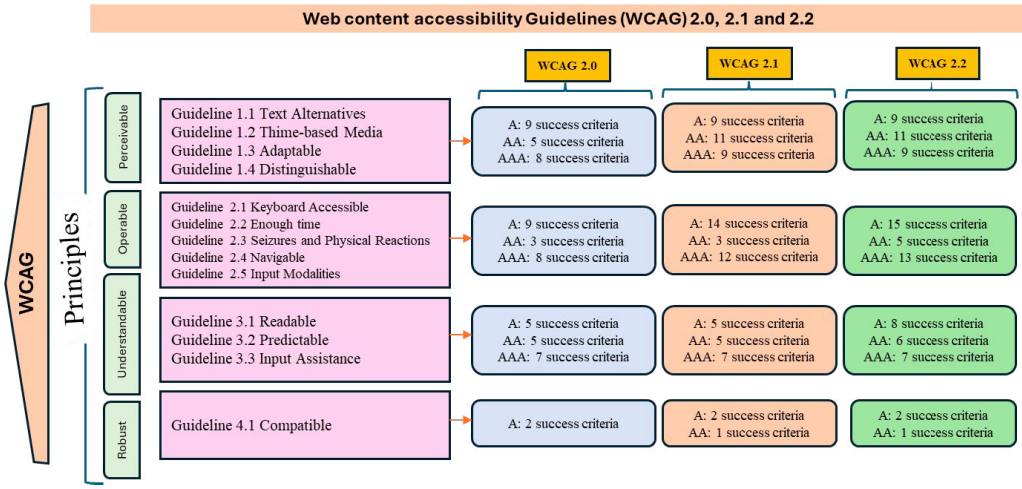



**Figure 2:** Overview of web content accessibility guidelines (WCAG) version 2.0, 2.1, and 2.2, adopted from (Ara et al., 2024)

A variety of guidelines and foundational principles exist for measuring and assessing the accessibility of systems, websites, and applications, with the specific choice often depending on the context of the design's development (Ara et al., 2024; Longley & Elglaly, 2021). While numerous studies have employed these WCAG guideline, their evaluation, involves expert reviewers assessing a website or application against a defined set of guidelines and success criteria to assess the accessibility of digital content (Campoverde-Molina et al., 2020). It is frequently employed in conjunction with other methods to efficiently determine usability and accessibility issues. While WCAG evaluation typically involves expert reviewers assessing a website or application against a defined set of guidelines and success criteria, it is frequently employed in conjunction with other methods to efficiently determine usability and accessibility issues (Acosta-Vargas et al., 2018; Debevc et al., 2015).

A second common method is heuristic evaluation, which involves a usability inspection combined with user testing techniques aimed at identifying usability problems within UI designs across various types of applications and websites. This approach provides valuable insights for enhancing the overall user experience (Thyvalikakath et al., 2009). Several researchers, such as Dharwaman et al., have utilized heuristic evaluation to identify challenges users encounter when interacting with a product and to propose solutions

for these issues (Dharmawan et al., 2024). This evaluation method, integrated into design-based learning activities, empowered students to identify potential accessibility barriers within their peers' design concepts (Debevc et al., 2015) and helped to ensure that products were aesthetically pleasing and provided a smooth and intuitive user experience (Thyvalikakath et al., 2009). In contrast, the initial heuristic evaluation framework was developed by Nielsen in 1990 (Nielsen, 1995) and subsequently refined by various researchers (Lima & Hwang, 2024) based on diverse criteria. These criteria include the scope of evaluation (e.g., specific aspects of the UI), the intended target audience (e.g., novice or expert users), specific user needs (e.g., accessibility requirements), the types of usability problems being investigated (e.g., navigation difficulties), and the specific context and domain of use (e.g., mobile or desktop environments) (Hermawati & Lawson, 2016).

Third, web automated tool testing has been serving as another important evaluation method for assessing the accessibility of various applications and websites. These accessibility tools, which can be freely available (such as WAVE[1], Axe, and Lighthouse) or commercial (like accessiBe), automatically scan for compliance with established accessibility guidelines (Acosta-Vargas et al., 2018). However, automated tools can sometimes generate false positives or negatives, necessitating human review to interpret the results accurately and identify issues that the tools might miss.

The last evaluation method is user testing, which involves observing real users of the system, including persons with disabilities, as they interact with the website or application to uncover accessibility barriers (Petrie & Bevan, 2009). While this method yields direct and invaluable feedback on the actual user experience, it can be resource-intensive to implement effectively (Ara et al., 2024).

Finally, many researchers and web accessibility testers employ combined evaluation methods, such as heuristic evaluation, WCAG compliance checks, and user testing, to comprehensively address both technical adherence to standards and real-world usability. This combined method allows for the

[1] WAVE Web Accessibility Evaluation Tools

efficient initial screening of large amounts of content (Debevc et al., 2015). For example, automated tools can rapidly identify code-level issues, hereas user testing can reveal practical usability problems that automated checks might miss (Acosta-Vargas et al., 2018). Therefore, a combination of automated and manual testing was recommended for a comprehensive evaluation.

Within educational settings, various researchers have utilized several evaluation tools and approaches to assess the design of systems and applications or in teaching students for digital technology accessibility. Freely available automated tools like WAVE are frequently used in teaching and learning practice in the design workflows (Massengale & Vasquez III, 2016). They provide immediate feedback on website accessibility issues, empowering students to address these concerns and refine their website (Acosta-Vargas et al., 2018). For instance, a study by Akgul et al (2024) and Mohammadi et al (2024) evaluated the quality and accessibility of university websites using automated web testing tools. While automated tools offer efficiency in quickly identifying apparent accessibility violations, they often require supplementation with manual reviews to detect accessibility issues that the tools may overlook (Mehralian et al., 2022). In addition, heuristic evaluation was used in academic settings while teaching and learning to evaluate the digital tools. For example, Debevc et al. employed a heuristic evaluation on different online educational websites to identify general usability issues and assess WCAG compliance, involving 14 students in the process (Debevc et al., 2015). Their findings highlighted that a combined evaluation strategy helps students develop a deeper understanding of both general usability principles and specific accessibility requirements. Our work, in paper III, used this combined approach: heuristic evaluation walk-throughs of students' UI design artifacts were used to evaluate and measure the accessibility of their designs to a diverse range of users.

While these strategies and standards are foundational, it is necessary to critically examine their conjunction with emerging technologies like AI. How GenAI could support the identification of accessibility issues is a promising area of research, even though significant ethical challenges require careful examination. Current debates in educational AI highlight risks such as algorithmic bias (Arora et al., 2023) and a lack of explainability (Afzaal et al.,

2024; Saqr & López-Pernas, 2024). Without understanding why an AI provides a specific accessibility suggestion, students and educators may struggle to critically evaluate and learn from its feedback, potentially hindering true learning and leading them to over-reliance on LLM outputs, and thus, new forms of digital exclusion.

Acknowledging these challenges, this dissertation aims to examine the role of LLM models as both a design assistant and a pedagogical partner within computing education. Rather than accepting these tools uncritically, it investigates how they can be used within frameworks of accessibility and inclusive design, focus on the pedagogical need for explainability in HCI and software engineering education. By doing so, it seeks to bridge foundational research on the specific challenges of accessibility and inclusive design practices with AI education.

# 3 ARTIFICIAL INTELLIGENCE IN EDUCATION

This chapter provides an overview of AI, particularly M-LLMs, in education. It examines how AI has been used to support teaching and learning practices, the paradigm shift of AI in education, and challenges and issues encountered in AI education. It also discusses accessibility and inclusive design within education, drawing from existing literature. Finally, it describes AI models with memory, detailing their operation and effects on educational practices.

Before diving into this section, it is useful to briefly explain the terms AI, M-LLM, and GenAI and their scopes as they are used within the context of our dissertation. In our study, the term AI serves as the overarching term and includes machines capable of performing tasks typically requiring human intelligence, such as problem-solving, learning, and decision-making (Jarrahi, 2018). Within AI, GenAI represents a specific subset focused on creating content, including text, images, audio, or code, by learning patterns from vast datasets (Samala et al., 2025). A prominent form of GenAI is the Multimodal LLM, which is an AI model primarily designed to understand, generate, and process human language, trained on immense text, image, or other corpora (Lee, 2025). Our dissertation specifically utilizes these distinctions to explore the unique capabilities and implications of large language models as a GenAI application within the broader context of AI.

## 3.1 Overview of Artificial Intelligence in Education

Artificial intelligence in education (AIEd) is one of the growing fields in education technology that presents several opportunities for learners, educators, and institutions (Sanusi et al., 2024). AIEd refers to the application of AI technologies and techniques to support educational practices. With the advancement of computing techniques, AI has been widely applied in educational practices (Mustafa et al., 2024), including personalized AI tutoring systems (Baillifard et al., 2025) and learning analytics (Misiejuk et al., 2025); for instance, systems that track student progress and identify struggling areas to provide additional

support (I. Khan et al., 2021); and adaptive learning systems, such as platforms that adjust content based on student performance (Sayed et al., 2023).

The literature has extensively explored the ways AI has been employed in the academic domain (Mustafa et al., 2024). Different scholars have categorized the application areas or roles of AI in education settings. For instance, Crompton and Burke (2023) categorize AI's role based on specific applications, such as prediction (e.g., identifying students at risk of dropping out), AI assistants (e.g., chatbots providing student support), intelligent tutoring systems (e.g., adaptive platforms for specific subjects), and managing student learning (e.g., AI-powered tools for scheduling and progress tracking). Other scholars, however, categorize AI's applications based on their domain tasks, such as administrative or teaching functions. For example, Chen et al. (2020) identified three such categories: administration, instruction, and learning. Chiu et al. (2023) further identified assessment and evaluation as a fourth area. Based on these tasks, we have provided the details of the categories as follows based on domain area.

***AI for learning:*** This is the first area where AI was used in the education domain. The adaptivity of technology plays key role in learning enhancement (student-facing); thus, nowadays, AI adapts the educational experience to the individual student by being able to meet students where they are, build on their strengths, and grow their knowledge and skills (Fitria, 2021). A range of AI applications have been designed to optimize how students acquire knowledge and skills through their design process. For instance, AI enhances content accessibility by providing tools (Ayeni et al., 2024) such as AI translation for multilingual learners and text-to-speech functionalities for students with different abilities, which promotes inclusivity and equity in education (Miranda & Vegliante, 2025). Furthermore, AI tools have been designed to support student engagement and improve their learning outcomes by creating more interactive and adaptive learning environments (Ayeni et al., 2024).

***AI in Instruction:*** A primary task for educators is creating instructional designs, a time-consuming process now being streamlined by AI (educator-facing). AI enhances instructional quality by introducing pedagogical tools (Pedro et al., 2019). A key example is personalized learning, which involves adapting learning pathways and content delivery to individual student

needs. AI-driven platforms like DeepTutor and AutoTutor personalize content (Rus et al., 2013), whereas other intelligent web-based systems analyze student behaviors to customize instruction (Brusilovsky & Peylo, 2003). These platforms recommend specific learning materials to improve the overall learning experience and help students achieve their goals. More recently, GenAI has expanded these capabilities by generating customized content for individual learners (Ayeni et al., 2024). More recently, GenAI has expanded these capabilities by generating customized content for individual learners. Furthermore, AI is creating more interactive and supportive learning environments. For instance, computer-assisted language learning provides customized language instruction, supported by AI-driven writing and translation assistants (Pokrivčáková, 2019). AI-powered chatbots can address routine student queries (Pokrivčáková, 2019), whereas collaborative robots, or cobots, can assist with teaching tasks such as reading and pronunciation (Timms, 2016). Ultimately, the integration of AI into instruction has demonstrably improved the effectiveness, efficiency, and quality of educators work (Chen et al., 2020), making learning more adaptive, engaging, and effective.

***AI in Assessment and Evaluation***: The third dominant domain area of AI in education is assessment and evaluation (Mustafa et al., 2024). AI has been used for various roles in the assessment and evaluation, notably in automating marking and forecasting student performance (Chiu et al., 2023). Alghamdi et al. (2020), Fu et al. (2020), and Kumar and Boulanger (2020) indicated that the integration of AI in education has improved and automated assessment processes, leading to more effective grading. Specifically, for example, AI-enhanced grading systems have been designed for diverse tasks (Tobler, 2024), such as language writing and essay scoring (Suresh et al., 2023), resulting in automated grading, efficient feedback, and reduced human bias.

Nevertheless, the application of automatic grading and marking has been largerly within a limited discipline, such as language learning, suggesting that this application of AI is still in its nascent stages of development in other domains like design education. Another role of AI in assessment, including predicting learning patterns, involves predicting student performance, dropout, at-risk, and satisfaction (Crompton & Burke, 2023). Therefore, the

AI tools have demonstrated the ability to support educators by providing student assessment scores and forecasting student success; for instance, they can analyze the depth and quality of students' engagement in learning activities, including discussion forums.

***AI in education Administration****:* Furthermore, AI is used in teaching and learning practices and in administrative activities (Qian, 2025). For example, AI has a role in educational administrative activities, such as facilitating administrative tasks, including managing and handling paperwork and curriculum planning and development, leading to enhanced efficiency for educators and a reduction in their administrative workload (S. F. Ahmad et al., 2022). This shift in traditional educational practice to adaptive modern technology supported educational practices and allowed instructors to focus on core responsibilities, such as student teaching and delivering instructions (Chen et al., 2020). Furthermore, specific AI-powered programs like plagiarism checking (TurnItIn) have contributing to academic integrity, and checking grammar (Grammarly) has facilitated administrative duties, easing the burden on instructors by assisting with tasks. These functionalities ultimately contribute to improved efficiency and effectiveness in educational processes and reduce the overall workload on instructors (Chassignol et al., 2018).

Although AI's primary applications in education have historically centered on learning, instruction, assessment, and administration, its potential roles are considerably broader. The range of AI functionalities highlights its pervasive influence throughout educational processes, encompassing several aspects ranging from administrative duties to individualized learning. Consequently, a comprehensive understanding of AI in education necessitates exploring its conceptual frameworks and practical implementations (F. G. Deriba & Sanusi, 2025). As detailed in Section 3.2, the evolving paradigms of AI, namely, AI-directed, AI-supported, and AI-empowered, offer insights into AI's varied contributions and the corresponding shifts in the learner's role.

## 3.2 The Paradigm of AI in Education

In educational practices, a fundamental understanding of the frameworks that guide teaching, learning, and educational philosophy—referred to as the paradigm of education (See Figure 3)—is necessary, as it facilitate informed decisions about teaching and learning, aligns practices with philosophical beliefs, and helps educators adapt to diverse student needs (Munna & Kalam, 2021). Consequently, there is an ongoing discourse among scholars regarding the enhancement of educational quality, particularly amidst the rise of AI. Grasping the paradigm shifts induced by AI is, therefore, vital. The most widely known paradigms are those of Ouyang and Jiao (2021), who identified three paradigms in AI education used to address educational and learning issues.

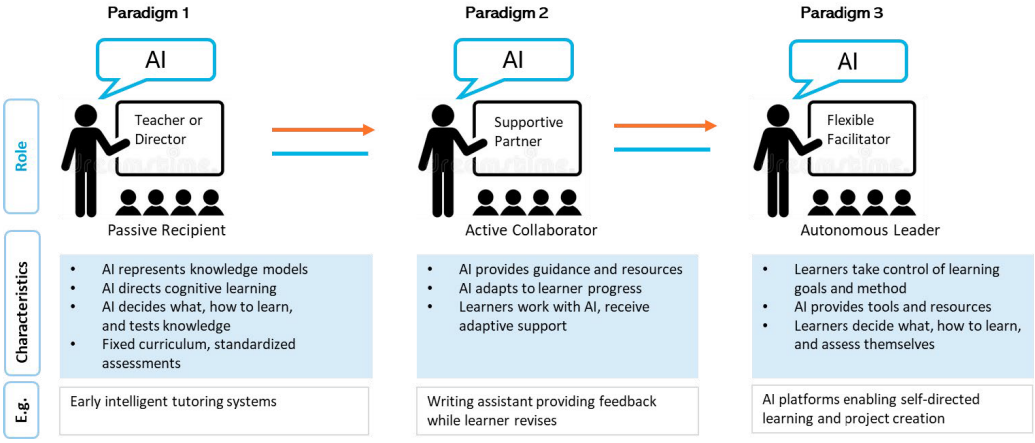



**Figure 3:** The three paradigms of AI in education

The first paradigm is AI-directed, whereas the learner is the recipient. AI is used to represent knowledge models and direct cognitive learning, with learners acting as recipients of AI services. In this early stage, AI acts like a teacher who directly instructs the learner. The AI decides what you learn, how you learn it, and tests your knowledge (Fitria, 2021). The learner is a receiver of information and instructions from the AI. An early intelligent tutoring system that provides a fixed curriculum and assesses student understanding through standardized tests exemplifies this paradigm.

The second paradigm is AI-supported, learner as collaborator (Kim et al., 2022)there is a lack of studies that inform how to design learning when AI is introduced as a collaborative learning agent to classrooms. The present study, therefore, aimed to explore teachers' perspectives on what (1. In this paradigm, AI supports learning while learners collaborate with AI. AI becomes more of a helpful partner in the learning journey. In this paradigm, AI still provides guidance and resources, but it also adapts to learners' progress and offers support when they need it (Darvishi et al., 2024). The learners actively work with the AI, and it assists them based on their needs. For example, a writing assistant that provides feedback on grammar and style while allowing the student to make their own revisions (McCarthy et al., 2022) is an example of this model.

Finally, in the third paradigm, AI empowers learning, and learners take agency in their learning as leaders. In this advanced paradigm, AI empowers students to take control of their learning (Ouyang & Jiao, 2021). The AI provides tools and resources, but they decide what they want to learn, how they want to learn it, and even how they want to assess their understanding. The AI acts as a flexible facilitator, helping them achieve their self-defined learning goals (Ouyang & Jiao, 2021). AI-powered platforms that provide access to a vast library of resources and tools, allowing students to pursue their learning paths and create their projects are examples of this paradigm. Therefore, this background on the applications and evolving paradigms of AI in education sets the stage to highlight the increasing interest in integrating AI in diverse educational contexts. However, further investigation is needed to understand its specific impact on design and accessibility education.

## 3.3 Issues and challenges of AI in education

While the integration of AI into education has improved pedagogical practices, it also presents several issues spanning ethical, pedagogical, and technical dimensions.

The ethical dimension of AI in education is a multifaceted challenge. A primary concern is algorithmic bias, which can occur when training data

reflects societal prejudices, leading to gender or racial discrimination (Akgun & Greenhow, 2022). Privacy is another major concern, as the collection of sensitive student data through surveillance applications like facial recognition can expose individuals to misuse (Zhai et al., 2021). Further ethical dilemmas include the legal implications of using AI tools without a clear understanding of their terms and conditions (Whalen & Mouza, 2023). Despite these concerns, a study by Schiff (2022) found that national policies have often overlooked the ethical application of AI in education, instead prioritizing the training of an AI-ready workforce. In response, the authors have proposed frameworks focusing on professional responsibility, fair governance, transparency, and inclusivity to guide the ethical development and deployment of AI in educational settings (Borenstein & Howard, 2021; Nguyen et al., 2023).

From a pedagogical standpoint, the integration of AI faces challenges. Over-reliance on AI tools is problematic, which could hinder students' development of critical thinking and knowledge processing skills (Adeshola & Adepoju, 2024). Furthermore, maintaining student engagement and motivation can be more challenging with AI systems compared to human teachers (Zhai et al., 2021). These challenges require reconceptualizing of the teacher's role and providing user training on the risks of unrealistic overreliance (Zhai et al., 2021). A related issue is academic integrity, as the misuse of AI in educational activities leads to plagiarism (Adeshola & Adepoju, 2024; Giannakos et al., 2024). Addressing these pedagogical gaps requires solutions like integrating AI ethics education into curricula, an area that remains less explored (Giannakos et al., 2024).

Finally, technical limitations pose another obstacle. AI models can suffer from hallucinations, inaccuracies, and biases, making them unreliable for educational domains (Lin et al., 2024; Zhai et al., 2021). The trustworthiness of AI is a primary concern, as fabricated content can spread misinformation and infringe upon copyright laws (Giannakos et al., 2024; Whalen & Mouza, 2023). Thus, AI-based technologies present a double-edged sword; they can be used for positive outcomes when applied carefully, but they also create serious risks if their effects are not fully understood.

# 4 REVIEW OF PREVIOUS LITERATURE ON AI, ACCESSIBILITY, AND EDUCATION

This chapter provides an overview of literature related to intersections of AI, accessibility, and design education. Specifically, it examines how AI addresses accessibility within the context of design education, exploring the long-term impact on learners and the associated pedagogical implications.

## 4.1. Role of Artificial Intelligence in Accessible Design Education

In recent years, AI applications have been used for several real-life domains to make or enhance accessibility for all users (Fitsum Gizachew Deriba et al., 2024). For example, AI has been used to identify barriers to digital content accessibility, such as low color contrast, missing alternative text for images, and improper heading structures (Firth, 2019).

Advanced AI tools, such as AccessiBe[2] and UserWay[3], use AI to scan websites, identify accessibility issues, and automatically suggest solutions related to screen reader optimization, keyboard navigation, and image description. Additionally, several AI tools generate text descriptions for images, such as Microsoft's Seeing AI app and the "Be My AI" feature from My Eyes (Eyes, 2020), which uses AI to describe visual content to individuals with visual impairments.

AI-powered speech recognition is another tool that allows impaired individuals, for instance, to navigate the internet control device and interact with digital content using voice commands (Jampala et al., 2024). For instance, voice assistants like Siri and Google Assistant utilize AI for voice control (Hoy, 2018). Live Transcribe by Google provides real-time transcription of spoken content for individuals with hearing impairments. AI algorithms are also used

[2] accessiBe: Web Accessibility - Web Accessibility by accessiBe | Shopify App Store

[3] UserWay | Web Accessibility AI Solution for Compliance

in translation to automatically generate captions and transcripts for video and audio content, thereby making multimedia accessible (AlShaikh et al., 2024). For instance, Netflix uses AI to generate custom captions.

Consequently, AI has become an important tool in design education, improving traditional pedagogical approaches by enabling the generation of diverse design concepts from a given set of parameters (Chandrasekera et al., 2025). This capability allows students to, for example, explore numerous design variations for a product based on specific requirements (Ghorbani, 2023). AI further enhances the learning experience by providing personalized feedback and guidance on design projects, offering explanations, and reinforcing design principles (Dirin & Laine, 2024). Notably, AI tools are instrumental in helping students understand and implement accessibility principles by providing real-time feedback (Paper II). Moreover, adaptive technologies have been suggesting design content tailored to a learner's specific needs, offering affordances that directly align with the demands of accessibility and design education (Katz et al., 2021).

The recent advancements in GenAI, particularly Multimodal LLMs, have broadened these pedagogical strategies. These technologies offer immersive learning experiences, including simulating real-world design scenarios and intelligent design suggestions. For instance, conversational AI agents have been used to prompt students to justify their design choices in relation to accessibility principles (Liu et al., 2024; Morgan et al., 2020). Similarly, researchers have integrated universal design principles with AI-based adaptive assignments to foster a more inclusive educational environment (Hyatt & Owenz, 2024).

Furthermore, studies have shown AI's effectiveness in supporting novice designers throughout the design process (Xu et al., 2020) and examined its potential to address accessibility shortcomings through real-time compliance checks and personalized feedback (Kubullek & Dogangün, 2023). This body of work underscores the capacity of GenAI to enhance adaptive learning by providing contextually relevant design recommendations.

Table 2 summarises studies on how AI technologies have been utilized in the teaching and learning practices of computing education and the software development process.

**Table 2:** AI in the design and development of digital technologies or teaching and learning in computing education

| Authors | Focus of the study | The AI used in the study |
|---|---|---|
| Dirin & Laine (2024) | Examined the effect of AI tools on student performance in a software engineering project course and explored student perceptions of AI tool usage in software engineering education. | M-LLM (ChatGPT and Copilot) |
| Tan & Luke (2024) | Explored the use of the ChatGPT-3.5 textual GenAI model to digitize and accelerate future thinking techniques within a design workflow, to generate future scenarios for architectural concept development. | ChatGPT 3.5 |
| Chandramma et al. (2025) | Focuses on enhancing e-learning accessibility through the integration of AI and inclusive design principles to provide learners with equal access to educational resources. | NLP and speech recognition |
| Wu (2024) | To explore the AI-enhanced adaptive UIs to examine web accessibility for users with visual impairments. | Machine learning with image-based reCAPTCHA tools |

Even though AI has been used to overcome accessibility issues in design and computing education, its integration for supporting inclusive design within computing education remains underexplored. For example, a review by Pisoni (2021) on the use of AI technology for accessible museum and cultural heritage experiences emphasized the need for inclusive design, including interaction, pedagogical, and participatory, and highlighted AI's potential to improve online and on-site accessibility.

While M-LLMs show promise for design education, especially in recommending inclusive design practices aligned with WCAG guidelines and addressing accessibility issues, there is limited empirical research on

how these AI tools are used in the teaching and learning of design and the development of educational technology within a pedagogical context. Specifically, few studies have empirically investigated how M-LLMs support accessibility education and inclusive design learning within computing education, and they have not thoroughly evaluated their impact on students' conceptual understanding of accessibility and their self-directed learning abilities in this field. Flechtner and Stankowski (2023) have emphasized the critical need to actively engage software design students with AI tools to enhance guidance for their inclusive design practices. Our study builds upon this recommendation by providing empirical evidence for the pedagogical value of AI for accessibility and inclusive design education.

## 4.2 Long-Term Impact of AI on Learning and Accessibility in Design Education

The integration of AI, particularly LLM, into educational settings offers promising avenues for accessible learning design. However, insights into the nature of AI memory and its short long-term effects on learners in academic domains remain nascent. Understanding the nature of AI systems has implications for how they process, retain, and utilize information across various modalities (Chandrasekera et al., 2025), which is relevant for comprehending their cognitive abilities within educational contexts (Du et al., 2025). Thus, a careful consideration of LLMs is necessary, particularly when evaluating their lasting impact on learners.

Although studies are limited in context, existing research on the memory of GenAI offers valuable insights. For instance, Rolls (2024) elucidated how these models generate subsequent tokens in a sequence by forming compressed representations of their training data. Rolls clarified that this process shares some similarities with human semantic memory retrieval, where AI could utilize schemas. Nevertheless, a distinction lies in the AI's limited ability to update information and its tendency towards stochastic parroting. This means it relies on relationships learned from massive datasets to convincingly generate human-like text while lacking a true semantic understanding of

the word patterns. This contrasts with the continuous updating inherent in human semantic memory, which stores general knowledge, including facts and concepts (Rolls, 2024). This indicates that, unlike human memory or the data storage of traditional AI-based educational platforms, LLM inherently lacks true long-term memory of individual learner interactions. On the other hand, it is important to understand how students' interaction with LLMs affects their long- or short-term learning journey in improving their learning outcomes, particularly in our case of design and accessibility education.

Further illuminating this characteristic, Hoskins (2024) discussed how LLMs, by scraping vast amounts of data, represent and reimagine the past in novel ways, generating AI-prompted memories that are constructed (creating the memory on the fly using its learned patterns) rather than retrieved (accessing back a pre-recorded piece of information). The author emphasizes that this might differ from that of humans (who recognize the past from the present), with AI potentially fabricating a past that never truly existed and altering human agency in the processes of remembering and forgetting (Hoskins, 2024). Consequently, Hoskins introduces the concept of a *third way of memory* to address these intricate interactions between humans and AI in shaping both individual and collective recall.

Paaß and Giesselbach (2023) further clarified the operational mechanism of GenAI, stating that it functions by predicting subsequent elements within a sequence of input data. In NLP applications, such as ChatGPT-4, these elements, known as tokens, can represent words. These models achieve this capability through extensive training on diverse and massive datasets, including text, image, and audio. By analyzing these data, the AI determines the most statistically probable next token following a given input sequence (Paaß & Giesselbach, 2023). However, the responses generated by AI may not always constitute factually accurate answers to user queries. Instead, they represent the most statistically likely continuation based on the patterns discerned from their training data. Recent work by Guo et al. (2025) presents an interesting development with the AI system, featuring a self-growing memory network specifically designed to enhance self-efficacy, mitigate loneliness, and foster social interaction among the elderly. This

system analyzes user-written diaries and generates personalized, empathetic responses, suggesting potential avenues for AI memory in specific contexts.

Thus, while ongoing discussions explore the potential of AI technologies, including their role in supporting inclusive design and accessibility, particularly through AI feedback rather than to human feedback, the on impact of GenAI on students' long and short-term learning of design education remains to be explored.

# 5 METHODOLOGY

This chapter provides an overview of the research approach and methodologies employed in this dissertation, focusing on investigating accessibility and inclusivity in learning for inclusive design and development education. The ultimate aim of this research is to contribute to a more equitable and accessible digital future for all learners and educational stakeholders. First, this chapter outlines the research design, then discribes the research context and the participants involved in this study. Subsequently, it details the procedures undertaken in each included study of the dissertation. Thereafter, the methods of data collection, including the pre-processing and analysis techniques used, are presented. Finally, the research ethics and ethical procedures followed throughout this dissertation are described.

## 5.1 Research Approach and Methodology

The general objective of this study is to address the challenges of digital accessibility by investigating how AI-based tools, particularly M-LLMs, can be leveraged to support inclusive design practices. This involves evaluating the current state of accessibility in existing educational technologies, such as V-Labs and educational websites, and exploring potential solutions to overcome identified accessibility gaps.

To address the multifaceted nature of the research objectives, this study utilized a systematic literature review methodology alongside a mixed-methods approach, integrating both qualitative and quantitative research paradigms. This integration allows for a more robust understanding of bridging accessibility divides in digital education through AI-enhanced inclusive design practices.

The methodologies of this study are guided by three research questions (see section 2.3). The four interconnected studies collectively sought to answer the overarching dissertation question: How can design accessibility and inclusivity education be enhanced in digital technology design and

development—particularly using AI-based tools and interventions—to create more equitable learning environments and better prepare students to develop inclusive design skills and knowledge? Each research question is explicitly mapped with its purpose and research design, as presented in Table 1.

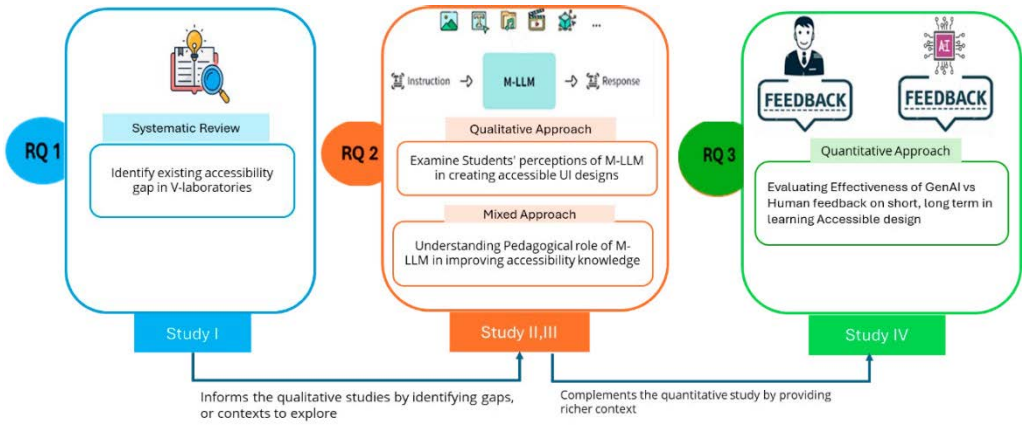



**Figure 4:** Research questions with each study design used in the dissertation

The dissertation follows a sequential research design, where each study methodologically builds upon the results of the previous one, as depicted in Figure 4. The trajectory began with Study I, a systematic review that addressed RQ1 by identifying critical gaps in the existing literature on V-Lab accessibility. The limitations of a literature-only perspective necessitated a contextual investigation. Therefore, Studies II (qualitative) and III (mixed-methods) were designed to address RQ2 by exploring these gaps empirically examining students' perceptions and the pedagogical role of M-LLMs in improving accessibility knowledge. The rich insights from these studies then created the foundational evidence needed to test a formal intervention. This trajectory comes to an end within Study IV, a quantitative experiment that addressed RQ3 by directly comparing the effectiveness of GenAI versus human feedback on student learning, a central question that arose from earlier studies. The following sections will provide a detailed overview of the research methods and approaches utilized in each phase of this trajectory.

## I Systematic Review

According to Post et al. (2020), literature review is "a study that analyzes and synthesizes an existing body of literature by identifying, challenging, and advancing the building blocks of theory through an examination of a body of prior work." Depending on the purpose of the review, researchers can use different strategies, standards, and guidelines developed specifically for conducting a literature review (Snyder, 2019). These definitions helped researchers to understand the prior work in their field, enabling them to easily identify gaps in the body of literature and potential avenues for future research (Kraus et al., 2022). More importantly, a literature review may be the best methodological tool to answer to different research questions. For example, researchers might want to evaluate evidence in a certain area or to examine the validity or accuracy of a particular theory (Tranfield et al., 2003).

Building upon previous work by Kitchenham and Brereton (2013), this dissertation chose the systematic review approach to conduct literature reviews and identify research gaps in the literature on a relevant topic to our study (see Figure 1 for an overview). According to Kitchenham and Brereton (2013), this methodological framework requires transparency; for instance, a detailed description of the involved review steps is required to improve the validity of the systematic review that our study followed. This step comprises the following three main phases:

**Phase 1- Planning the Review:** In the planning phase, we first identified the purpose of the review, which is to assess the accessibility features and inclusiveness of V-Labs in addressing RQs. While the RQs were defined in Section 2.3, we aimed to examine the effectiveness of V-Labs in terms of accessibility and inclusivity by synthesizing empirical studies. The next steps were developing the review protocol, which included defining the scope of the review, the search strategy (databases, keywords, and search strings), inclusion and exclusion criteria for selecting studies, methods for quality assessment of the selected studies, the data extraction strategy (what information was collected from each study) based on defined codebooks, and the data synthesis strategy (how the extracted data would be analyzed and combined). This information is presented in Table 3, which we have used

for our systematic study. Thus, the protocol ensured the reproducibility and consistency of the study (See Figure 5).

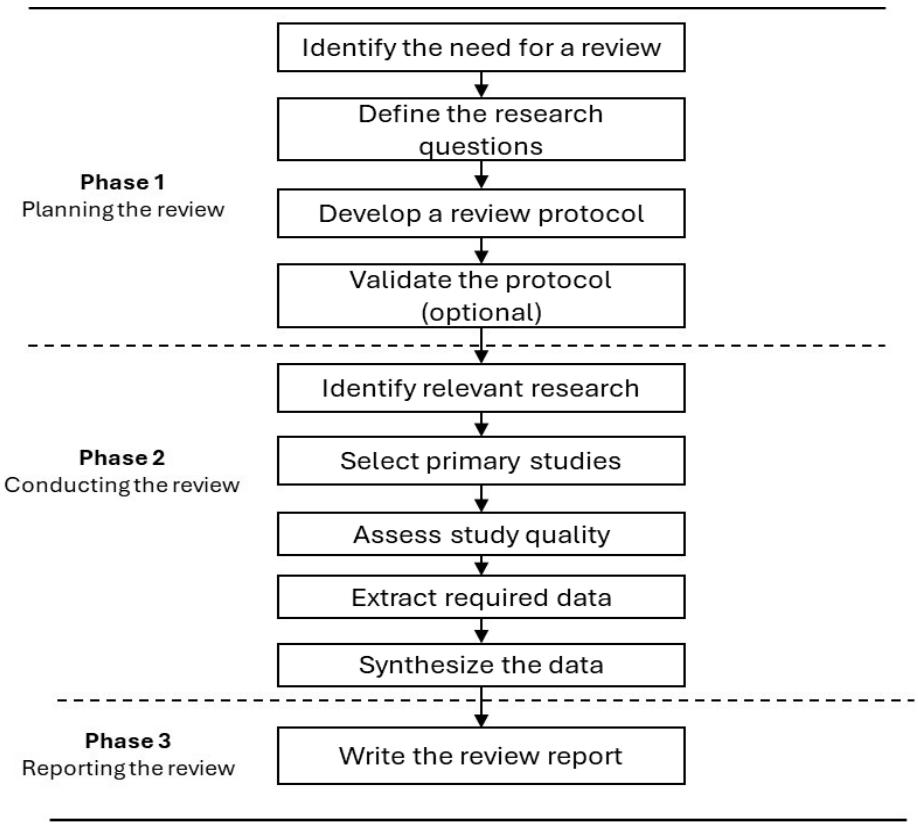



**Figure 5:** Overview of systematic review development adopted from (Kitchenham et al., 2009)

**Phase 2 - Conducting the review**: In this phase, the relevant studies were identified from three prominent databases, namely Scopus, Web of Science, and ERIC, using relevant keywords. While Scopus and Web of Science provide coverage of peer-reviewed literature across various disciplines, ensuring that a wide range of relevant studies are captured (Mongeon & Paul-Hus, 2016), ERIC is the premier database for education-related literature (Hertzberg & Rudner, 1999), which is the primary reason behind choosing these databases. Applying the inclusion and exclusion criteria, after importing all articles into the Rayyan web-based system to remove duplicates and facilitate the selection and screening of the studies (Ouzzani et al., 2016), the primary studies relevant

to the study were identified by screening titles, abstracts, and full-text articles. This was conducted in two phases: first studies were assessed based on the abstract and title on inclusion criteria, then screened by reading the full paper. This was conducted by evaluating the included studies using predefined criteria and collecting all the relevant information from each included study based on the developed codebook. Subsequently, we thematically analyzed the extracted information, which involved quantitative analysis and thematic analysis using deductive coding, in which researchers start with a set of themes, then code the data according to these pre-determined categories (Fereday & Muir-Cochrane, 2006). In this stage, in-depth data coding on selected full-text articles was conducted in a spreadsheet by the first author.

**Phase 3- Reporting the Review**: In this stage, the extracted information is meticulously reported. The extracted information consisted of various categories, including study characteristics, participant details, types of V-Labs, disciplines, educational settings, development environment, evaluation measures, and accessibility design features. Subsequently, the findings were synthesized and reported.

**Table 3:** Overview of systematic literature review

| Study 1: Deriba et al (2024) | |
|---|---|
| Research question | **RQ1.** What essential accessibility features are integrated into V-Labs to cater to diverse user needs?<br>**RQ2.** How do existing studies assess the effectiveness of accessibility of V-Labs and which aspects do they assess? |
| Databases | Web of science, Scopus, ERiC |
| Search string | ("Virtual Lab*") AND ("Accessib*") |
| # of paper identified | 392 |
| Inclusion criteria | Timeline: 2000-2023<br>Peer-reviewed published in journal or conference<br>Language: English<br>Topic and focus: Study discusses V-Lab, their accessibility, considers diverse needs in the design and implementation of V-Labs |
| # of paper included | 36 |
| Coding process | Deductive coding with iterative refinement. |
| Method | Descriptive statistics, Thematic analysis |

### II Empirical studies

This section outlines the empirical studies used in this dissertation, specifically the research design of Papers II, III, and IV.

## 5.2 Overview of Research Design

While several research designs and approaches exist, including qualitative, quantitative, and mixed methods, this dissertation selected a mixed sequential research design. Several studies support using mixed research

methods to understand complex educational phenomena (Creswell, 2013; Meyer & Schutz, 2020). This approach is particularly effective because such phenomena cannot be fully understood through a single research method alone. Furthermore, according to Gregar (2023), mixed research is defined as "a methodology for collecting, analyzing, and mixing both quantitative and qualitative data in a single or series of studies to better understand research problems."

The term mixed methods, though descriptive, has not always been used consistently; it is somtimes used interchangeably with other terms. For example, multimethod or multitrait (Campbell & Fiske, 1959), integrating qualitative and quantitative approaches (Fielding & Fielding, 2008), combined quantitative and qualitative research (Creswell, 1994), or mixed methodology approaches (Tashakkori & Teddlie, 1998).

For our dissertation work, we used a mixed approach context stated by Creswell (1994). According to Creswell and Creswell (2003), this method, which combines quantitative and qualitative research, further involves three strategies, namely sequential, concurrent, and transformative procedures.

Sequential study designs involve using the results from one research method to inform or build upon a subsequent method (Creswell & Creswell, 2003). For instance, a researcher may first use a qualitative approach to explore a topic, then follow up with a quantitative study using a larger group to see if the initial findings hold true for a wider population. Conversely, a study could start with quantitative methods to test specific ideas and then use qualitative methods to delve deeper into the experiences of fewer participants (Lund, 2012).

In concurrent designs, the researcher gathers quantitative and qualitative data simultaneously and subsequently merges them to develop a complete analysis of the research issue (Östlund et al., 2011). This simultaneous data collection allows for the integration of information during the interpretation phase, leading to a more holistic understanding of the results. Additionally, this approach can include nesting one data form within a broader data collection process to address varied research questions or analyze different units within an organization (Leech & Onwuegbuzie, 2009).

Finally, transformative procedures employ a specific theoretical perspective as a guiding framework throughout a study that incorporates both quantitative and qualitative data (Leech & Onwuegbuzie, 2009). This lens shapes the research focus, data collection methods, and the anticipated outcomes or changes. The data collection itself within this framework can utilize either a sequential or concurrent approach.

Considering foregoing, our study used a sequential mixed research design. Study I served as the foundational phase, identifying a gap in the accessibility of educational technology. This gap prompted the subsequent exploratory work in Study II (qualitative) and Study III (mixed-methods), which investigated how M-LLMs could address these accessibility challenges in an educational context. The findings from these studies then provided the necessary rationale to design the conclusive quantitative experiment in Study IV, which tested the impact of human versus M-LLM feedback on students' learning of accessible design.

## 5.3 Research Context and Participants

In this section, we provide an overview of the study contexts and participants. While the systematic study (Paper I) focused on the global accessibility of digital resources like V-labs, the other studies (Papers II, III, and IV) were conducted within Ethiopian higher education institutions. We chose Ethiopia, to understand an African or Global South context, and explore the effectiveness of AI-based educational tools in a resource-constrained setting. Specifically, we aimed to investigate the adoption and impact of AI feedback in an environment where limited local expertise in accessibility education presents a significant barrier to equitable learning. For our experimentation and study design, we chose undergraduate computer science program courses that contain content related to design and accessibility.

Different scholars have suggested suitable courses to equip students with knowledge of accessibility and inclusivity in the development of digital technology, including websites and applications. These include HCI (Elglaly et al., 2024; Shinohara et al., 2020), software engineering (Baker et al., 2020;

El-Glaly, 2020), web design and development (Ferati & Vogel, 2020). For our study, we used HCI and web design and development courses for 4th and 3rd-year undergraduate computer science students. In each empirical study, within the research context, we considered the demographics of participants, the teaching and learning setting, and the pedagogical approach used.

In Paper II, we utilized an HCI course that included 42 fourth-year undergraduate students, comprising 36 men and 6 women, with ages ranging from 19 to 33 years. The mean age of the students was 24 years. This course, carrying three credit hours, was delivered during the first semester of 2024, from March to June, over 16 weeks. The curriculum incorporated six chapters, with the primary aim being to offer insight into and promote the development of computer science applications. CS graduates should cultivate a mindset that recognizes the paramount importance of users and their organizational contexts. Consequently, they should use user-centered method throughout the development, evaluation, and deployment phases of applications. This necessitates that graduates develop a knowledge of HCI, areas such as user and task analysis, ergonomics, accessibility standards, human factors, and cognitive psychology.

The course objectives explicitly state that, upon completion of the course, each student should be able to describe the relationship between cognitive principles and their application to interfaces and products. They should also be able to explain the conceptual terms essential for analyzing human interaction with products, such as affordance, conceptual model, and feedback. Furthermore, students are expected to analyze different user populations to determine their abilities and characteristics with regard to the use of both software and hardware products. Another key objective is to describe the connection between the development of a user interface and the user's knowledge of an application domain and, finally, to list several emerging alternative input or output devices for computers.

After students had grasped the basic concepts of the course, they were divided into nine groups. Each group was tasked with designing Uis for learning tools, with their design preferences guiding the process. While various methods exist for forming student groups for class projects, including random assignment and self-selection, as noted in reference (Aivaloglou &

Meulen, 2021), for the UI design group projects in this study, we opted for instructor-selected groups. This approach was chosen to ensure a balanced distribution of students skills and to account for students' previous academic grades, as also supported by (Aivaloglou & Meulen, 2021). In this study, we used collaborative project-based learning (Baser et al., 2017) to examine their understanding of accessibility and inclusive design through group-based project tasks after they had learned basic concepts. The group project involved designing a UI, contributing 30% to the total course grade, and was to be completed within four weeks. The remaining assessment comprised a 50% final exam and 20% in tests. Detailed study procedures for the project and course overview are presented in section 4.4 of the dissertation. The overarching aim of this study was to understand how students could acquire knowledge of accessibility by designing a UI using M-LLM, and subsequently, to ascertain their perspectives after completing their design work.

Similarly, in Paper III, we used the HCI course and the demography of 4th-year undergraduate computer science students. However, the study design and data used were different from those of the second study (Paper II). In Paper III, we first examined their familiarity with design tools and M-LLMs, then conducted an experimental study using a single-group pre-post design. The pre-familiarity design tool and AI knowledge assessment revealed that they had limited prior exposure to accessibility principles (M = 5.29, SD = 2.28) but intermediate familiarity with design tools (M = 7.24, SD= 2.47). Two instructors facilitated the study, one with five years of HCI teaching experience and one lab assistant with three years of experience. In this study, we aimed to examine and evaluate the impact of GenAI on students' accessibility-aware UI design skills based on M-LLM suggestions and to determine their psychological constructs based on Self-Determination Theory.

In Paper IV, we used the Web Design and Development programming course (CoSc-M3101) for third-year undergraduate computer science students. Students (N=81, Female = 25, Male = 56) with a mean age of 22.73 participated in study. This study used an experimental design with control and experimental groups, to understand how GenAI compared to human feedback, affects learning accessibility design through a given intervention

to ensure accessible web design and development (details are presented in section 3.4).

Across the empirical studies (Papers II, III, and IV), the sample sizes were determined by the number of specific undergraduate courses, consisting of participants registered for the course during the study period. In design-based educational research, these practices aimed to elucidate learning processes within authentic educational settings, rather than statistical generalization (Barab & Squire, 2016). While a priori power analysis is relevant, it was not conducted in our studies since we did not use a sampling instead included the actual registered CS 3rd, and 4th year student population; however, the sample sizes used are consistent with established conventions in computing education research. Similar intervention studies in HCI and accessibility education use varying sample sizes, focusing on in-depth qualitative and mixed-methods analysis (e.g., Bhatia et al. (2023)). The focus was on ensuring the studies were feasible and could yield meaningful insights into the complex phenomenon of how students learn accessibility design, which aligns with the exploratory and design-based nature of this dissertation (Collins et al., 2016).

## 5.4 Research Study Procedures

The studies included in this dissertation employed varied procedures to achieve their respective objectives. For instance, Study 1 adopted a systematic review approach guided by the PRISMA protocol (Kitchenham et al., 2009). The subsequent studies utilized different research procedures relevant to each specific study (Opie, 2019), as detailed below:

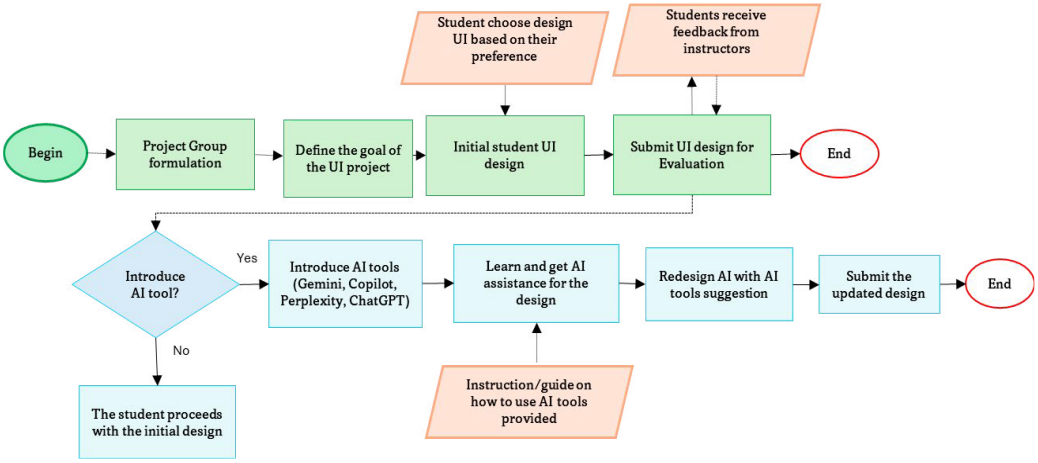



**Figure 6:** Study Design procedure of paper II (adopted from Article II)

Paper II commenced with the formation of project working groups. Thereafter, the study's purpose was defined, specifically focusing on designing a UI for a digital application chosen based on student preferences (See Figure 6). Subsequently, the students received instructor feedback, based on their initial designs created, to refine their UI. This was followed by an intervention introducing a GenAI to be used in their design process and incorporating the received feedback. Instructions and manuals on utilizing the M-LLM for design were provided, along with an hour of guidance by the researcher on using design tools like Figma. Consequently, students used these models to enhance their designs, seeking learning and assistance from the model. Finally, the improved UI designs were submitted to the instructor for evaluation, and students were interviewed by the researcher using guided questionnaires to gather their perspectives on the M-LLM and the process of designing an accessible UI for their application.

While the procedures in Paper III shared similarities with those in Paper II, this study involved a more in-depth intervention, and the pedagogical and experimental contexts differed. This study incorporated student training, specifically providing an intervention on GenAI tools and training on designing accessible UIs for various applications. The study comprised two design phases. In the initial pre-design phase, students were instructed to design their UIs based on classroom lessons, integrating concepts of system

design and accessibility (See Figure 7). Course materials and handouts were provided. The HCI course lessons were conducted in accordance with the Ethiopian Public University Harmonized Computer Science Undergraduate Curriculum, which structures the HCI course into six chapters delivered over 16 weeks (totaling 48 hours). These pre-designed student UIs were evaluated using a pre-survey assessing their self-determination in evaluating the specific implementation of accessibility elements and their awareness of assistive technology. In the post-design phase, students improved their designs using a M-LLM, specifically Gemini. We hypothesized that the LLM would function as a metacognitive scaffold (Matsuda et al., 2020), offering real-time suggestions (e.g., compliance checks) to foster autonomy (e.g., exploring solutions). In this study, two authors prepared manuals on using M-LLMs and conducted a 2 hours lecture session using Zoom meeting on querying Copilot, Gemini, Perplexity AI, and ChatGPT; however, students were restricted to using Gemini to minimize inconsistency and complexity (Burger et al., 2023). Finally, we examined the influence of the M-LLM on students' self-efficacy, motivation, and focus on the student's cognitive engagement with the AI tool, an interaction that implicates the psychological needs of autonomy (the student's sense of agency in the design process), and competency (their developing of accessibility skills) in UI design through the lens of Self-Determination Theory. This evaluation involved both low-achieving and high-achieving students, and the student UI Design Projects were further subjected to Heuristic Evaluation. Notably, while the students project in our study was conducted in groups—a context where relatedness is certainly at play—our intervention and measurement were specifically targeted at the human—AI interaction. Therefore, our investigation focuses on the autonomy and competence pillars of SDT.

The study design in Paper IV involved two groups of 3rd-year students: an experimental group and a control group. The study commenced with pre-accessibility and design-related knowledge assessments for both groups. Subsequently, both groups received a two-week intervention, comprising two two-hour sessions per week (See Figure 7). The first two-hour session of the initial week involved a discussion of various types of persons with disability and their implications for technology utilization.

Furthermore, legal and ethical considerations were examined, including international legal frameworks, such as the Americans with Disabilities Act (ADA), the European Accessibility Act (EAA), Accessible Rich Internet Applications (ARIA), Section 508, and EN 301 549. The ethical imperatives for accessibility, namely inclusion, equity, and universal usability, were also addressed.

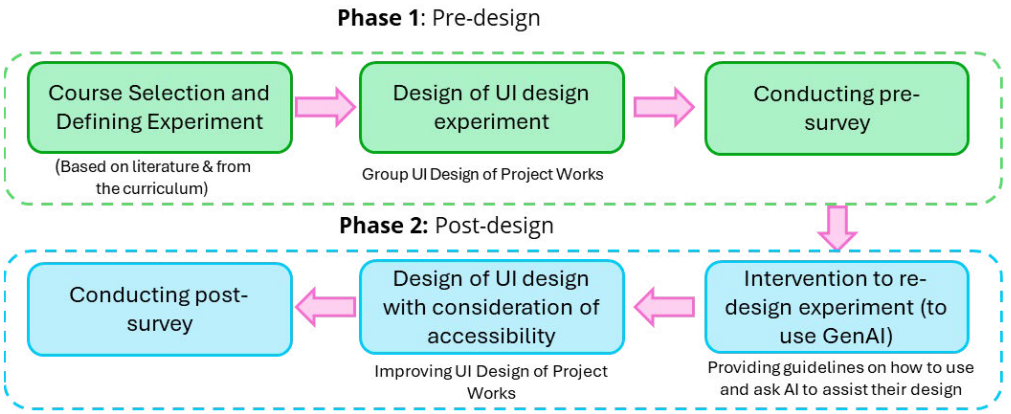



**Figure 7:** Study design procedure of study III (adopted from Article III)

Session two focused on accessibility guidelines, specifically the Web Content Accessibility Guidelines (WCAG) principles of perceivability, operability, understandability, and robustness. This session further elaborated on WCAG conformance levels (A, AA, AAA) and key success criteria. The subsequent two-hour session in the second week centered on the role of assistive technologies, such as screen readers, screen magnifiers, and alternative input devices. Additionally, the session explored accessibility evaluation tools, including WAVE and Lighthouse, color contrast analyzers, and manual testing protocols.

The concluding session addressed technical facets of accessibility, including HTML accessibility (semantic HTML, appropriate heading usage, and form and image accessibility), CSS accessibility (e.g., color and contrast, focus indicators), and JavaScript accessibility (e.g., ARIA attributes for dynamic content, focus management using JavaScript).

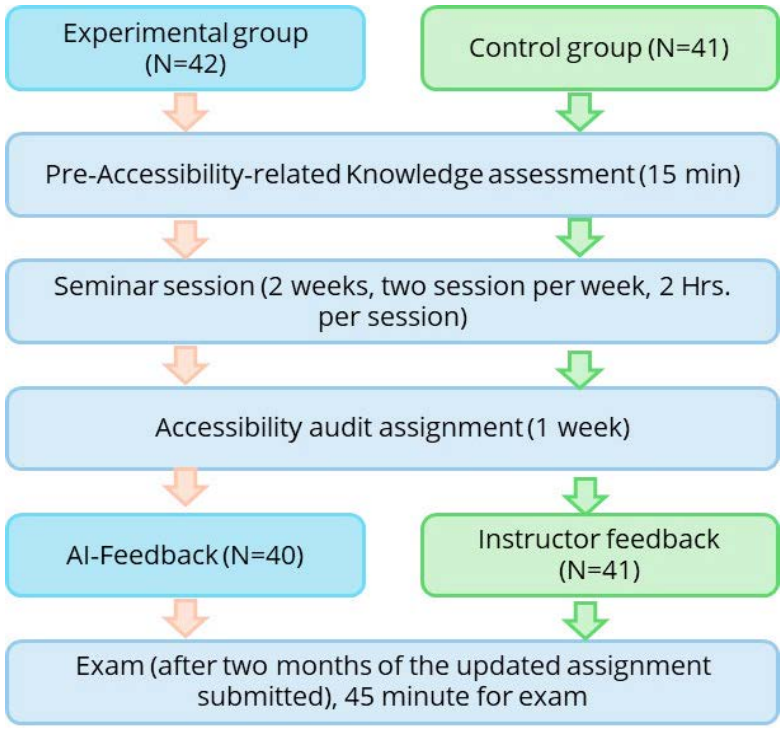



**Figure 8:** Experimental study procedure of paper IV (adopted from article IV)

Both the experimental and control groups received the same instructional content for each session in a face-to-face format but at different times, ensuring students acquired foundational knowledge (Suskie, 2018). Following the intervention, both groups of students were given assignment tasks and exams. The assignments involved evaluating the accessibility of Ethiopian higher education websites, and the exams were related to the assignments; however, they also included critical analysis questions on problem-solving and reflective open-ended questions aimed at assessing students' analytical abilities (Ku, 2009). The assignments were conducted based on instructor-provided web evaluation criteria and guidelines on evaluating the accessibility of digital technologies, after which students submitted their evaluation reports. Subsequently, each group received feedback on their reports from both the instructors and the M-LLM, with the same evaluation metrics used for both, although the feedback process differed between groups.

Feedback for the control group was provided directly by the course instructor, who delivered digital written comments on the students' reports.

In contrast, participants in the experimental group utilized ChatGPT to generate their own feedback. The primary course instructor, along with a trained research assistant, subsequently acted as the two independent graders for the final assignment and exam scores.

To ensure the AI-generated feedback was both consistent and of high quality, a specific prompt was created to guide the experimental group. This prompt, designed using the Basic Prompting technique (Marvin et al., 2023), required students to supply their complete report to ChatGPT (4.0) and ask for suggestions aligned with their learning goals. Students in the experimental group were instructed to use this provided prompt followed by the full text of their report (see paper IV, section 3.2). The first author validated the fidelity of the AI's output by randomly sampling and examining 25% (10 out of 41) of the AI-generated feedback. This examination confirmed that the AI consistently followed the prompt's directives and delivered feedback consistent with the course's grading rubric. To guide this process, students in the experimental group received a detailed instruction sheet, the specific prompt, and a brief 20-minute orientation on the AI tool. The protocol required a single interaction with ChatGPT using the given prompt, and students had to submit the AI feedback with their revised assignment, which enabled researchers to verify protocol adherence and perform the fidelity check. Both groups then had one-week to revise their assignments based on the feedback they received before the assignments were graded.

To assess both short-term and long-term learning impacts, a 45-min exam was administered eight weeks (two months) after students submitted their revised assignments. This interval was chosen to evaluate knowledge retention beyond the immediate revision cycle. All 81 students who participated in the initial assignment and feedback phase also took the final exam, resulting in an attrition rate of 0%. This was ensured because the exam was a mandatory component of the course curriculum. To control for confounding variables, the same exam was administered to both the control and experimental groups simultaneously under identical proctored conditions. The graders remained blind to the group assignments of the students during the exam evaluation, maintaining the integrity of the experimental control for this long-term measure.

## 5.5 Data collection

Data collection in computing research encompasses different categories, including primary, and secondary data methods (Mazhar et al., 2021). Primary methods involve firsthand data acquisition through techniques such as surveys, interviews, observations, experiments, and usability testing to directly address research questions (Driscoll, 2011). Secondary methods leverage existing data sources like literature reviews, government publications, online databases, and web analytics to gain insights from previously collected information (Stewart & Kamins, 1993). A range of other methods is also vital to the field.

Based on our dissertation's research questions (RQs), we employed both primary and secondary data for the qualitative and quantitative studies (See Table 4). While we used secondary data, specifically literature reviews of existing peer-reviewed studies, for Paper I, the remaining studies (Papers II, III, and IV) utilized different primary data sources. In Paper II, we used researcher-guided interviews to gather primary data directly from study participants, based on instruments adopted from existing studies (Bi et al., 2022; Cheng et al., 2024; Li et al., 2024). In Paper III, primary data collection involved mixed data sources, including a scaling survey, design artifacts, and interviews. Survey data, rated on a 5-point Likert scale, were used to measure students' perceptions of the pre- and post-UI design, accessibility, and self-determination. The design artifacts were used to evaluate and measure the accessibility of their designs for diverse users. The evaluation instrument for the design artifacts was adopted from existing studies (Alomari et al., 2020; Yeratziotis & Zaphiris, 2018), and a walkthrough heuristic evaluation of the designs was performed. The survey instrument was adopted from Ryan & Deci (2017) and refined by the researchers within the study context.

Finally, in the last paper (Paper IV), we used student exams and assignments as tools to evaluate their understanding of accessibility and inclusive design concepts. We then evaluated the student exams and assignments from both the experimental and control groups to compare score distributions between the two independent feedback groups (Human vs. AI) and to evaluate changes within groups over time.

**Table 4:** Studies data source and research context

| Study | Data Types | Data collection method | Data source | Research context |
|---|---|---|---|---|
| Paper 1 | Secondary Data | Searching on academic Databases | Scopus, Web of Science, ERiC | Globally |
| Paper 2 | Primary Data | Researcher leads interview | Interviewed data | In Ethiopia |
| Paper 3 | Primary Data | Researcher-led interviews, survey questions, | Design artifact, Interview, Scaling survey | In Ethiopia |
| Paper 4 | Primary Data | Students score | Assignments, Exams | In Ethiopia |

## 5.6 Data Preprocessing and Analysis

Data preprocessing is an essential step that prepares raw data for actual data analysis (Romero et al., 2014). In our study, before analysis, each study's data was preprocessed using different tools such as atlas.ti. According to Famili et al (1997), data preprocessing is performed for two main reasons: to overcome problems with the data, including handling missing data, correcting errors, removing duplicates, and making preparations for data analysis.

The preprocessing of study I, including data collection and identification, extraction, and categorization, was presented in Section I. However, the other three empirical studies (Paper II, Paper III, Paper IV) used different data analysis methods, including data cleaning and data coding of textual analysis based on quantitative and qualitative data. The details of each study's data preprocessing and analysis were presented as follows.

### 5.6.1 Preprocessing

To improve data quality, this dissertation performed data cleaning and removed errors and inconsistencies from the information (Rahm & Do, 2000). The audio-recorded interview data in Paper II were first transcribed and imported into a spreadsheet for each guided question. Each transcribed sentence with missing or incomplete data points was deleted. While we

requested their consent for the use of their data for research purposes, data from non-consenting groups, as well as from groups with missing values, were removed. Similarly, while in Paper IV, preprocessing was conducted on quantitative data, in Paper III, it was conducted for both qualitative and quantitative data.

### 5.6.2 Coding of Data

This study used both qualitative (in Papers II and III) and quantitative coding (Papers III and IV) of data. Coding in qualitative research is the analytical process of organizing raw data into themes that assist in interpreting the data (Baralt, 2011). Coding is, thus, the activity that the researcher (coders) engages in, while codes are the names used denote a group of similar items or ideas that the researchers have noticed in their dataset. This coding was further classified into two methods of analysis, particularly when performing thematic coding: deductive coding—derived from the philosophical framework or theory, using pre-existing themes or frameworks to analyze the data—and inductive coding, which is data-driven, allowing themes to emerge from the data itself; for instance, themes were emerging from participant's discussions (Fereday & Muir-Cochrane, 2006).

While Paper I, the systematic review, uses deductive coding to extract relevant information from the primary studies, in Paper II, we used inductive coding of textual data on students' perspectives regarding the use of M-LLMs in the design of accessible UIs. These data were then imported into Atlas.ti software to identify themes, patterns, and concepts, using coding and categorization techniques to perform thematic analysis (Lester et al., 2020). The identified themes, necessary concepts, and quotes were then included in the study report.

Quantitative data coding, which is numerical, typically involves labeling, counting, frequency reports, and statistical analyses. Sometimes, quantitative methods are also used to code the data using NLP algorithms, sentiment analysis, or topic modeling (Albalawi et al., 2020). Paper III used quantitative coding to prepare the datasets for statistical analysis, particularly for the ENA. Similarly, in Paper IV, quantitative coding of students' exam results and

assignment scores was for perform the statistical analysis (the details are presented in section 4.6.3).

### 5.6.3 Data analysis and tools

While several quantitative data analysis methods exist, including regression analysis, and correlation analysis, we used descriptive data analysis to explore the data in both (Papers III and IV) studies. In addition, thematic analysis (in Papers II, III) and ENA were further utilized as an additional data exploration tool in Study III.

Furthermore, in Paper III, we used heuristic evaluation conducted by the researchers to analyze the students' UI designs. This method was selected over automated WCAG evaluation tools for two primary reasons. First, the students' designs were static Figma prototypes, not live, coded websites. Automated tools (e.g., WAVE, Lighthouse) require implemented HTML/CSS to analyze and are, therefore, incompatible with design-stage artifacts. Second, and more importantly, heuristic evaluation allows for the assessment of broader usability and inclusive design principles that extend beyond technical WCAG compliance, including evaluating concepts like the match between a system and the real world and aesthetic and minimal design, which are relevant for understanding of accessibility from a perspective that cannot be captured by automated checkers. Only students' designs specifically developed for pedagogical purposes, including an Amharic language learning application, a virtual meeting or video conferencing application, a digital note-taking application, a learning management system (LMS), and a virtual lab environment, were evaluated. A three-point scale, ranging from 0 (not incorporated) to 1 (fully incorporated), facilitated the heuristic evaluation process. Pre-design scores indicated the initial level of incorporation of usability heuristics, whereas post-design scores of students reflected the extent of improvements achieved after design iterations (See paper III). The evaluation focuses on seven key variables, including match between system and real world, visibility of system status, user control and freedom, consistency and standards, error prevention, aesthetics and minimal design, and accessibility, which were adopted from an existing study (Alonso-Ríos et al., 2018).

In Paper IV, rigorous assessment scoring and inter-rater reliability assessment were performed on students' assignments and exam grading. To ensure consistent and objective evaluation, all assignments and exams were graded using a detailed scoring rubric. Assignments were assessed on a 100-point scale with five criteria directly linked to the course's learning objectives and WCAG principles, whereas exams were scored out of 10 points to measure conceptual understanding. To ensure reliability and mitigate bias, two independent raters, who were blind to the students assigned feedback groups, graded all submissions. The inter-rater reliability was validated on a sample of assignments using Cohen's Kappa, which yielded a value of K = 0.83, indicating almost perfect agreement (Chaturvedi & Shweta, 2015) between the raters before they proceeded with grading the remaining papers.

#### 5.6.3.1 Descriptive and inferential statistics

In our dissertation work, particularly in Papers III and IV, we used both inferential and descriptive statistics. Descriptive statistics summarize data to describe the pattern or relationship within a sample or population (Kaur et al., 2018). In Paper III, different descriptive measures were used, including mean, standard deviation (SD), and Cohen's d. Pre-post-intervention means and SD were used to describe the average scores and their variability at each time point, respectively, whereas Cohen's d was used to quantify the magnitude of the effect observed in the study. In Paper IV, mean was also used to measure the baseline score, after-feedback score, and exam score of students, and SD was used to measure the spread or variability of a dataset around its mean scores.

Inferential statistics, the second statistical measure used, can be classified as either parametric or non-parametric. Non-parametrics method are used with nominal or ordinal data, which lack of a normal distribution, and for situations where other assumptions for nonparemetric tests cannot be met (Hollander et al., 2013). This includes Chi-Squared, Fisher's Exact, Mann-Whitney U, Wilcoxon Signed-Rank test, and Spearman Rho (Allua & Thompson, 2009). Statistical significance is calculated using information contained only in the sample (rather than the population).

In our study III, we used Mann-Whitney U to compare exam score distributions between the two independent groups. This non-parametric test was used to determine if one group tended to achieve significantly different scores from the other. To compare related measures, specifically the change in scores from the pre-feedback assessment to the post-feedback assessment among the same participants, the Wilcoxon Signed-Rank non-parametric paired test was performed. This test assesses whether a change in scores over time within each group. Parametric statistics are one of the most common approaches for interval/ratio data, assuming a normal distribution of data (Vrbin, 2022). This includes t-test, ANOVA, MANOVA, Pearson's r (correlation), and regression (Allua & Thompson, 2009). We used paired t-tests in Paper III to assess the statistical significance of the observed difference between pre- and post-intervention means, allowing us to determine if this difference could be attributed to the intervention or to simple chance. The SPSS tool, Jamovi, was used for statistical analysis in Papers III and IV.

#### 5.6.3.2 Thematic Analysis

We followed the six steps of thematic analysis described by Maguire and Delahunt (2017) in our study, specifically in Papers II and III. First, the researcher became familiar with the data by conducting and documenting the records of follow-up interviews with the nine student groups. Second, the researcher generated initial codes from the interview data using Atlas.ti software to organize similar ideas and concepts (Friese et al., 2018); examples, remembering design elements, and accessibility and usability awareness. The third step, searching for themes, involved categorizing these initial codes into specific themes; in our case, we created five themes that corresponded with the interview questions. Fourth and fifth, these themes were reviewed and defined, resulting in the clearly structured categories presented in the findings, such as AI impact on UI design for an HCI project and the challenges of using AI tools for accessible UI design. Finally, the write-up phase is represented by the findings section of the paper, where the analysis, organized by the defined themes, is presented and supported with the necessary quotes from the students.

#### 5.6.3.3 Epistemic Network Analysis

Shaffer and Ruis (2017) introduced ENA as an innovative method for modeling connections between codes, displaying them as a non-directional network in two dimensions. One standard method of using ENA involves assessing the intensity of relationships between codes to compare different groups and to map their average network structures. Beyond this, ENA facilitates statistical analysis to determine differences between the groups. In Study III, ENA was utilized to visualize the co-occurrences of different groups to map learning patterns across achievement levels. Specifically, ENA was used to investigate students' perceptions of AI tools in UI design projects, focusing on associations between codes such as accessibility, knowledge, design, and improvement for low-achieving and high-achieving groups of students.

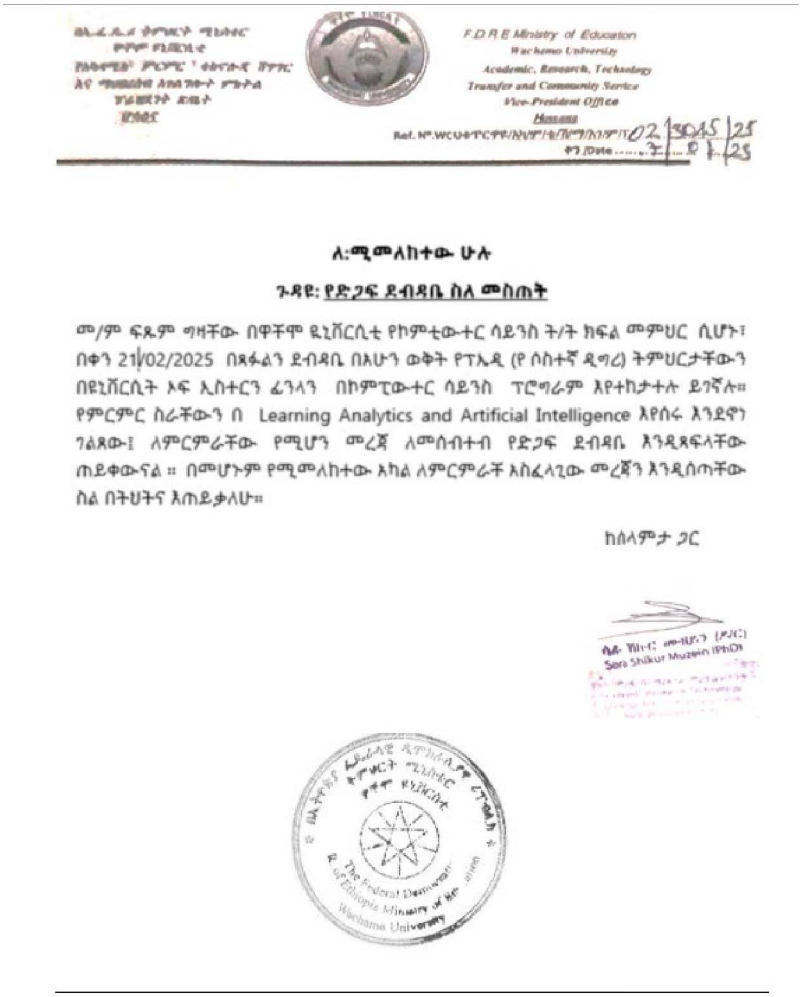
## 5.7 Research Ethics

Ethical conduct is vital in scientific research for several reasons. Primarily, it ensures respect for the privacy of individuals participating in studies (Weinbaum et al., 2019). Furthermore, adherence to ethical principles is indispensable for upholding the integrity and trustworthiness of research findings (Weinbaum et al., 2019). Although specific ethical considerations are context-dependent, varying across nations and academic disciplines, fundamental tenets of research ethics underpin all responsible research practices (Sikes & Piper, 2010). These core principles encompass several important aspects: ensuring the voluntary and informed consent of individuals before their enrollment in a study, rigorously safeguarding the anonymity and confidentiality of their data, explicitly informing participants of their right to withdraw from the research at any juncture, according due consideration to the specific vulnerabilities of certain participant demographics, and diligently striving to mitigate any potential risks associated with participation (Midgley et al., 2013).

In our dissertation work, we meticulously adhered to ethical procedures and principles across all constituent studies. In Paper I, a systematic review was conducted. While not involving direct interaction with human

participants or primary data acquisition in the conventional sense of empirical research, we integrated all requisite ethical considerations throughout its methodological framework. This encompassed the rigorous application of clearly defined review protocol guidelines (Page et al., 2021), including explicit inclusion and exclusion criteria for study selection (See Table 3), a transparent search strategy detailing databases, keywords, and the search process, and a systematic study selection process involving screening and quality assessment. Furthermore, data extraction was conducted using a predefined codebook, and data synthesis was performed using appropriate methodological approaches.

In Paper II, prior to the commencement of researcher-guided interviews, the interviewer explained the purpose of the interview, and assured their anonymity and confidentiality. Explicit informed consent was then requested from each participant to record the interview. Moreover, to ensure the fidelity of the transcribed data, the audio-transcribed text was provided to the interviewees to verify its accuracy in representing their verbal contributions, after the data were transcribed. Similarly, in Paper III, informed consent was obtained from all participating students, and their anonymity was preserved throughout the study. Finally, in Paper IV, an acknowledging support letter was granted by the Academic Vice President of Wachemo University (one of the Ethiopian higher educational institutions: See Figure 9) to collect the data from their institution before the study initiation. Subsequently, participants were informed about the study's objectives, and informed consent was obtained following a thorough brief to them about the study. We informed them of their right to withdraw from the study at any point during the data collection process. To rigorously protect their confidentiality, all identifying information, including names and ID numbers, was anonymized prior to data analysis.



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ከሰላምታ ጋር

Sora Shikur Muzein (PhD)

The Federal Democratic Republic of Ethiopia Ministry of Education Wachemo University

**Figure 9:** Acknowledging support letter to collect data

Furthermore, we recognized the ethical implications related to bias in AI tools and their accessibility. To promote the ethical use of AI, in emperical studies, students were provided with targeted training and awareness sessions aimed at developing their understanding of how AI systems function, including their inherent limitations and potential biases in data and algorithms. This training was designed to equip students with the prompt and critical skills needed to evaluate AI-generated outputs and to mitigate overreliance on potentially biased systems.

# 6 RESULTS

## 6.1 Overview of the research results

This chapter presents the findings of the research conducted in this dissertation, focusing on the accessibility and inclusivity of digital technology design. It presents the findings of the studies through four interlinked papers (Papers I–IV), one systematic and three empirical, that collectively make a meaningful contribution to the body of knowledge in the context of accessibility and inclusivity education.

## 6.2 Identification of Accessibility Problem: Findings from the Systematic Review and Empirical Studies

Paper I focused on identifying relevant literature concerning the implementation of accessibility in digital educational technologies over the years, including the key areas emphasized in published scientific papers. Utilizing three pertinent academic databases (see Table 3), 36 articles were retrieved and analyzed to answer the RQs of the studies. The systematic review aimed to answer the guiding research questions: (i) What are the current focal points of accessibility design features in digital learning technologies, and how have they catered to diverse user needs? (ii) How have these technologies incorporated accessibility design features? and (iii) What assessment approaches and evaluation perspectives were used for accessible learning technologies? Furthermore, through empirical studies (detailed in Papers III and IV), this research explored the needs and awareness of participants and analyzed the identified knowledge gap in accessibility and inclusive design education and practice before undertaking empirical studies to understand their background knowledge about the domain.

The details of the selected articles in the review were also highlighted, including the year of publication, education level, design used, assessment

approach followed, and incorporated accessibility design features. The analysis revealed that 69% of the studies were conducted in higher education and encompassed a broad range of learning environments. Notably, only two labs were specifically designed for visually impaired students, and whereas 47.3% of the studies focused on various engineering subjects, there was a limited number of labs developed specifically for educational purposes.

While the systematic review identified a gap—a lack of accessibility and inclusive design in digital learning technologies for all learners, particularly evident in the synthesis of existing V-Labs—it also revealed several issues that contribute to this accessibility problem (see Figure 10). These issues included a lack of incorporation of accessibility design features, such as limited diverse representation in lab scenarios; absence of responsive design; infrequent use of multiple modalities (e.g., text, images, video) for information delivery; unclear and inconsistent visual design; non-intuitive navigation; and a lack of specialized features. Our finding also revealed that various challenges hinder the accessibility of learning labs, including a lack of essential design components and the difficulty in balancing functionality with ease of use. The implementation of inaccessible design features in these technologies, such as the inability to navigate, low color contrast, inconsistent layout and navigation, images or links lacking descriptive text, and videos without captions, had consequences for these problems.

Furthermore, the review highlighted several overarching challenges. First, insight into the actual accessibility of the learning technologies studied was limited. Despite their widespread adoption and potential to enhance learning across different subjects, an understanding of their current accessibility status remains elusive. Second, a key concern was limited inclusivity in digital technology for persons with disabilities. The prevailing sentiment present accessibility barriers for students with cognitive, visual, and auditory limitations. Third, the integration of advanced assistive technologies in digital technologies was notably limited, as was the accommodation of accessibility elements within the learning environments.

While assistive tools to support diverse learners exist, their incorporation into learning tools is minimal. In our review of the V-lab case, for instance, features like robust keyboard navigation, clear visual focus indicators, compatibility with screen readers, and voice recognition software were not

consistently implemented. Fourth, complex challenges in the design and compatibility of V-Labs impede universal accessibility. These challenges include technological limitations, pedagogical considerations, and infrastructural issues. Fifth, the pedagogical design, comprising the learning activities within these environments, was not created to accommodate the diverse needs of students, which could limit the full participation of all learners. For instance, providing equivalent visual and auditory feedback in a meaningful manner for visually impaired learners remains a design challenge. Sixth, limited multilingual support and a lack of exploration into the role of online tutoring in addressing accessibility challenges are evident. While linguistic diversity was a relevant aspect of equity, the integration of multilingual features in V-Labs was infrequent. Seventh, insufficient attention to interaction design details further contributes to accessibility challenges. Features like clear error indication, intelligent suggestions, and mechanisms that destigmatize errors through trial-and-error are not consistently implemented or thoroughly explored for their accessibility benefits. Unlike the well-established guidelines for web accessibility, the absence of specific standards for V-Labs makes it difficult to consistently evaluate and compare their accessibility.

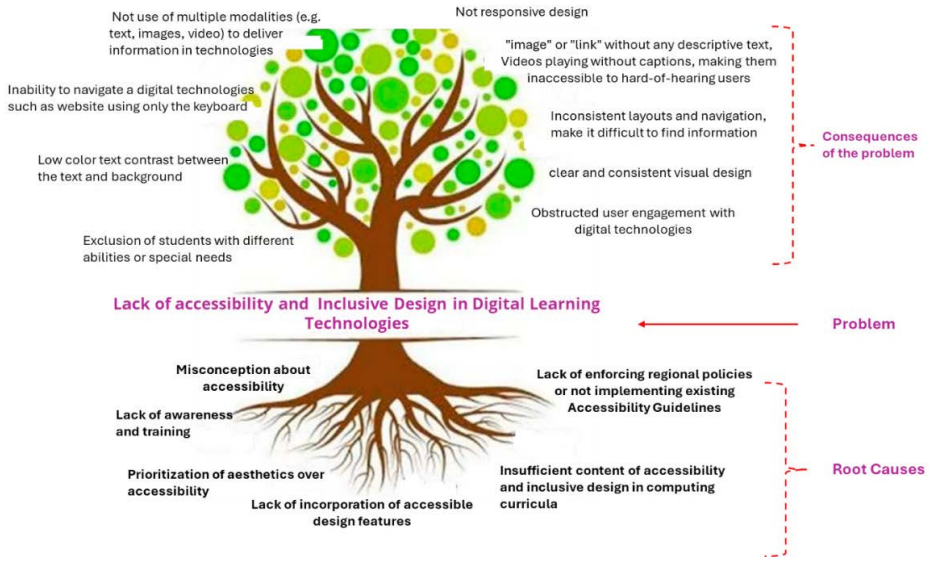



**Figure 10:** Digital Inaccessibility and Lack of Inclusive Design: Cause and Consequence [Paper I, III, and IV]

Moreover, from the pre-knowledge assessment of students' views in Papers III and IV, we identified several accessibility challenges, including a lack of awareness regarding accessibility and inclusive design in students' training, and misconceptions about accessibility. Paper IV reveals that most students (N = 63, 76.8%) reported limited experience or knowledge of individuals with diverse abilities interacting with technology, and their self-assessed accessibility knowledge in web design and development was predominantly basic (N = 57, 69%). Furthermore, familiarity with specific accessibility issues in web design was low (N = 68, 81%), and a majority (N = 60, 73%) reported not considering these issues in their previous design works. While over half of the students (N = 41, 55%) had prior involvement in designing or developing websites or software, a smaller proportion (N = 30, 37%) affirmed having knowledge of accessibility standards and principles, suggesting a gap between general development experience and specific accessibility expertise.

Therefore, the primary research phase provided insight into the challenges to be addressed, as well as an understanding of the existing situation and possibilities for further improvement in the design of accessible technologies. These findings underscore the need for greater attention, research, and application to develop accessible and inclusive learning technologie. Accordingly, we conducted the second study by focusing on how students develop accessible learning technologies with the support of emerging adaptive technologies, such as AI.

## 6.3 Artificial Intelligence in Accessible Design Education

This section presents the findings of empirical studies (Papers II, III, and IV) included in this dissertation, with the focus being on student perceptions of GenAI in design and their accessibility awareness and self-efficacy. Furthermore, the findings explore the impact of AI on students' UI project design and that of network connections on how different groups (low- and high- acheiving students) perform with the basic codes of UI design.

In ENA, for our study context, code represents specific concepts of knowledge that were being analyzed within a dataset related to students' beliefs, understanding, or knowledge structures on GenAI and accessibility.

### 6.3.1 Qualitative insights into Students' Experiences in Designing Accessible UI and the Role of Multimodal LLM in their Design Process

A qualitative study explored students' experiences in designing accessible UIs and their perceptions of GenAI tools within the design workflow. The research focused on undergraduate computer science students engaged in a UI design project as part of their HCI course. Data were gathered and analyzed through thematic analysis of student feedback and follow-up interviews, providing insights into the role of AI tools in UI design education.

The study revealed that GenAI tools played several constructive roles (see Figure 11). Notably, students found AI tools to be effective as reminders and suggestion providers, aiding in the design process by suggesting UI design elements and improvements, as a collaborative design partner. The finding revealed that AI tools served as a valuable source of quick and clear accessibility and inclusive design information, delivering design suggestions grounded in design principles and international standards, with students appreciating the tools' ability to provide simplified and time-saving responses.

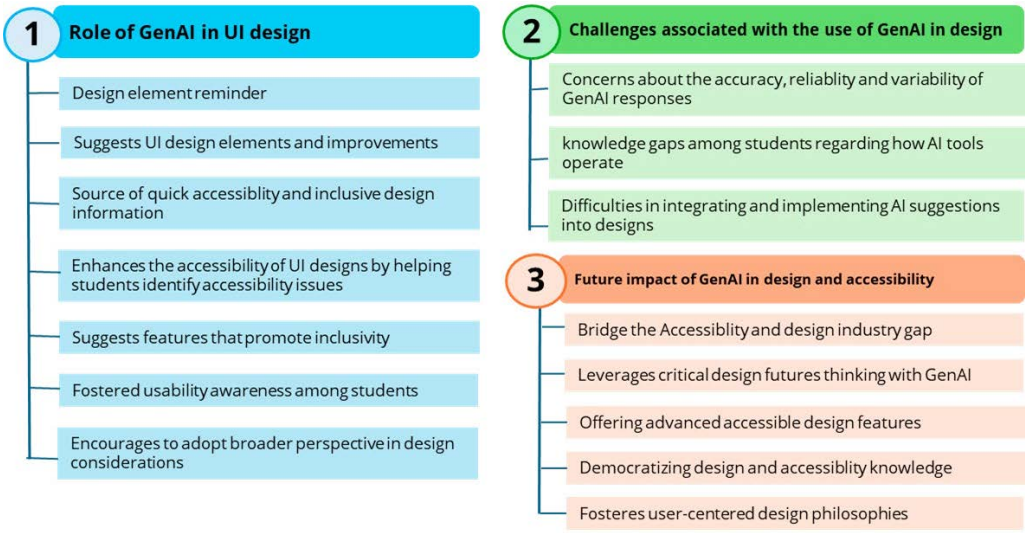



**Figure 11:** Students' perception of GenAI in accessible and inclusive UI design (see Paper II)

Furthermore, AI tools have contributed to enhancing the accessibility of UI designs by helping students identify accessibility issues and suggesting features that promote inclusivity. By considering the students' view, we observed that the use of AI tools also fostered usability awareness among students, encouraging them to adopt a broader perspective in their UI design considerations.

However, the study also identified challenges associated with the use of GenAI tools. These challenges include limitations related to the tools themselves, such as concerns about the accuracy and variability of AI-generated responses, as well as knowledge gaps among students regarding how GenAI tools operate. Students also encountered difficulties in integrating and implementing AI suggestions into their designs, with some finding that an over-reliance on these suggestions could lead to cluttered designs and expressing a need for clearer implementation guidance. Group dynamics and decision-making processes in collaborative settings presented further challenges, particularly when students had differing levels of familiarity with AI tools and conflicting views on which suggestions to incorporate in their project design.

Furthermore, students demonstrated varying degrees of comfort with using AI tools. While some students reported feeling comfortable due to the tools' ease of use and positive impact on their design skills, others expressed partial comfort, due to the perception that AI tools sometimes required prior knowledge, could be confusing, and had the potential to induce over-reliance, potentially hindering independent thought.

Looking ahead, students generally held positive views on the potential of AI tools to improve the accessibility of software design for diverse user groups. They anticipated that AI would continue to evolve, offering advanced features, democratizing design knowledge; filling the accessibility expert gap in the industry, particularly in developing regions like Ethiopia, and supporting user-centered design philosophies.

### 6.3.2 Quantitative Findings on Student Accessibility Awareness, Self-Efficacy, and the Impact of AI in the Design Process

The quantitative study, as presented in Paper III, explored the impact of using M-LLMs on students' accessibility awareness, self-efficacy, and the UI design process. The study involved undergraduate third-year computer science students who participated in a 12-week teaching experiment within an HCI course focused on accessible UI design. A mixed-methods approach was employed, utilizing pre- and post-intervention design projects, surveys, and statistical analyses to evaluate the effectiveness of LLMs in enhancing students' understanding and implementation of accessible UI design.

The study's quantitative findings, derived from pre- and post-intervention surveys and analyses, indicated improvements in students' understanding and application of accessible UI design principles following the intervention of M-LLMs. Specifically, the research revealed students' abilities to apply UI design principles increased, with statistically significant ($p<0.01$) results being observed in their consideration of user needs, application of design principles, and integration of keyboard and screen reader compatibility; See Paper III appendix E, pre-post students UI design. Furthermore, the study highlighted positive changes in students' implementation of accessibility features; enhancements were observed in their use of alt text and color contrast, incorporation of speech and audio options, acknowledgment of the importance of accessible design, and understanding of the impact of inaccessible design.

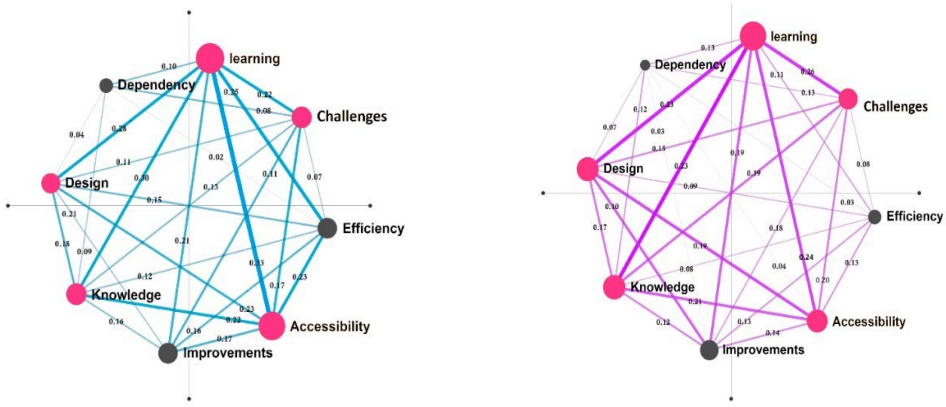
In terms of self-determination, students reported greater autonomy post-intervention, demonstrating increased freedom in applying design principles, including accessibility features, and considering assistive device compatibility. Additionally, students' self-efficacy was also positively impacted, per their reports; they had confidence in overcoming design challenges, designing accessible UIs, and improving their designs. However, while these gains were evident, the study also indicated areas of less improvement; specifically, changes in foundational accessibility knowledge, competence, and motivation were limited ($P>0.01$).

### 6.3.3 Insights from ENA of Student Experiences among Low- and High- acheivers with GenAI

In Paper III, we further investigated student perceptions of GenAI tools within UI design projects through ENA to understand the differences between low- and high-achieving groups in their UI project work achievement and to understand the groups' patterns of associations between key codes—accessibility, knowledge, design, and improvement. Participants were categorized into lower-achieving and higher-achieving groups based on their final project evaluation scores. Out of the 30% allocated for project work, students' marks were classified into two groups: "low achievers" if they scored below 21%, and "high achievers" if they scored between 21% and 30%. Semi-structured interviews provided qualitative data that enriched the quantitative ENA results, revealing distinct patterns in how these groups interacted with GenAI tools.

The ENA highlighted differences in network connections between the two achievement groups. In ENA, the strength of a connection (edge weight) signifies the frequency of co-occurrence of elements. Thicker, darker lines or higher numerical weights indicate more frequent co-occurrences, while thinner, lighter lines or lower numerical weights represent less frequent co-occurrences (M. Khan, 2021). We did not use a predetermined threshold for occurrence values of ENA; instead, we present and compare the connection values between two lower and higher achievers.

Regarding accessibility, lower-achieving groups exhibited a stronger association between accessibility and the challenges they faced compared to higher-achieving groups (See Figure 12). This suggested that lower-performing students encountered more difficulties in integrating accessibility principles into their designs, with some expressing uncertainty about applying relevant guidelines. Conversely, higher-achieving students demonstrated a stronger link between learning and accessibility, indicating their ability to effectively utilize AI-generated insights to inform accessible design practices. While both groups demonstrated a similar connection between their foundational knowledge and accessibility considerations, the application of this knowledge appeared to vary.




A: High achiever network

B: Low achiever Network

**Figure 12**: Epistemic network analysis of ;ow-acheiving vs. high-acheiving student groups

In the domain of knowledge, higher achievers displayed stronger connections between knowledge and their learning processes, as well as between knowledge and their design activities. This reflected their capacity to synthesize AI-provided knowledge with established design standards, with some explicitly mentioning cross-referencing AI suggestions with established guidelines. In contrast, lower achievers showed dependence on AI tools, lacking a similarly strong knowledge-related connection observed in higher achievers. This pattern potentially indicates an over-reliance on AI without a critical evaluation of the generated information.

Concerning design, lower-achieving students experienced more challenges linked to the design process, often struggling with iterative refinement. Some participants described feeling overwhelmed by AI suggestions and uncertain about prioritizing changes. Conversely, higher achievers effectively used AI feedback for design improvement through iteration, suggesting a more strategic approach to incorporating feedback. Their stronger connection between design considerations and accessibility further under-scored their commitment to adhering to international standards throughout the design process.

Finally, regarding improvement, higher-achieving students closely associated improvement with accessibility, indicating they used AI suggestions to enhance the inclusivity in their designs. Lower-achieving students, on the other hand, linked the challenges they faced to improvements, implying that AI served more as a tool to address existing difficulties.

### 6.3.4 Findings from a Heuristic Evaluation of Student-Developed UI in Project-Based Learning

In our dissertation, specifically in Paper III, we used a heuristic method to identify design problems in UIs created by students through project-based learning in their HCI courses. A heuristic evaluation is a usability inspection method where experts assess a UI based on a set of predefined usability principles.

The heuristic evaluation revealed varied areas of progress across the different student projects. The Amharic language learning application improved in incorporation of UDL principles, enhancing features like font and color adjustments, auditory prompts, and gamification, but error prevention and accessibility needed further attention. The LMS showed advancements in accessibility, icon clarity, and user progress indicators, but error prevention and visibility of system status did not progress (see all the students' pre- and post UI designs in Appendix 5). The virtual meeting application introduced relevant updates, such as font and color adjustments for low-vision users, and multi-input chat features were included. The digital note-taking application demonstrated improvement with the addition of braille support, audio input, and accessibility enhancements. Finally, the virtual lab application incorporated valuable features like experiment simulations and improved accessibility inputs.

Overall, the heuristic evaluation of the students' initial and revised UI designs indicated a positive trend towards better usability, with an average post-design score improvement of 51.7% across all projects, suggesting an enhancement in their understanding and application of UI design principles following the use of GenAI in their design process. This improvement highlights the effectiveness of the pedagogical approach in supporting better UI design skills among the students, which leads us to our final study (Paper IV) on examining how this GenAI has a lasting effect on design and accessibility education.

## 6.4 Feedback in Accessibility and Inclusive Design Education

### 6.4.1 Comparative analysis of AI and Instructor Feedback Impact on Accessibility Learning

In paper IV, the researchers performed a comparative analysis to empirically investigate the impact of AI-generated and instructor-delivered feedback on student performance in learning accessibility and inclusive design through practical website accessibility audits. The analysis compared assessment outcomes between students who received AI feedback and those who received instructor feedback across three stages. First, a pre-feedback baseline assessment of students' digital audit and accessiblity assignment were performed. Thereafter, a post-feedback performance assessment was conducted after the intervention or feedback from the instructor, and GenAI feedback was provided to them. Finally, a long-term knowledge assessment via a two-month delayed exam was performed. Non-parametric statistical tests, namely the Wilcoxon Signed-Rank test and the Mann-Whitney U test, were employed for this analysis.

According to the baseline student performance analysis there were no statistically significant differences between the AI feedback group ($M = 7.63$, $SD = 0.85$) and the instructor feedback group ($M = 7.13$, $SD = 1.32$; $W = 969.0$, $p = 0.156$), which indicated that both groups possessed comparable initial knowledge levels in the subject matter. Following the feedback intervention, the analysis of the post-feedback performance assessment also showed no statistically significant difference between the AI feedback group ($M = 8.03$, $SD = 0.83$) and the instructor feedback group ($M = 7.86$, $SD = 1.54$; $p = 0.745$), see Table 5. This suggested that both feedback modalities had a similar immediate impact on improving student performance after the practical website accessibility audits task. However, the analysis of the long-term knowledge score outcomes, as measured by the exam scores, indicated a statistically significant difference between the two groups ($p < 0.01$). Descriptive statistics revealed that the mean exam score for the instructor feedback group (Mean = 8.04, SD = 1.53) was higher than that of the AI feedback group (Mean = 6.00, SD = 0.82). This finding suggests that instructor-delivered feedback had a more long-term effect on students' knowledge on learning accessibility.

**Table 5:** Descriptive statistics of Human vs AI feedback

| Scores | Feedback | N | Mean | SD | P-Value |
|---|---|---|---|---|---|
| Baseline score | AI-feedback | 40 | 7.63 | 0.85 | P = 0.156 |
| | Human-Feedback | 41 | 7.13 | 1.32 | |
| After Feedback score | AI-feedback | 40 | 8.03 | 0.83 | P = 0.745 |
| | Human-Feedback | 41 | 7.86 | 1.54 | |
| Exam score | AI-feedback | 40 | 5.27 | 1.66 | **P< 0.001** |
| | Human-Feedback | 41 | 7.21 | 2.52 | |

Further analysis of gender revealed no significant differences between the AI and human feedback groups for either males or females in the baseline or after-feedback scores (see Table 6). However, a statistically significant difference in exam scores was observed for both male and female students ($p < 0.01$) within each group. Specifically, within both genders, students who received instructor feedback showed higher mean exam scores than those who received AI feedback.

**Table 6:** Descriptive statistics of students' performance with each group across gender

| Gender in each group | N | Score at different points | Mean | SD | W | Z | P-Value |
|---|---|---|---|---|---|---|---|
| AI feedback - Male | 27 | Baseline sore | 7.611 | 0.923 | 16.0 | -3.323 | **< 0.001** |
| | 27 | After Feedback sore | 8.00 | 0.82 | | | |
| AI feedback - Female | 13 | Baseline sore | 7.654 | 0.718 | 6.0 | -2.191 | 0.013 |
| | 13 | After Feedback sore | 8.077 | 0.886 | | | |
| Human feedback - Male | 29 | Baseline sore | 7.198 | 1.317 | 0.0 | -4.107 | **<0 .001** |
| | 29 | After Feedback sore | 8.043 | 1.527 | | | |
| Human feedback - Female | 12 | Baseline sore | 6.979 | 1.384 | 0.0 | -2.521 | 0.013 |
| | 12 | After Feedback sore | 7.417 | 1.55 | | | |

These findings underscore the potential importance of considering feedback modality and its interaction with individual learner characteristics when designing educational interventions focused on both immediate learning and long-term knowledge effects in accessibility education.

# 7 DISCUSSION AND CONCLUSION

This chapter highlights the importance of the findings of the research reported in this dissertation and reflects on them in the context of our research question. In addition, we explain how this dissertation contributes to different bodies of knowledge and scientific communities. Potential future directions stemming from this study are also covered in this chapter.

## 7.1 Reflection on the research Questions

> ***RQ1**: What accessibility elements are implemented in learning technologies, specifically in V-Labs, and how do they accommodate users with diverse needs?*

Section 1.2 established the lack of accessibility in digital educational tools and underscores the primary deficiency that often emerges among web and digital technology designers. This problem arises from a lack of understanding of accessibility and inclusive design principles among these designers, leading to limited accessibility elements being incorporated into digital learning technologies. Furthermore, a needs assessment of computer science students' experiences, as presented in Paper III, revealed the lack of foundational knowledge on accessibility and inclusive design among students.

To address the limited inclusion of digital accessibility in educational technologies, it is essential to first identify key accessibility issues and gaps within current digital platforms. This identification is necessary for equipping students with knowledge of these important design and inclusive concepts, thereby preparing them for future careers in the digital technology industry. Consequently, understanding these accessibility issues by answering RQ1 is vital.

RQ1 addresses how developers and researchers might currently design and develop these educational technologies (focusing on Virtual Lab) and underscores the importance of ensuring equitable learning environments.

Thus, RQ1 highlights the accessibility of educational technologies, particularly V-Labs. By review of secondary studies, RQ1 examined how learning technologies incorporate accessibility for wider user bases. Kumar and Owston (2016) suggested that to understand the current state of accessibility in digital learning technologies, it is necessary to evaluate and identify accessibility features to eliminate learning barriers. Following this recommendation, our investigation in Paper I reveals a varied landscape of accessibility features being integrated into V-Labs. As V-Labs are an example of digital educational learning technology, we used them as a case study in our research. The accessibility features identified in V-Labs were broadly categorized into instructional design support, interaction support, virtual service accessibility, and subject-specific accessibility.

Based on these findings, although the implementation of various accessibility elements in existing literature is a positive step, their efficacy in accommodating the vast spectrum of diverse user needs highlights a greater design challenge. Paper I highlights the potential of V-Labs, and the limited insight into reviewed studies on accessibility and inclusivity, particularly in the context of design education. These findings lead us to to explore further study on how to incorporate accessibility and inclusivity in design by making learners aware of these needs.

> ***RQ2**: How do Multimodal LLMs support software development students' awareness of accessibility concepts and their ability to integrate inclusive design practices into UI design?*

The importance of digitally inclusive experiences has positioned accessibility as a fundamental tenet of quality software development and UI design. In this context, the emergence of sophisticated AI, particularly GenAI, presents both exciting opportunities and pertinent challenges for educating the next generation of software developers. Thus, answering RQ2 requires a synthesized examination of student experiences with GenAI tools (Paper II) and investigates the pedagogical efficacy of M-LLMs, along with broader insights from the field (Paper III).

The journey of a software development student toward advancing accessibility often begins with awareness, recognizing the diverse needs of users and the principles that support inclusive design. In this dissertation, through Paper II, the researchers examined how GenAI contributes to the initial design stage by acting as design elements and requirement reminders as well as suggestion providers. The finding revealed that students valued AI for making them aware of accessibility and usability aspects. Paper III reinforces this, demonstrating through an experimental study that structured engagement with an M-LLM led to better understanding of accessibility principles among computer science students.

Paper III further quantifies the impact, showing enhancements (51.7%) in the heuristic evaluation scores of students' post-intervention UI designs, indicating a tangible improvement in applying these concepts. Both studies (Papers II and III) highlight AI's role as an efficient information retrieval resource; students in Paper III specifically appreciated the GenAI for providing access to design principles and accessibility guidelines like WCAG, allowing them to get concise answers instantly. This efficiency could enable students to focus more on iterative design refinement. Furthermore, M-LLMs serve as powerful tools to expose students to potentially accessible designs and thereby broaden their understanding of accessible design patterns.

Beyond raising awareness, the basic aim of this study is improving students' skills to *integrate* inclusive design practices into their UI design workflows such that they can incorporate the implimentatation of accessibility in future the when they join technology industries. Paper II highlighted that students perceived GenAI tools as helpers in designing accessible UI for different users. Paper III provides empirical support for this, with findings revealing significant positive effects ($p<.001$) on students' actual implementation of accessible UI features and their self-efficacy after using an M-LLM. The qualitative data from Paper III describes the M-LLM acting as a collaborative design partner, providing iterative feedback and suggestions for optimizing UI elements and incorporating accessibility features like keyboard navigation alternatives and audio recognition. Students in Paper III also reported an improvement: from a compliance-based mindset to prioritizing usability for diverse populations

after M-LLM interaction, suggesting a broader integration of inclusive design thinking.

However, the integration of M-LLMs into accessibility education was not without challenges, as several issues were raised across both Paper II and III studies and in the broader literature. For instance, Paper II identified concerns about the accuracy of AI responses, variability in suggestions, students' knowledge gaps regarding AI, and difficulty in implementing suggestions. Paper III quantifies some of these limitations by examining the practical implementation skills and self-efficacy with M-LLM use, as well as gains in foundational accessibility knowledge, competence, and motivation.

Both studies (Papers II and III) point to the critical importance of prompt engineering; Paper III explicitly states that students' ability to formulate effective prompts influenced the quality of feedback received from the M-LLM, and Paper II noted issues with question accuracy. The black box nature of GenAI can also hinder learning if students passively accept AI outputs without critical evaluation. Paper III's ENA demonstrated that high-achieving students strategically used the M-LLM to augment their design expertise, particularly in accessibility standards compliance, while lower-achieving students tended to utilize it more for reactive task correction and showed greater dependence. This highlights a risk of AI supporting cognitive offloading, referring to reducing their mental effort to perform in learning design activities if not managed properly.

we examined the lasting impact of GenAI on supporting software development students in accessibility and inclusive design within computing education (RQ3) based on students' perspectives on the support that GenAI offer in understanding their design accessibility principles and how their designs improve with LLM suggestions.

> ***RQ3**: To what extent does AI-based feedback improve students' proficiency in identifying and resolving accessibility issues in design education compared to instructor feedback in the short- long term?*

This dissertation addresses RQ3 by examining the feedback modalities on students' learning of digital accessibility education, particularly technical

skill and an understanding of UX through web auditing of Ethiopian higher educational websites' design accessibility (Paper IV).

In the study, the researcher provided students with GenAI intervention and assessed how they could incorporate the tools to get support through its feedback in their web-auditing and provided an intervention through a seminar on key topics on accessibility and design concepts. Based on the analysis, the study's findings offer valuable insights, suggesting an interplay between the efficacy of AI feedback and the lasting importance of human instruction. The study revealed the dichotomy between short-term and long-term learning accessibility and inclusive design outcomes. The research finding in Paper IV indicates that AI-generated feedback was as effective as instructor feedback in facilitating immediate improvements in students' ability to identify accessibility issues. This might be due to AI's capacity to provide consistent and standards-aligned evaluations, particularly in technical domains like web accessibility, where adherence to guidelines (e.g., WCAG) is relevant. According to Gao et al (2025), GenAI tools can efficiently identify specific design issues and offer targeted suggestions for improvement, thus aiding students in achieving design competence.

However, the study reveals a significant difference when in long-term outcomes. Students who received feedback from instructors indicated better performance in delayed assessments designed to measure the outcome of accessibility concepts. This suggests that while AI can effectively guide students in the application of accessibility principles in specific tasks, it may fall short in supporting the broader cognitive processing necessary for the internalization and long-term recall of these principles (Hatalis et al., 2023). Instructor feedback, with its adaptability to students' learning backgrounds, appears to be more conducive to creating a deeper understanding.

This difference was due to the contextual nature of human feedback. Instructors can guide individual learning styles, address specific misconceptions (Fakour & Imani, 2025), and offer broader insights that connect accessibility principles to user experiences. Accessibility involves understanding the diverse ways in which various people interact with technology and the social implications of inclusive design. Thus, human educators are better equipped to provide these complex insights, thus improving the critical thinking

and problem-solving skills that are essential for long-term competence in accessibility.

On the other hand, in different educational domains, several studies comparing AI and human feedback across various academic and cognitive contexts have yielded multifaceted findings. For example, a recent meta-analysis by Kaliisa (2025) indicated no statistically significant differences in learning performance or student feedback perception when comparing AI-generated feedback against human feedback in learning performance. However, high heterogeneity persisted across the findings, indicating wide variability depending on context, specific AI models (e.g., rule-based vs. generative AI), and study design. The study noted that limited research on hybrid feedback (AI plus human) demonstrated some positive effects on performance gains, learning dispositions, and engagement when compared to human-only or AI-only conditions. However, Steiss (2024) identified specific qualitative differences, finding that well-trained human evaluators provided higher-quality formative feedback on student writing across different dimensions (e.g., clarity of directions, accuracy, prioritization, and supportive tone); in contrast, GenAI was better only in providing criteria-based feedback. Similarly, in a programming course, Leite's (2020) work demonstrated that human feedback improved students' comprehension of algorithmic logic by connecting code syntax errors to underlying logical issues. The course grade distribution further indicated that students who received human feedback performed better, with this effect being most noticeable among students in the middle two quartiles of the class. Most of these studies focused on effects observed during the intervention period or immediately following it, detailing short-term learning. This aligns with our finding that human evaluators can perform better than AI in providing short-term feedback. Furthermore, while there is a lack of long-term comparative studies on the effects of these assessments, our study found that the benefits of human feedback are sustained. Future research is required to validate this finding in similar or different contexts.

## 7.2 Study Reflection

Ensuring the universal accessibility and ultimate acceptability of information technologies by anyone, anywhere, at any time, and by any media and device are primary goals of the information society (Stephanidis & Savidis, 2001). This aligns with the concept of universal access; in the context of this study, it refers to the effort to proactively apply methods, principles, and tools of universal design (Persson et al., 2015). The aim is to develop information technologies that are accessible and usable by all citizens, including the very young, the elderly, and diverse persons with disabilities (Fonseca et al., 2018). Thus, while various accessibility frameworks (such as UDL), principles (like W3C's Web Accessibility Initiative), guidelines (e.g., WCAG), and other initiatives have been developed globally to mitigate accessibility and inclusive design challenges and barriers, their adoption and implementation remain nascent in developing regions, particularly in Africa. While this study began by the assessment of learning environments, particularly V-labs, for their accessibility, it identified gaps that need to be addressed through design education.

Addressing accessibility in design is an intrinsic component of the design lifecycle, instead of an eleventh-hour remedy. Consequently, integrating accessibility into the design of digital technologies is paramount (Gilbert, 2019). This integration is achieved through various methods, including providing awareness regarding accessibility and inclusive design principles among software development students. Indeed, industry experts have argued that software development students should acquire the skills of inclusive design and knowledge of accessibility early on, before joining the industry (Martin et al., 2022). In addition, Papers II and III (which show students gained skills in accessibility and knowledge of inclusive design through their UI design process) confirm our rationale and contribution for students entering the technology industry sector, where accessibility skills are often lacking.

Moreover, while this dissertation focuses on accessibility awareness and inclusive design education, it acknowledges that recent participatory design research emphasizes the active involvement of disabled users throughout the design process. Studies in participatory and co-design approaches (e.g.,

Van Dijk et al. (2016), Bennett and Rosner (2019), Guedes et al. (2024)) have argued that meaningful collaboration with disabled participants has moved beyond designing for users toward designing with them. Contrasting with our focus on student learning and design process improvement, these perspectives highlight the importance of incorporating user participation and lived experience into accessibility education to foster empathy, ethical sensitivity, and real-world relevance.

Furthermore, it is important to equip them with the practical skills necessary to conceptualize and develop universally accessible digital learning technologies. Such education can be delivered through formal academic programs or informal training initiatives. Within formal education, this can be actualized via dedicated accessibility-focused courses (Sonka et al., 2021) or by embedding these principles and practices within existing computing curricula (Baker et al., 2020; Elglaly et al., 2024; Whitney, 2020). Disciplines such as HCI, Web Design and Development, Software Engineering, and Programming are widely recognized as pertinent areas for accessibility education; however, their inclusion and depth often remain inadequate (Baker et al., 2020). Despite these subjects being expected to incorporate accessible content, observations reveal a tendency to either moving accessibility to an elective status or provide insufficient coverage. Therefore, to effectively equip students with skills and awareness in software development, there is a need for pedagogical interventions.

This dissertation contributes to computing education by presenting an empirical investigation of the application of GenAI technologies within the context of design education. The findings from this research indicate that AI, and M-LLMs in particular, can serve as a potent tool for curriculum developers seeking to integrate accessibility into their teaching practices within design and computing programs. Our findings further align with broader discussions in the inclusive design field, which consistently emphasize the imperative of adhering to established standards, such as the WCAG, in the development of any learning technology (Caldwell et al., 2008). Furthermore, the principles of UDL are advocated, promoting the creation of flexible learning environments to incorporate diverse individual needs from their inception (L. Zhang et al., 2021). Evidence from this dissertation, including qualitative insights regarding

GenAI tools (Paper II) and focused experimental results with an M-LLM (Paper III), demonstrates the considerable promise of M-LLMs in supporting software development students. These models have been shown to enhance students' awareness of accessibility concepts and improve their capacity to implement inclusive design practices, leading to better UI design outcomes and supporting better student autonomy and self-efficacy.

However, this potential necessitates a critical and reflective pedagogical approach. Educators need to address several challenges. This includes inculcating prompt literacy among students; mitigating the risk of over-reliance on AI tools, a particular concern for students who may be lower-achieving; supporting the skill of evaluating AI-generated suggestions; and acknowledging and navigating the potential for biases within AI systems. We also acknowledge that many AI tools present accessibility challenges, especially for users with visual, auditory, or cognitive impairments. While our study did not directly assess accessibility features of the AI tools used, we recognize this as a critical ethical concern and recommend that future research explicitly evaluate and address the inclusivity and accessibility of AI technologies in educational contexts.

## 7.3 Implications of the findings

This dissertation includes studies on AI's role in enhancing UI accessibility (Article II), the use of M-LLMs for teaching accessible UI design (Article III), and the impact of AI-generated feedback (Paper IV). It presents the implications for various stakeholders in education and technology (See Figure 13). These findings collectively highlight the potential of AI in promoting inclusive digital environments while underscoring considerations for its effective and ethical integration.

**For Educators and Academic Institutions:** our research emphasizes the relevance of integrating accessibility as a core competency within computing education, rather than treating it as an optional topic. AI tools, particularly M-LLMs improve how students learn about and implement accessibility in UI design. Incorporating these tools into HCI and software

engineering curricula can equip students with practical skills and bridge the gap in the case of many practitionering acquire accessibility knowledge outside formal education. In practical terms, these findings suggest that accessibility education can be operationalized within HCI curricula through clearly defined learning outcomes, scaffolded activities, and transparent assessment criteria. For instance, accessibility principles can be threaded across existing design and programming courses by aligning project tasks with inclusive-design checkpoints and using rubrics that evaluate both usability and accessibility dimensions. Through this approach, accessibility education can be treated as an integral component of students' design practice instead of as an isolated topic.

However, institutions should provide training on how AI tools operate, ethical principles, and accessibility standards for their students and educators to maximize their utility and avoid over-reliance on AI. While AI can offer scalable, standards-aligned feedback, especially because it is beneficial for resource-limited settings like in Ethiopia or large class sizes, human instructors remain essential for supporting critical thinking, providing conceptual understanding, and facilitating long-term knowledge outcomes. Therefore, pedagogical frameworks should explore models that combine AI's efficiency with the irreplaceable cognitive and contextual insights of human educators, and support PjBL environments augmented with AI to promote self-determination and practical skills.

**For UI or UX Designers, Developers, and Future Practitioners:** AI tools can serve as valuable collaborative design partners and information resources, facilitate design workflows, and foster user-centered approaches by helping designers consider diverse user needs and bridging theoretical knowledge with the practical implementation of accessibility standards like WCAG. Students' ability to apply UI principles and implement accessibility features improved with AI assistance. However, designers and developers should be aware of AI tool limitations, such as response variability and the need for critical assessment and the selective integration of suggestions. Foundational knowledge in accessibility and the critical evaluation of AI-generated suggestions remains relevant. Therefore, practitioners need to use AI to augment their skills rather

than replace their understanding. Developing prompt engineering skills is also highlighted as essential for effective interaction with AI tools.

**For AI Developers and Researchers:** This dissertation provides empirical evidence of M-LLMs' pedagogical value in domains like accessible design education. It also underscores critical challenges, including the limitations of free-tier AI tools, variability in response accuracy, and the necessity for users to possess foundational knowledge for effective tool utilization. This suggests opportunities for developing more sophisticated AI-based pedagogical tools that offer context-aware guidance and can better support the development of both procedural skills and conceptual understanding in complex areas like accessibility. Future research needs to examine AI feedback systems beyond surface-level error suggestion toward tools that can better support cognitive engagement, understand individual learning contexts, and promote lasting learning.

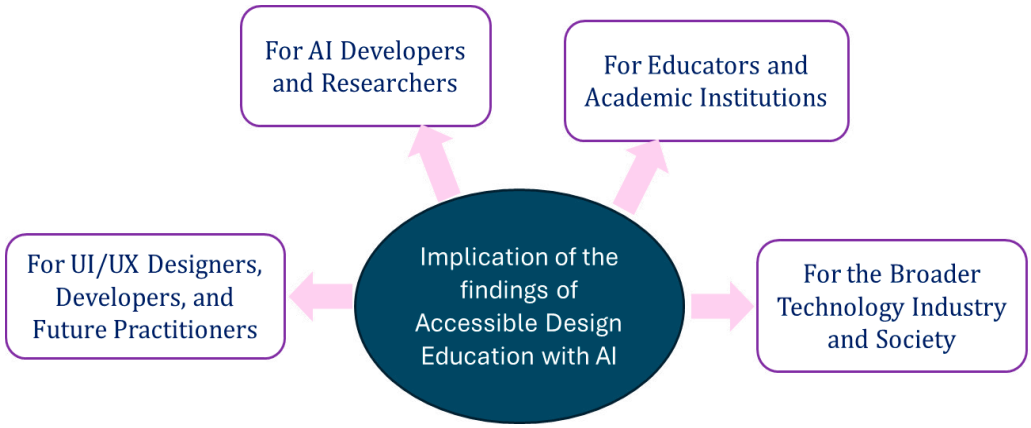



**Figure 13:** Implications of the dissertations' findings

**For the Broader Technology Industry and Society:** This study points toward AI's potential to democratize design knowledge and promote more inclusive digital products. By making accessibility considerations more prominent and easier to address, AI can contribute to a digital environment that is usable by everyone, including individuals in developing regions with limited access to resources or inclusive and accessible design training. Realizing this potential, however, requires ongoing research and development to improve

AI tool accuracy, address knowledge gaps in their usage, and ensure that AI supports the design of accessible and equitable technology for all users.

## 7.4 Contribution of the Research

Scientific literature identifies diverse research contributions, including theoretical, practical, methodological, empirical, conceptual, and technological, which Kok and Schuit (2012) categorized based on how they advance knowledge, methods, or applications. A theoretical contribution, according to Whetten (1989) extends beyond adding or subtracting factors (the whats) in a model. It requires demonstrating how such changes occur in existing relationships (the hows) and reorganizing causal maps, given that theory concerns relationships rather than simple lists. The impactful theoretical developments often arise from borrowing external perspectives to challenge a theory's underlying rationales (the whys), thereby leading to broad reconceptualization. Moreover, applying a theory in new contexts (addressing who, when, and where) constitutes a contribution only if it yields novel insights about the theory itself, thus improving it rather than reaffirming its use.

In terms of practical contributions, Bartunek and Rynes (2010) emphasized their role in bridging the research–practice gap. By examining common recommendations (e.g., increasing awareness, implementing training, refining design elements) and how these are communicated, studies can provide valuable insights into making academic findings relevant and actionable for practitioners, ultimately enhancing their real-world usage.

Methodological contributions, as outlined by Bergh et al. (2022), take several forms: introducing research conduct for a broad audience, providing incremental yet widely relevant changes, creating new methods, transferring existing methods to new contexts, refining and extending current techniques, or presenting modest advances that complement a theoretical contribution. Other scholarly works provide overviews of additional contribution types, such as empirical and technological advancements.

Therefore, this dissertation offers several contributions, primarily practical and methodological, which have implications for the scientific community, developers, and other stakeholders engaged in design and accessibility education.

The contributions of Paper I are described as a synthesis and review of empirical research that results in a conceptual structuring of knowledge, the identification of knowledge gaps, and an agenda for future research and practical improvement in V-Lab accessibility. Specifically, Paper II delivers empirical and practical contributions with implications for pedagogical methodology. Concurrently, Papers III and IV present both practical and methodological implications for educational frameworks. Table 7 summarizes of the contributions from each constituent paper of this dissertation.

**Table 7:** Contributions of the studies

| Articles | Contributions of the included studies | | | |
|---|---|---|---|---|
| | Practical | Methodological | Empirical | Conceptual |
| Article I | | | | ✓ |
| Article II | ✓ | | ✓ | |
| Article III | ✓ | ✓ | | |
| Article IV | ✓ | ✓ | | |

### 7.4.1 Practical Contributions

The practical contributions of this research are demonstrated through an investigation of the effectiveness of AI tools in enhancing student learning of UI design and digital accessibility. Paper II empirically examines the application of AI tools within an HCI course. It reveals their functions as design elements, requirement reminders, suggestion providers, and sources of accessibility and usability awareness. This paper highlights real-world student experiences and perceptions, including practical challenges such as concerns about AI response accuracy and student knowledge gaps, while

also suggesting avenues for further study aimed at promoting user-centered UI design.

Building on these findings, Paper III provides empirical support for integrating M-LLMs into HCI and software engineering curricula. Through a 12-week teaching experiment, this study examined the effectiveness of LLMs in enhancing students' understanding and implementation of accessible UI design skills.

Finally, Paper IV contributes to practical pedagogy by investigating the comparative effectiveness of AI-generated feedback versus instructor-delivered feedback in the context of teaching digital accessibility. It provides empirical evidence suggesting that while AI feedback can provide comparable short-term learning outcomes, instructor-delivered feedback is effective for supporting long-term outcomes.

### 7.4.2 Methodological Contributions

The methodological contributions of this research focus on rigorous approaches to evaluating the pedagogical value of AI in accessibility education. Paper III employs a mixed-methods approach, utilizing a single-group pre-post quasi-experimental design. This study incorporates different analytical techniques, such as ENA, to map student learning patterns and their interactions with AI tools. This approach elucidates AI's pedagogical value and suggests a reconceptualization of educational strategies to support self-efficacy and address universal digital accessibility challenges within computing education.

Similarly, Paper IV makes a methodological contribution by employing a quasi-experimental design to compare AI-generated and instructor-delivered feedback in teaching digital accessibility. This study offers insights into both the potential and limitations of AI in supporting accessibility education, highlighting the importance of human mentorship for critical thinking and adaptive learning skills. Together, these methodological approaches guide future research in the domain of AI-enhanced education.

## 7.5 Limitations of the research

This dissertation comprises several empirical studies, each contributing to our understanding of AI's role in HCI and accessibility education. However, it is important to acknowledge the limitations across these studies, including generalizability, methodological scope, and the dynamic nature of AI technology.

The first limitation across the studies is the generalizability of the findings. Paper II, with its qualitative approach and small sample size of students, offers rich insights into student experiences but limits the broader applicability of its conclusions to a wider population of HCI students or diverse design contexts. In this study, we only identified students' perspectives to understand the role of AI in design; however, it is necessary to obtain the educators' and designers' perspectives to obtain more insights.

Similarly, Paper IV's quasi-experimental design, while practical, involved a small sample of undergraduate computer science students within a particular institutional context. This restricts the direct applicability of its results to diverse student populations, different educational levels, or varied pedagogical settings, especially in global contexts with differing technological infrastructures and learning cultures. The selection of instructor-selected groups in Paper II might also introduce a bias that may not reflect the dynamics of self-selected student teams in real-world project scenarios, further impacting generalizability.

Third, the evolving nature of GenAI tools has limitations. The effectiveness of the specific AI feedback mechanisms and tools studied in Papers II, III, and IV might change frequently with technological advancements. This implies that some findings may be time-bound, as changes in AI tools and their capabilities are continual. While Paper III highlighted AI's strength in practical skill development and compliance with established accessibility standards, it also suggested that AI's impact on foundational learning, such as accessibility knowledge, competence, and motivation, was limited. This underscores that while AI is a potent enhancer of practical skills, it does not necessarily supplant the need for a holistic educational framework that addresses all facets of learning.

Fourth, while the individual studies contribute valuable insights, these limitations highlight the need for future studies that employ larger, more diverse sample, utilizes mixed methods approaches, and consider advancements in AI technology to elucidate AI's role in design and accessibility education.

Lastly, our study focused on the SDT needs of autonomy and competence, as our AI intervention targeted individual student interaction, not group collaboration. We, therefore, did not use "relatedness." Studies that compare individual versus collaborative AI use to explore the tool's impact on competence, autonomy, and social connectedness are further required.

## 7.6 Future Directions

As indicated in the section 7.4 of this dissertation, the limitations of our study present opportunities for researchers. The role of AI in enhancing UI or UX accessibility by examining advanced AI tools and pedagogical strategies should be further explored. This necessitates research that extends beyond short term project improvements to the long-term impact of AI integration on students' critical thinking and problem-solving skills in design, even where preliminary long-term evaluations were of insufficient duration.

The findings from this dissertation underscore that GenAI and M-LLMs are valuable tools in accessibility education. Building on these findings, AI's capacity for generating diverse accessibility solutions for complex scenarios and understanding user context needs to be first understood. Second, while AI can enhance practical implementation skills and self-efficacy, there was limited evidence for its impact on foundational accessibility knowledge, competence, and motivation, indicating the need for longitudinal analyses to understand how engagement with GenAI tools affects students' ability to internalize design principles and develop critical evaluation skills with regard to AI-generated content. This could be acheived by tracking student performance over several semesters to observe their academic growth and solving accessibility challenges without over-reliance on AI. Building on this, future research should aim to bridge the gap between AI-supported

education and participatory design methodologies by investigating how students, after being trained with AI tools, engage in co-design sessions (starting from collaborative ideation and refinement, and even testing) with users with disabilities.

Third, based on Paper II's exploration, the computing and design curricula should integrate advanced GenAI tools and teach students how to engage with them effectively and ethically. Although Paper III used a limited case (a student's Amharic language UI application design) to understand the impact of AI feedback on local amharic context improvement, it is relevant to examine this topic in a broader context to examine the limitations of generative AI models regarding underrepresented cultures and low-resource languages, such as the Amharic language in the Ethiopian context.

The dissertation identified several challenges in accessibility and inclusive design. Other studies need to address these challenges. First, well-equipped or developed curricula on inclusive design and accessibility education are lacking. Hence, research into effective professional development programs for educators, equipping them with the skills to teach accessible design with AI, is relevant. Second, the dissertation underscores that ensuring accessible learning technologies is paramount to preventing the creation of new digital divides. Investigating how AI can contribute to educational equity by developing accessible learning technologies, based on identified accessibility issues, that effectively accommodate all users, particularly those with cognitive, visual, and auditory limitations, represents another vital area for future work.

**Thank you!**

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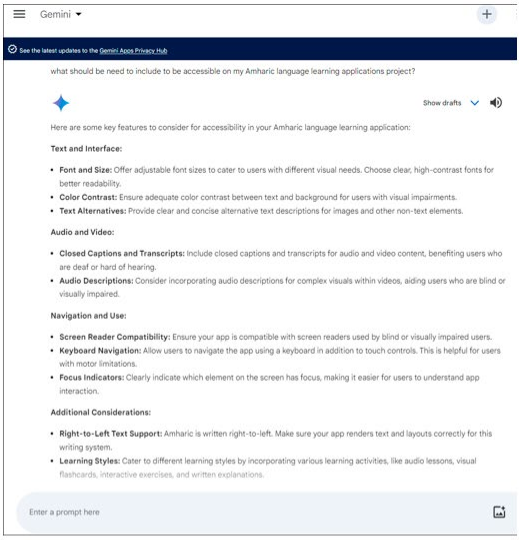
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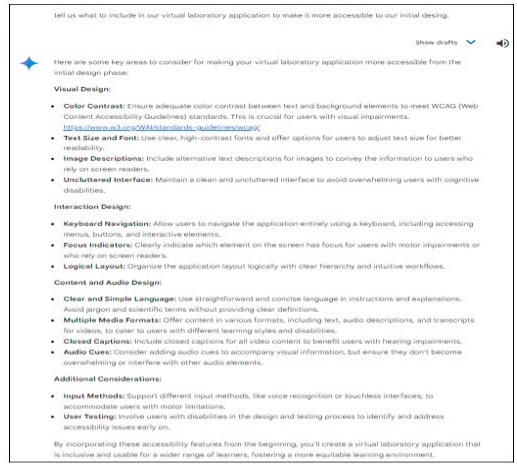
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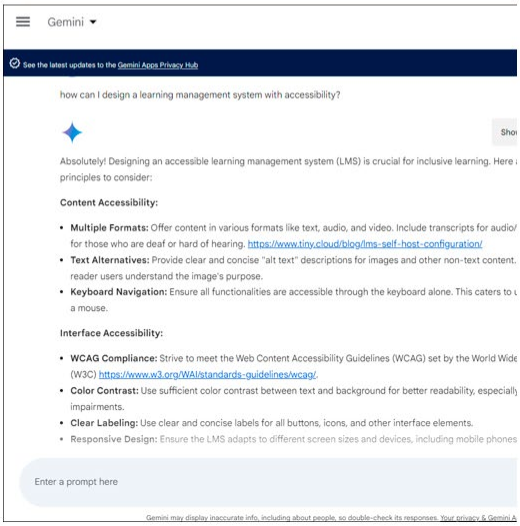
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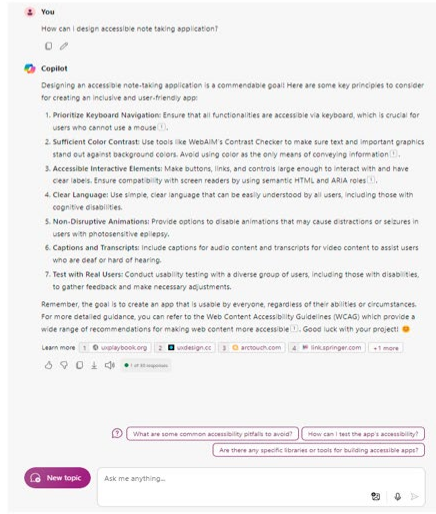
# APPENDICES

**Appendix 1** (from paper III)
Sample of student requests and AI Tools respond to students to refine their UI design project



**Appendix 2** (from paper III)

UI design Heuristic evaluation variable and items (Alomari et al., 2020, Yeratziotis & Zaphiris 2018)

| Heuristic Evaluation Form for the Students' UI Design Project+<br>Project Name: _____<br>ID No. of Students (separate with coma): _____ | |
|---|---|
| Heuristic Evaluation Variables | Heuristic Evaluation Items |
| *1) Visibility of System Status* | *Q1. Is the current system status clearly communicated to users?*<br>*Q2. Is clear what is happening on the user interface?*<br>*Q3. Are users provided with immediate feedback after performing actions?* |
| *2) Match Between System and the Real World* | *Q4. Does the system use language and concepts familiar to the target users?*<br>*Q5. Are metaphors and icons used in the interface intuitive and easy to understand?* |
| *3) User control and freedom* | *Q6. Are the navigation elements clear and consistent throughout the interface?*<br>*Q7. Can users easily move between different sections of the interface?* |
| *4) Consistency and standards* | *Q8. Is the terminology consistent throughout the interface?*<br>*Q9. Are visual elements (fonts, colors) consistent across all interface sections?*<br>*Q10. Is the page and button layout consistent across different pages of the site?* |
| *5) Error Prevention* | *Q11.Does the designed interface help prevent errors, if they occur?*<br>*Q12.Does it provide clear and helpful error messages?* |
| *6) Aesthetics and minimal design* | *Q13. Is the interface free from unnecessary or irrelevant information?*<br>Q14. Is the color scheme effective, pleasing to the eye, and provides sufficient contrast? |
| 7) Accessibility | Q15. Does the UI consider users with visual, auditory, and other disabilities?<br>Q16. Does the interface meet basic accessibility standards (e.g., color contrast, keyboard navigation)?<br>Q17. Are there text alternatives for non-text content? |

**Appendix 3** (from paper III)
Epistemic codes and their descriptions

| No | Codes | Description of codes |
|---|---|---|
| 1 | Accessibility | Discussions or responses on making accessible and inclusive design. |
| 2 | Challenges | Issues or challenges faced during the design process. |
| 3 | Learning | knowledge and insight gained through learning GenAI suggestions or functional aspects of the tools |
| 4 | Design | Mentioned design elements like layout, color, interactivity, and adherence to international standards and guidelines. |
| 5 | Improvements | Specific improvements were made to the UI design based on AI suggestions. |
| 6 | Dependency | Dependence on AI tools for tasks, including in the design. |
| 7 | Knowledge | General support or knowledge gained from AI tools in the design process. |
| 8 | Efficiency | Comments/views on time-saving and ease of use in AI tools and their comfort. |

**Appendix 4** (from paper III)
UI Design Pre-post test Questions

- **UI Design Principles & Application**

1. User needs and preferences were considered throughout my design process.
2. I have applied design principles effectively in my user interface design project.
3. My interface demonstrates a well-structured layout.
4. I integrate keyboard navigation/screen reader compatibility.

- **Accessibility Knowledge & Compliance**

5. I am familiar with WCAG guidelines.
6. I am familiar with assistive technologies.
7. I understanding design considerations for assistive devices .
8. I understand the impact of design choices on assistive technology usability.

- **Accessibility Implementation**

9. I have integrated features that support accessibility goals.
10. I include alternative text for images and ensure color contrast ratios.
11. I incorporate speech/audio output options for users with visual impairments.
12. I acknowledge the importance of designing interfaces accessible to all users.
13. I understand the impact of inaccessible design on user experience.

- **Competence**

14. I am knowledgeable about accessibility principles and guidelines.
15. I am knowledgeable about assistive technologies.

- **Autonomy**

16. I had the freedom to apply design principles effectively.
17. I had the autonomy to include accessibility features.
18. I considered compatibility with assistive devices.

- **Self-Efficacy**

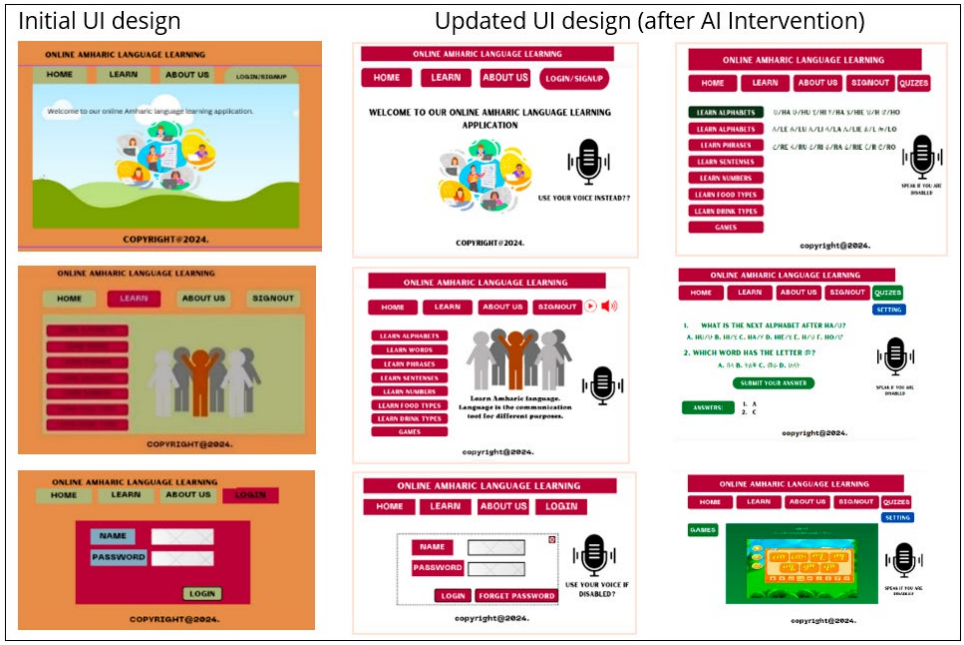
19. I am confident in my ability to apply  design principles and concpets effectively.
20. Confidence in overcoming design challenges.
21. Confidence in designing UIs accessible to individuals with disabilities.
22. Confidence in evaluating/improving UI designs.

- **Motivation**

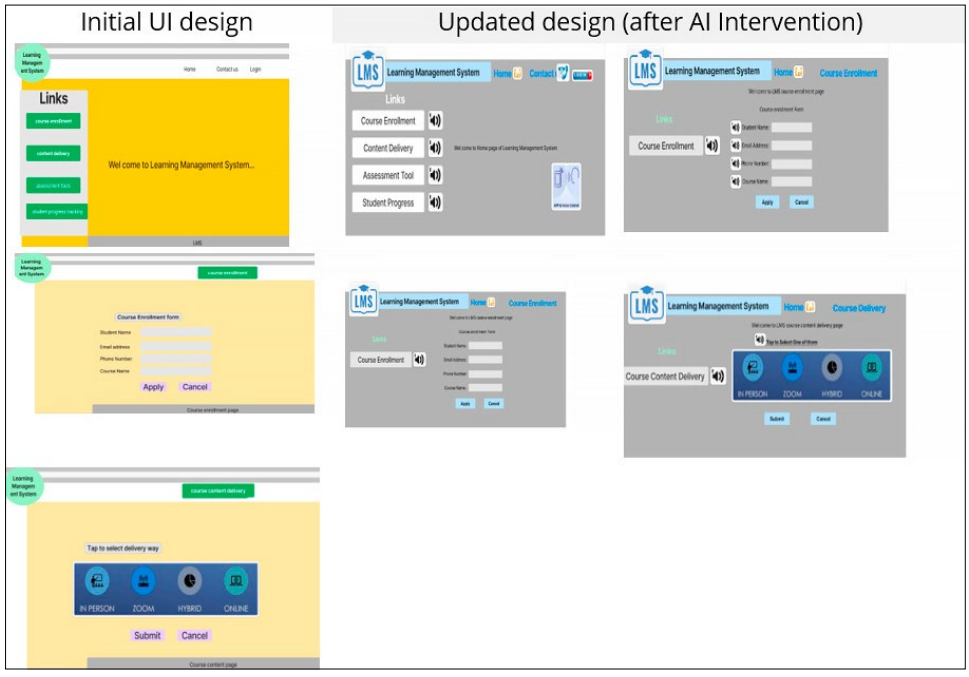
23. I find software design interesting and valuable to my professional growth.
24. Engaging in software design tasks stimulating and beneficial for my skill development.
25. I perceived capability to complete design tasks.

**Appendix 5** (from paper III)

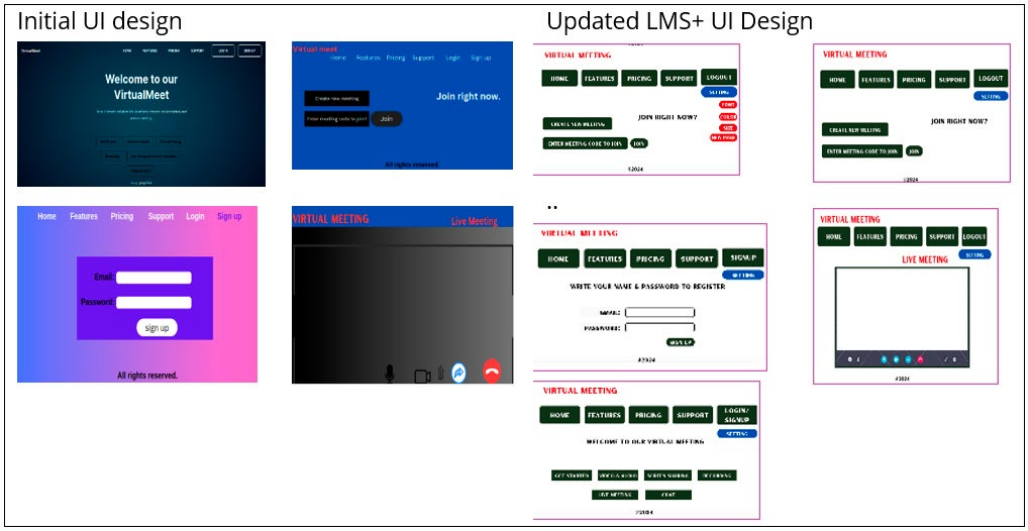
Students' UI design class project (before and After using AI tools as assistance)



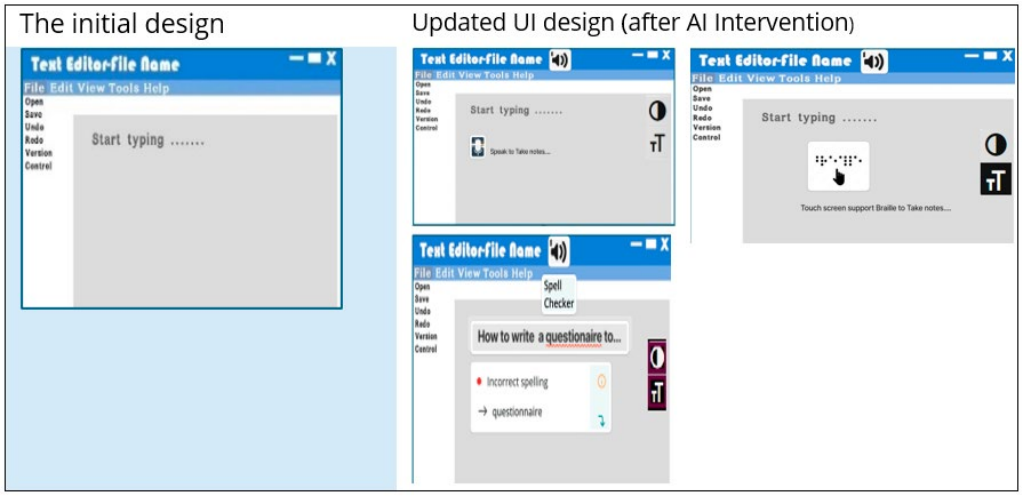

Group-1: Amharic Language Learning Application




Group 2: Learning management system UI design




Group 3: Virtual meeting application UI design




Group 4-Digital note-taking application UI design
before and after intervention

**Appendix 6** (from paper IV)

**Pre-Accessibility related knowledge Assesement**

1) I have knowledge of, or personal experience with, differently abled individuals (such as those with low vision, blindness, hearing loss, learning or intellectual differences, motor challenges) and older people interacting with technologies

○ No Experience ○ Limited Experience ○ Moderate Experience ○ Extensive Experience

2) I have accessibility knowledge in design and development, specifically in website design and development.

○ No Knowledge ○ Basic Knowledge ○ Intermediate Knowledge ○ Advanced Knowledge

3) I am familiar with accessibility issues in web design (e.g., alternative text,screen-reader interfaces, color contrast, etc.), and considered them when I design websites.

○ Not Familiar ○ somehow Familiar ○ Moderately Familiar ○ Very Familiar

4) I have previously been involved in designing or developing websites or software.

○ Yes ○ No

5) I have knowledge of accessibility standards and principles.

○ Yes ○ No

**ARTICLE I**

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# Assessment of accessibility in virtual laboratories: a systematic review

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In an era of rapid evolution in educational technologies, Virtual Labs (V-Labs) have emerged as a promising solution, fundamentally altering how learners engage with scientific concepts and experiments. Despite their potential, ensuring their effectiveness and inclusivity in terms of accessibility to diverse students remains a challenging task. Currently, there is limited insight into the accessibility of V-Labs, a gap that our study aims to address. This study seeks to ascertain the effectiveness of V-Labs in terms of accessibility and inclusivity. We synthesized empirical studies, reviewing 36 articles published between 2000 and 2023. Of these articles, 69% of the studies were conducted in higher education and covered a wide range of learning environments. Our study revealed that 47.3% of the studies focused on various engineering subjects. Our findings provide insight into V-Labs' accessibility from four key perspectives: (a) students with limited abilities, (b) diverse cultural and linguistic backgrounds, (c) instructional design features and content availability, and (d) interaction supporting features. We also identified existing gaps in the accessibility of the V-Labs from the four perspectives. Furthermore, we examined the assessment methods of V-Labs, shed light on the aspects that are evaluated, and underscored the need for future work on assessment strategies.

KEYWORDS

**virtual laboratory, accessible educational environment, inclusive technologies, online learning, online experiment**

## 1 Introduction

The emergence of the Virtual educational environment, specifically Virtual Laboratories (V-Labs), has transformed the traditional teaching approaches to modern scientific experimentation for students and researchers, opening new and innovative ways to explore and understand complex phenomena. A virtual lab is a digital simulation environment in which experiments or simulation activities are conducted virtually without the need for a physical lab. Recently, virtual labs have been widely used across the primary, secondary, and higher education sectors due to their ability to engage students and educators through simulation in teaching and learning settings.

While V-Labs have been found effective in enhancing student learning outcomes when compared to traditional labs, they provide special benefits for enabling experimental analysis across several disciplines (Al-Nakhle, 2022). The benefits attributed to cost efficiency by eliminating physical space and equipment (Apiola et al., 2022; Elmoazen et al., 2023), providing a safe environment for experiential learning (Potkonjak et al., 2016), time efficiency by elimination setup, and global accessibility (He et al., 2022). Despite its potential, access to the V-Labs relies on reachability, and connectivity of the internet, which may

create a gap in disparities of access and hinder inclusivity. Research by Van den Beemt et al. (2023) suggests that V-Labs promote active student engagement; improve learning outcomes and academic performance (Wahyuman et al., 2021), facilitate self-directed learning (Reginald, 2023), and motivate learners through gamified interactions.

To fully utilize V-Labs for teaching and learning, a multifaceted strategy is needed. For instance, the need to ensure proper functionality, regular maintenance, and use of the latest technology are essential. While V-Labs should be designed to captivate learners, make experimentation fun, and integrate into curricula emphasizing the practical application of course content; studies focus, for example (Sapriati et al., 2023), on evaluating the effectiveness and outcome of using V-Labs in various disciplines. Besides, incorporating accessible features in V-Labs for diverse learners, including those with visual or other impairments, is essential. Due to this, the effectiveness of V-Labs accessibility depends on how they are implemented and integrated into the learning environment.

Several reviews have recently explored many facets of V-Labs. For example, some studies address barriers, challenges, and possible advances in their area (e.g., Alnagrat et al., 2021; Deriba et al., 2023). Other reviews have highlighted trends and advancements in the academic landscape of V-Labs (Raman et al., 2022). In addition, specific studies examined the implementation and benefits of V-Labs within various disciplines and courses (Fan et al., 2021; Reeves and Crippen, 2021). The remaining examined the application of learning analytics and assessment methods within V-Labs' environment (Elmoazen et al., 2023). However, it is important to note that there are limited studies that focus on examine the accessibility of V-Labs and exploring the current state V-labs environment. To address these gaps, our study poses two key research questions:

(RQ.1) What essential accessibility features are integrated into V-Labs to cater to diverse user needs?

(RQ.2) How do existing studies assess the effectiveness of accessibility of V-Labs and which aspects do they assess?

This paper provides insights from a systematic review, making at least the following contributions:

- Identifying diverse virtual learning environments across various disciplines, focusing on V-Labs to cater for those with specific educational needs.
- Examining the assessment of V-Labs with a particular focus on accessibility features usability, user experience, and engagement of educators and students.
- Examining and discussing the accessibility features of V-Labs and addressing gaps in accommodating users with disabilities, diverse cultural and linguistic backgrounds, instructional design characteristics, and interaction design support.
- Examining the gaps for future research and development of enhanced inclusivity in virtual learning environments.

The remaining section of our study is organized as follows: Section two reviews the existing state-of-the-art focusing on accessibility, inclusiveness, and the existing challenges of the V-Lab. Section three outlines the methods employed to investigate the study. Section four presents the synthesized results from the studies, and discussions are presented in section five. Finally, we provide the concluding remarks in section six.

# 2 Background

## 2.1 Virtual labs across various fields of study

Across various academic disciplines, V-Labs offers an immersive and interactive approach to understanding complex concepts (Esquembre, 2015). In life science, for instance, V-Labs are used to simulate complex biological processes, such as mitosis and other dynamic cellular process phases (Muhamad et al., 2012). These simulations help demonstrate how data on these phases are collected and analyzed. Additionally, V-Labs enables understanding of the experimentation procedures in genetic expressions (Akhigbe and Ogufere, 2020), the structure of DNA, enzyme-controlled reactions, and cell reproduction behavior (Flowers, 2011), and assists in analyzing and visualizing microscopy in teaching (Pylvänäinen et al., 2023). These indicate students and researchers can understand life science's deep processes without the need for expensive laboratory equipment.

Additionally, V-Labs have been widely used in the physical sciences, including physics, where they are used to simulate a wide range of phenomena such as the motion of a projectile, and to observe how different variables like launch angle, initial velocity, and air resistance affect its trajectory (Chen, 2010; El Kharki et al., 2021). Furthermore, V-Labs in physics employ the use of unobservable light rays as a means to assist students in comprehending the behavior of light (De Jong et al., 2013). This includes understanding how light rays travel, the outcomes when light rays encounter obstacles or colored acetate, and the colors of the rays involved. Similarly, in chemistry, V-Labs are utilized for various activities (Tatli and Ayas, 2010). For example, to experiment with the titration process, observe how chemical changes are made (Tatli and Alipasa, 2012), and how different factors, such as the pH of the solution and the indicator used, affect the results of the experiment (Harvey et al., 2022). This indicates that V-Labs enable students to manipulate different chemical processes and principles, observe reactions, and analyze the outcomes without the need for substances that pose potential risks to human health, the environment, or property due to their physical, chemical, or biological properties.

Furthermore, in engineering disciplines, V-Labs have been used to design, analyze, and test different systems (Balamuralithara and Woods, 2009). For example, in Mechanical Engineering, they enable students to observe the behavior of materials and fluids under different conditions and analyze their properties (Uribe et al., 2016). In Electrical Engineering, V-Labs are utilized for analyzing electrical circuits, and measuring voltage, current, and resistance (Dhang and Kumar, 2023). They also serve to assemble and test electronic circuits, including amplifiers, filters, and digital systems (Hernandez-Jayo and Garcia-Zubia, 2016). In Civil Engineering, V-Labs are employed to analyze forces and deformations in structures such as beams, trusses, and frames (Turkan et al., 2017). Additionally, they are used to experiment with concrete mixtures, strength, and durability in materials (Ramezanianpour and Jovein, 2012).

The field of medicine and healthcare has also witnessed the integration of V-Labs to complement traditional medical practices (Cheung et al., 2023). For example, in anatomy, V-Labs are used for medical students to practice surgical simulation, helping them to understand surgical techniques and procedures (Shen et al., 2006; Prentice, 2016). In pathology, V-Labs are employed to analyze and identify tissue histology structures, cellular abnormalities, and diseases (Barisoni et al., 2020; Kainulainen et al., 2023). V-Labs are also used to diagnose virtual patients, explore the effects of different

drugs (Abutaleb, 2020), and learn about various medical procedures through sophisticated virtual simulations (Zare Bidaki, 2018). These simulations offer a safe and controlled environment for honing medical skills and enhancing decision-making abilities.

Although studies show that V-Labs are less commonly used in social science disciplines (Marks and Thomas, 2022), they are beginning to gain traction as a valuable tool. In economics, for example, V-Labs are used to simulate market behavior and test economic theories (Innocenti, 2017). While they remain a relatively new learning environment in many disciplines, they hold the potential to revolutionize the way experiments are conducted and theories are tested. Thus, V-Labs have opened doors across various disciplines, providing a versatile and accessible platform for students and researchers to explore and comprehend concepts, from simple to complex, fostering a practical understanding of knowledge. In this study, we will explore and comprehensively analyze V-Labs across various fields of discipline labs.

## 2.2 Virtual labs: accessibility features

Different researchers have employed diverse terminology to articulate accessibility attributes in distinct manners when referring to digital tools. For instance, Aljuhani et al. (2018) used usability features to identify their experimental tools' ease of use, clarity of the experiments' steps, navigation of the screens, and the interaction and enjoyability of their tools. Chaiyarak et al. (2021) mentioned the universal design element to express the tools easiness of accessing materials, offering customizable options, and helping physical, visual, auditory, sensory, communication, and interest impairments. In Robinson et al. (2017), functionality features were indicated to express the compatibility, portability, and multimedia capability of the tool. The remaining studies termed these as accessibility options. These terms were all interchangeably used to express the accessibility features of digital tools, specifically for V-learning environments. Few studies have attempted to equip and use their digital learning environment with accessibility features. For example, Ertugrul (2000) identified and used 11 criteria (or accessibility features) to consider while building V-Labs, such as user-friendliness, multi-platform portability, compatibility, extendable library, debuggable features, and more.

V-Labs' features may vary depending on the nature of the proposed field (or subject), the specific aim, and the type of V-Lab. Accessing the accessibility of the lab environment is essential to make sure all students are served equally. A study by Reginald (2023) identified that, without appropriate features of digital tools, user engagement may be obstructed, making it difficult for learners to fully participate in the learning processes. One contributing factor to these issues is the lack of well-defined or established guidelines to evaluate digital learning platforms, particularly V-Labs. In this study, we focus on a comprehensive analysis of accessibility features that are considered in the implementation of V-Labs.

## 2.3 Inclusive and accessible virtual labs

The true potential of a V-Lab is realized through the integration of inclusivity and accessibility. However, identifying accessible features and developing V-Labs for diverse learners is a challenging task due to various factors. For example, lack of incorporating accessible design features in the lab, such as diverse representation in lab scenarios, responsive design, and the use of multiple modalities (e.g., text, images, video) to deliver information. Similarly, Liu et al. (2022) states that clear and consistent visual design, intuitive navigation, and responsive design are essential features for accessibility to special needs students. The Report of Purdue University (Anderson et al., 2022) identifies one of the challenges faced by individuals with disabilities is the lack of incorporating specialized features in the design. To overcome these issues, Elfakki et al. (2023) developed and evaluated a 3D V-Lab for students with disabilities. Their study demonstrates the effectiveness of the 3D virtual environment in enhancing cognitive skills related to physics subjects.

On the other hand, inclusive V-Labs are required to encompass considerations for various disciplines as well as diverse needs and participants who have different characteristics, backgrounds, abilities, and learning styles. For example, students with different learning modalities (e.g., visual, auditory, kinesthetic), students with diverse cognitive development and abilities, and students with disabilities or special needs. A few studies have investigated diverse learners and students with visual or other impairments. For instance, Thompson et al. (2022) explored inclusive V-Labs for diverse learners and identified that students' knowledge and confidence improve when they participate in the labs. Similarly, Gavronskaya et al. (2021) examined how inclusive online labs in courses improved students with special needs, especially those who had previously struggled in science classes.

This indicates studies highlighting the need for focused attention and a comprehensive analysis of both the accessibility features and inclusiveness of existing labs.

## 2.4 Challenges of accessibility in virtual labs

The use of V-Labs in education has grown significantly over the past few years, but making them accessible to all students presents a complex set of challenges, including technological, pedagogical, and infrastructure (Deriba et al., 2023). Technological and infrastructural challenges include the lack of essential design components; difficulty in integration and interoperability with multiple systems (or platforms); and issues with media, file, and executable standards (Vary, 2000). Balancing functionality with ease of use is also a challenging task. For example, according to Leporini and Buzzi (2019), developing V-Labs compatible with assistive technologies is challenging due to the need for consideration of different technologies and compatibility within the V-Lab platform. This indicates necessitating a deep investigation of the accessibility requirements and compatibility environment across different platforms. Similarly, Kapici et al. (2019) underlined that ensuring the accessibility of V-Labs across multiple platforms and devices can be difficult, especially if they depend on specific operating systems. Furthermore, Since V-Labs are widely accessed over the Internet, they are susceptible to issues related to network latency, bandwidth limitations, and connectivity disruptions (Jara et al., 2009). Maintaining a consistent experience across different network conditions and underlying resources required to support V-Labs accessibility (Potkonjak et al., 2016) is essential. For example, ensuring that V-Labs are hosted on reliable servers with sufficient bandwidth is important.

TABLE 1 Summary of related literature and their focus of study.

| Type of review | Conducted disciplines | The focus of the review |
|---|---|---|
| Systematic review (Udin and Ramli, 2020) | Biology lab | The utilization and analysis of V-Lab with a focus on examining the characteristics of their content and effectiveness in assessing specific variables. |
| Systematic review (Zhang et al., 2021) | Physiology labs | Assessing the educational impacts of transitioning from traditional to virtual alternatives by evaluating the effectiveness of V-Labs. |
| Systematic review (Chan et al., 2021) | Chemistry labs | Identifying the important elements related to technology and instructional design in V-Labs, as well as assessing their effectiveness compared to passive teaching methods and hands-on lab experiences. |
| Systematic review (Reeves and Crippen, 2021) | Multi subjects ($n = 10$) | Examining the use and effectiveness of V-Labs |
| Systematic review (Mercado and Picardal, 2023) | Biotechnology | Analyzing the utilization, effectiveness, and technological aspects of V-Lab simulations in biotechnology education |
| Systematic review (Elmoazen et al., 2023) | Multi subjects ($n = 7$) | The intersection of learning analytics and V-Labs aims to investigate the educational levels, subjects, applications, and methods of analysis used in research on V-labs. |
| Literature review (Byukusenge et al., 2022) | Biology labs | Exploring the effectiveness of V-Labs in teaching and learning biology and identifying the topics often taught and the linked learning outcomes in the field. |
| Literature review (Sypsas and Kalles, 2018) | Multi subjects ($n = 3$) | The applications of V-Lab and their complementary role to physical labs. |
| Preliminary review (Deriba et al., 2023) | Computing labs | Investigating the accessibility barriers in V-Labs and exploring potential solutions to overcome them. |
| Umbrella review (Sapriati et al., 2023) | Multi subjects ($n = 7$) | Determining students' self-regulated learning on the use of V-Lab and investigating whether the implementation of V-Labs enhances cognitive, metacognitive, motivational, behavioral, and contextual strategies |

Pedagogically, despite the need for V-Labs to be designed in a way that supports learning objectives and allows all learners to participate fully regardless of their abilities, most V-Labs neglect to do so. For instance, Ji et al. (2022) highlights the challenges in designing V-Labs that offer visual and audio feedback to enhance accessibility for learners with visual impairments. This is due to the need for precise synchronization between the visual and audio data elements (Misiejuk et al., 2023) and ensuring that the feedback is meaningful and effectively conveys the necessary information to the learners. Furthermore, Nedungadi et al. (2018) suggested that incorporating Universal Design for Learning (UDL) principles into V-Labs is necessary to make it more engaging and interactive in the learning environment.

To broaden the scope of designing, developing, and implementing V-Labs, and to address these challenges, our study investigates the identification of essential accessibility features. This endeavor aims to overcome these challenges and ensure that diverse learners have full access to the lab environment.

## 2.5 Related literature and gap analysis

There have been numerous efforts to synthesize V-Lab research in previous studies. Most researchers have extensively focused on exploring V-Labs' applications, highlighting their ability to offer superior practical learning experiences compared to traditional labs (Sypsas and Kalles, 2018; Sasongko and Widiastuti, 2019; Elmoazen et al., 2023). Another study (Zhang et al., 2021) assessed the transition from traditional methods to V-Labs, emphasizing their effectiveness in improving student learning outcomes. Specific studies (Byukusenge et al., 2022) attempted to identify commonly taught topics via V-Labs and associated learning outcomes. Additionally, investigations (Sapriati et al., 2023) into students' self-regulated learning behaviors in V-Lab settings aimed to understand their impact on cognitive, metacognitive, motivational, behavioral, and contextual educational strategies.

On the other hand, the design and effectiveness of V-Lab subject content have been other focal points, and few researchers have examined V-Lab content utilization and assessed specific learning variables (Chan et al., 2021; Reeves and Crippen, 2021). They identified critical elements related to technology and instructional design within V-Labs, comparing their effectiveness against traditional teaching methods and hands-on lab experiences. While several related literature are conducted as shown in Table 1, there is a lack of studies that explored the inclusivity of diverse populations of students including different abilities.

Recently, researchers have started to explore accessibility in V-Labs. For example, Deriba et al. (2023) have identified accessibility barriers within V-Labs and call for further study to explore accessibility features of the V-Labs. Thus, our study motivates us to systematically synthesize V-Lab research focusing on the inclusiveness and effectiveness of accessibility features.

# 3 Methodology

The authors followed a systematic literature review guideline (Page et al., 2021) and the eight basic steps of systematic reviews described by Okoli and Schabram (2015). These guidelines were used to identify the objective of the review, develop a draft protocol, establish inclusion and exclusion criteria, conduct a comprehensive search for relevant data, and assess the quality of selected articles before writing the review.

First, the authors identified the study's objective, which was to examine the accessibility features and inclusiveness of V-Labs in

addressing research questions. After defining the scope of the review, the authors collaborated regularly to develop a protocol, aimed at minimizing bias during publication selection and data processing. By meticulously planning the review strategy, as outlined by Fink (2019), the protocol ensured reproducibility and consistency. It encompassed various aspects, including information sources, literature search strategy, inclusion criteria, quality assessment of studies, data extraction, and data synthesis methods, following the guidelines established by Kitchenham et al. (2007).

The criteria for including and excluding studies [that] used were derived from the research questions (Kitchenham, 2004). To be considered for inclusion, the reviewed articles needed to address the accessibility features, and accommodate diverse learning environments in V-Labs. Table 2 shows the inclusion and exclusion criteria that were applied in the selection process.

## 3.1 Database and literature search

The authors selected three well-established databases for their search: Scopus, Web of Science (WoS), and ERIC. Both Scopus and WoS databases apply stringent inclusion criteria for journals and conferences, possess a robust meta-data system, and are widely recognized as reputable sources for literature searches (Kumpulainen and Seppänen, 2022). Additionally, ERIC is an educational database encompassing a diverse array of educational literature (Robbins, 2001).

We conducted multiple search iterations using different combinations of keywords. Initially, we searched by combining the keywords "virtual laboratory" with "accessibility," which severely restricted our findings, causing us to miss several relevant papers. This limitation occurred because some authors did not use the word "laboratory" in their study. They were represented in various forms, such as lab, labs, or laboratories, which resulted in those papers being overlooked by the initial keyword selection. To address this issue, we decided to broaden the search and retrieve any article containing the keyword "Virtual lab*" with a wildcard. As well as "accesssib*," for accessibility, accessible. Then, using the expertise of researchers, we qualitatively identified which of these articles were truly about virtual laboratories. So, the search formula that produced the most effective results: ("Virtual Lab*") AND ("Accessib*") are applied to the metadata of the papers in each database.

TABLE 2 Inclusion and exclusion criteria used in the study.

| Criteria | Inclusion | Exclusion |
|---|---|---|
| Topic and focus | Research work discussing V-Lab, their accessibility features, and considers diverse needs in the design and implementation of V-Labs | A research paper that does not specify accessibility themes |
| Publication stage | Peer-reviewed original work published in a journal or conference proceeding | Non-peer-reviewed articles and Articles in the press |
| Document type | Journal article and conference proceeding | Posters, review papers, books, dissertations, editorials, reports, policy papers, short papers reporting on research in progress |
| Language | English | Other languages |

The chosen combination of keywords was utilized to search the fields of article titles, abstracts, and author keywords. The search was conducted over 4 days, starting from the 10th of May, 2023. In total, 392 articles were retrieved from the three databases: 277 articles from Scopus, 92 articles from WoS, and 23 articles from the ERIC database. The retrieved studies were conducted from 2000 to 2023. Subsequently, all these articles were uploaded to the Rayyan Systematic tool (Ouzzani et al., 2016) for analysis.

Initially, 64 studies were excluded because they were not peer-reviewed original articles. Afterward, duplicate articles ($n = 115$) were removed, resulting in 213 unique articles. The first set of researchers reviewed and assessed the abstracts, titles, and keywords of the first 100 papers. During this process, studies that were categorized as "maybe" were discussed, and the full texts were examined for assessment to resolve any uncertainties. The remaining articles were then independently reviewed and filtered by both the first and second researchers. A meeting was held afterward, where all authors discussed any ambiguous cases and solved them by consensus. Based on the predefined inclusion and exclusion criteria, the initial title and abstract scan identified 107 publications that were deemed suitable for a full-text review (see Figure 1). After the full text was included and studies identified, data extraction was performed by the first author.

To acquire data from the included articles, relevant information was systematically collected using a codebook (Büchter et al., 2020) and followed the codification (Heikkinen et al., 2022), as presented in Table 3. This approach aimed to minimize individual differences between the reviewers. The first 10 studies were coded by the first author, and then a meeting was held to discuss any discrepancies with the second author and to finalize the codebook before proceeding with coding the remaining articles. Subsequently, a quality check was conducted on the retrieved papers before commencing the synthesis stage. During this phase, the authors organized all the gathered and discussed data within the framework of the review hypothesis (Webster and Watson, 2002).

Based on the codification schema, the extracted information consists of various categories. These includes extracting the characteristics of the study such as descriptive statistics, study design, participant details, types of V-Labs, the specific disciplines for which the V-Labs were developed, educational settings and levels, V-Labs development environment, V-Lab evaluation measure, and metrics. In addition, accessibility design features, and instructional design accessibility were extracted from the studies guided by Huang's study (Huang and Benyoucef, 2023).

# 4 Results

## 4.1 Study characteristics

The study characteristics are presented in the following sections, encompassing descriptive statistics of reviewed studies, the participants' educational levels, the topics covered in the V-Labs, the research designs employed, and details about the virtual learning environments developed. Subsequently, we have presented the aspect of accessibility and the evaluation methods utilized to measure the effectiveness of the V-Lab.

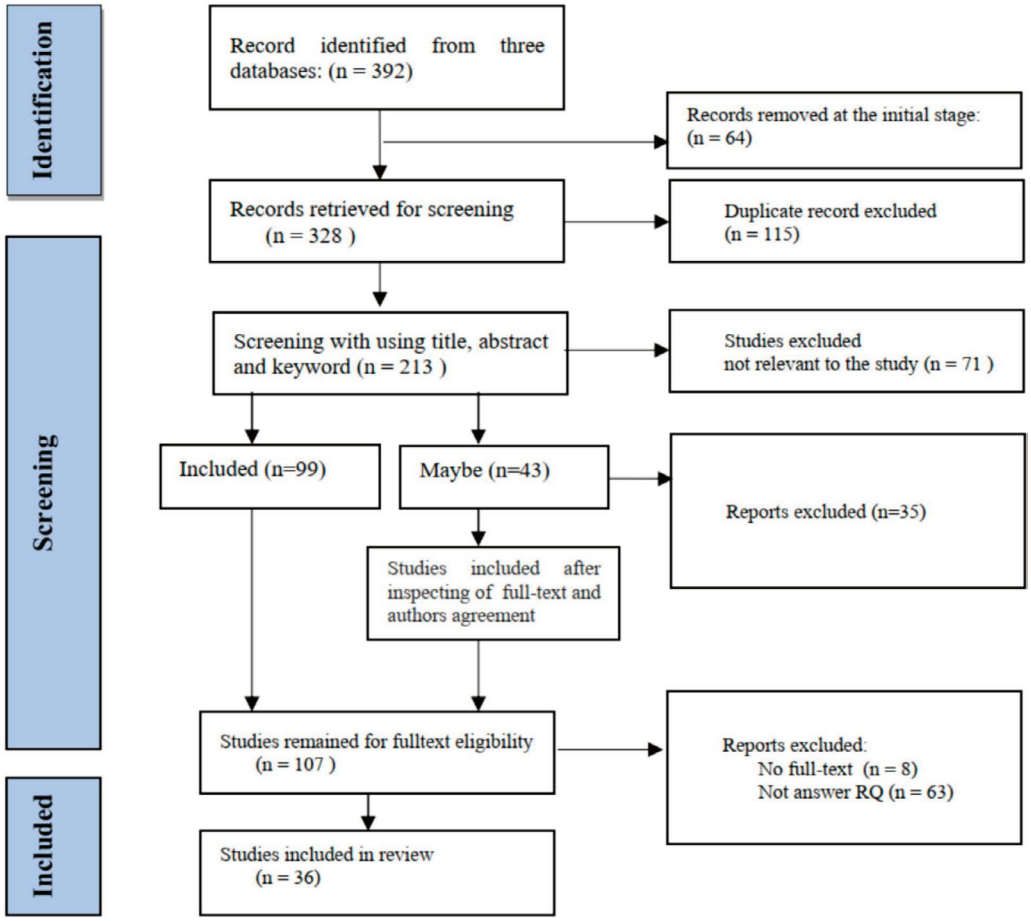



FIGURE 1
Study selection process.

| ID | References | Study design | V-Lab type | Accessibility features |
|---|---|---|---|---|
| S1 | Gómez-Chabla et al. (2017) | Case study, prototype development | Virtual robotics lab, computer science | Friendly and easy to use, Availability of 24/7, Easy interaction design (low complexity) |
| S2 | Bähring et al. (2004) | Experimental, case study | Microprocessor lab, computer electrical and computer engineering | Allow to edit and assemble the program, Provide a pull-down menu |
| S3 | Adamo-Villani et al. (2006) | Experimental | Photorealistic 3D lab, electrical and computer engineering | Portable on different web platforms, allows easy integration of additional content, incorporates universal software design features (keyboard equivalents, visual focus indicators, magnifiers, and specialized input devices for students), interface (simple, resizable window and a toolbox), display 3D objects (Zoom in and out, rotate and point of view), 24/7 accessible |

*(Continued)*

| ID | References | Study design | V-Lab type | Accessibility features |
|---|---|---|---|---|
| S4 | Drigas et al. (2006) | Case study | E-culture virtual lab, applied technologies department | Compatible with every operating system and Internet browser |
| S5 | Buddhiraju et al. (2018) | Experimental | Satellite image processing and analysis lab, geomatics eng. | Provides various input parameters, Incorporating a self-assessment tool, incorporates a feedback component |
| S6 | Guerra et al. (2009) | Design-based, prototype development | Control system lab, electrical and electronic engineering | Play, pause, and reset simulation tasks, Show simulation status to students |
| S7 | Drysdale and Braithwaite (2017) | Experimental | OpenSTEM lab, multi department | Ability to reconfigure the locations and settings of these items, Compatibility with a range of browsers and PCs, mobile phones, 24/7 accessibility but need booking |
| S8 | Reitinger et al. (2013) | Experimental | 3D engineering V-lab, department of cybernetics | Allow to set parameters, allow a user to modify the numerical input field, ability to zoom in/out, incorporates a special slider that allows for speeding up or slowing down the simulation, and enables users to run, pause, and stop the simulation |
| S9 | Dinc et al. (2021) | Experimental | AR-based chemistry lab, chemistry department | Compatible with a wide range of low-end mobile devices, provides tools for editing the reaction, incorporates a quiz module, incorporates student progress tracking, integrates gaming features |
| S10 | Savaş and Erdal (2010) | Design-based | Control lab, department of electronics and computer education | Allows to enter values and change parameters, View students' results (score), 24/7 availability |
| S11 | Trnka et al. (2016) | Case study | General engineering virtual lab, department of instrumentation and control engineering | Allow access to web browsers and smartphones |
| S12 | Ceccacci et al. (2022) | Design-based, Pilot study | 3D Museum V lab, computer science | Enable level adjustment, support multimodal language, provide video description, add keyboard control, start/stop of audio/video description by proximity and button, the possibility to choose the font of textual descriptions, play mode option (Oculus/ Gamepad) |
| S13 | Karingula et al. (2013) | Design-based | Physics lab, school of engineering | Provide navigation and game-based experiments, provide 2D or 3D simulations, provide collaborative tools, Integrate video conferencing, Allow to adjust the speed of instruction delivery, generate report and statistics |
| S14 | Tejedor et al. (2008) | Design-based | Electronic circuit lab, computer science | Accessed through a Web browser, a User-friendly graphical interface, Allows student to explore different parameters and observing their effects |
| S15 | Watters et al. (2020) | Design-based | Denver virtual lab assistant, general science | Provide smart speaker, functions like cancel, help, virtual assistant; get step, next step, previous step, remaining steps; begin the lab, exit lab; material list, feedback, and instruction documents; Readable Format Guide; compatible with other commonly accessible electronic devices |

*(Continued)*

| ID | References | Study design | V-Lab type | Accessibility features |
|---|---|---|---|---|
| S16 | Colak et al. (2014) | Experimental | RLC circuit simulation lab, electrical engineering | Compatible with other commonly accessible electronic devices, interactive simulation, remote-access real-time experimental system, zooming in and out, 3D rotation, provides user-defined parameters, notification to the user to correct experiments |
| S17 | Yalcin and Vatansever (2016) | Design-based | Power electronics V lab, electrical-electronics engineering | Online or offline simulation, description of animations, allowing preparation and submission of lab reports, allowing online support from instructors, incorporating a help menu, report support, subject descriptions, multi-language support, user-friendly interface |
| S18 | Galang et al. (2022) | Design-based | Chemistry lab, chemistry department | Integration of questions and branching points, provide feedback to users, provide online simulation, allow to download the exercise, report, and others |
| S19 | Rossiter and Shokouhi (2012) | Design-based | Matlab/Simulink V lab, control engineering | Allow modifying the input, Accessible 24/7 |
| S20 | Spanias et al. (2000) | Design-based | Electronic lab, electrical engineering, and computer science | Provides graphical editor, supports lab report submission and grading, user-friendly, universally accessible through browsers, allows users to enter filter parameters |
| S21 | González et al. (2001) | Design-based | Physics lab, department of thermodynamics and applied physics | Describe the physical phenomena involved or analyzed, describe the necessary materials, allow users to change the parameters of different elements, and measure the performance |
| S22 | Almaatouq et al. (2021) | Design-based, case study | Emperica/modular virtual lab, computer science | Provide parameterizable experimental designs; reusable protocols; and rapid development, allow monitoring progress in real-time, allow running experiments on any web browser; include instructions and quizzes, provide game, allows experimental trial and repeated play |
| S23 | Ghazinejad et al. (2021) | Experimental | Photon quantum lab, mechanical and aerospace engineering | Allow selection the experiment and material |
| S24 | Thoms and Girwidz (2017) | Design-based | Optical spectrometry lab, faculty of physics | Configurations for adjustable parameters, allows users to view the outputs in different contexts, supports two languages |
| S25 | de Jong et al. (2014) | Design-based, prototype development | Go-Lab, STEM education | Provide direct links to the curricula and educational objectives; allow the user to automatically adapt to the different devices; students to compose hypotheses by dragging and combining predefined terms or using self-defined terms or sentences; allow students to observe/measure |

*(Continued)*

| ID | References | Study design | V-Lab type | Accessibility features |
|---|---|---|---|---|
| S26 | Yuan and Teng (2002) | Design-based | CALSB lab, civil and structural engineering | Provide accessible main menu bar; offers options for file management, data input, modification, results, and help; |
| S27 | Redel-Macias et al. (2015) | Design-based | Noise and machine vibrations lab, electrical and electronic engineering | Provide video tutorial; provide result analysis; image interpretation, calculations, and the ability to repeat practical activities; provide features for animations; and a user-friendly |
| S28 | Li (2022) | Case study | Intrusion detection systems lab, American Society for Engineering Education | Provide learning outcomes, lab procedures, and questions for students to complete; incorporates automatic grading, and provides immediate feedback; allows a trial-and-error approach intended for improving student learning |
| S29 | Cois et al. (2010) | Design-based | VTE virtual lab, Department of Defense and U.S. Federal Civilian Agencies | Searching for available content; allowing them to save their task progress and exit the lab |
| S30 | Picovici et al. (2008) | Design-based | Wireless sensor networks lab, electronic mechanical and aerospace engineering | Incorporates elements such as message exchange between the client and server, message handling; providing a command toolbar and command panel for taking actions; provide a message panel for displaying confirmation messages and received data |
| S31 | Achuthan et al. (2011) | Experimental | Amrita V-lab, science and engineering | Interactive animations and simulations; 24/7 available free online; incorporate Self-Evaluation, Assignment, and Feedback |
| S32 | Dikshit et al. (2005) | Design-based | Medical imaging V-lab, biomedical engineering | Interactive computational simulations; User-adjustable parameters, online tutorials; online homework (or questions and answers), projects and exams; detailed text description of subjects |
| S33 | VijayKumar et al. (2002) | Design-based | Microelectronics V-lab, electrical engineering | Allowed to zoom in/out, navigation links to each module, easy to learn and use, 3D objects |
| S34 | Raineri (2001) | Experimental | Biology V-lab, department of microbiology | Animated web material, Web-based animations, fun and easy-to-learn, web-based tutoring |
| S35 | Subramanian and Marsic (2001) | Design-based | Virtual biology labs, department of cell biology and neuroscience | Save their lab reports, expert system-based automatic help and guidance in running the labs |
| S36 | Sánchez et al. (2012) | Design-based | Controller design V-lab, control engineering | Automatic grading platform, and interactive exercises, allow the students to perform simulations and visualize, allowing entering values for the control signal |

TABLE 3 Codification schema used for the study.

| Coding schema | Description |
|---|---|
| Paper title | Title of the included paper |
| Authors (year) | Authors of the paper with the year publication |
| Context | Primary education, Higher education |
| Focus | What type of Lab environment is developed/implemented and for which field of study? e.g., robotics V-lab, security V-lab; computer science, biomedical,… etc. |
| Methodology | What method was used in the study? e.g. design based, case study,… etc. |
| Description of accessibility | What aspect of accessibility is addressed in the study? e.g. offline/online accessibility, multi-language support, interactive user feedback, collaborative tools, trial error support,…etc. |
| evidence on effectiveness V-Labs | How is the effectiveness of accessibility of V-Labs assessed? e.g., Instructor or student feedback, pre and post-simulation or experimental test, … etc. |
| Aspect of assessment | Which aspects do the studies use to assess their lab? e.g., satisfaction, motivation,… etc. |

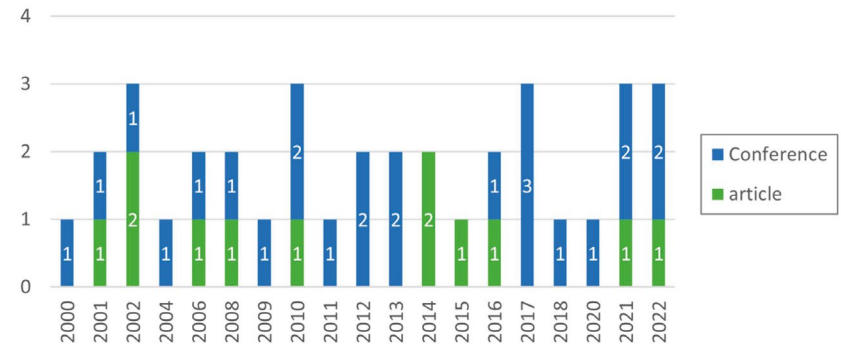



FIGURE 2
Type of studies and year distribution.

### 4.1.1 Description of included studies

A total of 36 studies were included in this review, which was conducted on four different continents and covering a total of thirteen countries. Figure 2 illustrates that the articles were published between 2000 and 2022, with notable peaks occurring in 2002, 2010, 2017, 2021, and 2022, each featuring three articles. This observation underscores the continued growth and widespread utilization of V-lab in educational sectors. Among the reviewed articles, 24 were presented at conferences, while the remaining 12 studies were publication in peer-reviewed journals.

### 4.1.2 Education level

The reviewed studies were conducted across primary, middle school, and higher education. Among the articles examined, the majority of developed labs (69%, $N = 25$) were tailored for higher education students, encompassing both graduate and undergraduate levels. Within this higher education context, two labs were specifically designed for visually impaired students. Six studies (17%) studies were focused on general education levels (see Figure 3), offering labs that catered to students across all educational levels while others with a specific emphasis on chemistry majors. Furthermore, other labs were developed for researchers and computational scientists. Finally, three studies ($N = 8$) were conducted for K-12 students.

### 4.1.3 Subject

We have looked at the particular field of study area [subject] that the reviewed studies' V-Lab development addresses. A total of 22 disciplines in science and engineering were included in the reviewed studies. A quarter of the total studies ($N = 9$; 25%) were conducted for electrical Engineering-related subjects (e.g., Electrical Engineering, electrical and electronic, electrical and computer). This was followed by eight (22%) studies for other engineering subjects (e.g., for Instrumentation and Control, Civil and Structural, Control, Geomatics Engineering, Mechanical and Aerospace Engineering).

Five (14%) studies were in Medical and Biology-related subjects (e.g., Microbiology, Biomedical, cell biology, and Neuroscience), Four (11%) studies were developed for Computer Science, and three studies were for multidisciplinary subjects (e.g., STEM). Additionally, there were also two studies for Chemistry subjects. Finally, the remaining three studies were conducted on Cybernetics, Applied Technology, Defense, and Federal Civilian Agencies subjects, listed as various categorizations (see Figure 4).

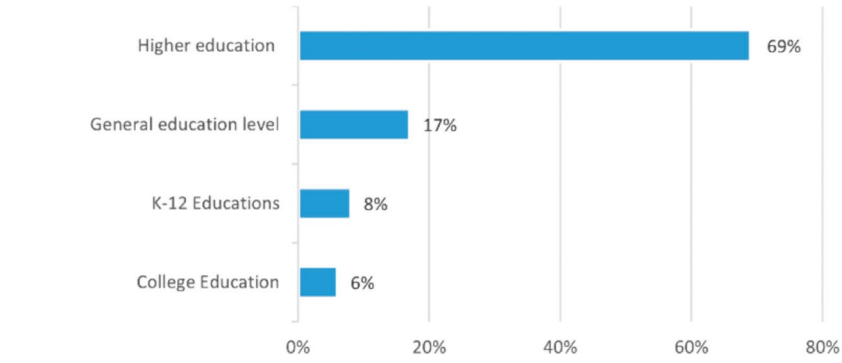



FIGURE 3
Level of education.

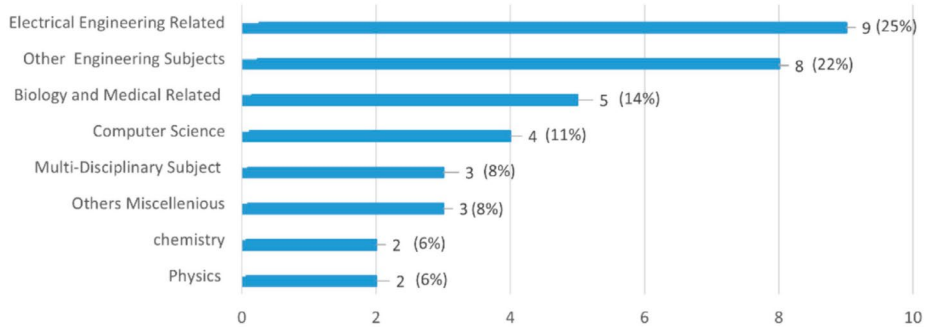



FIGURE 4
Studies V-Labs categories by subject.

### 4.1.4 Design used in the studies

Figure 5 presents the study design and evaluation method used in the reviewed study. The studies in this review included various study designs: 50% of the studies used design-based approaches ($N = 18$), 27.7% were experimental studies ($N = 10$), and three were case studies (Drigas et al., 2006; Trnka et al., 2016; Li, 2022). The remaining five studies were mixed methods, including design and prototype (Guerra et al., 2009), case study and prototype (Gómez-Chabla et al., 2017), design and pilot study (Ceccacci et al., 2022), design-based study and case study (Almaatouq et al., 2021), and experimental study and case study (Bähring et al., 2004).

Three different assessment types were used to evaluate their V-labs performance: (1) feedback, (2) self-assessment reports, and (3) pre-test and post-test scores. From the first type of evaluation, 14 studies (38.9%) used student feedback, three studies used student and instructor feedback with various evaluation metrics. From the evaluation of student feedback, six studies used the Likert scale, two used open-ended questions, two studies included attitudinal questions, and another two studies used both the Likert scale and open-ended questions. Notably, two studies only mentioned that their lab was evaluated based on student feedback, but did not provide specific assessment metrics. From instructors' and students' evaluation metrics, two studies used the Likert scale, while one did not report evaluation metrics; their study focused on the perceived effectiveness and usability of lab aspects.

The second type of reviewed evaluation assessment was self-testing or lab usage evaluation methods, including two studies (Colak et al., 2014; Almaatouq et al., 2021), that used accuracy of comparison and score rate of outcome evaluation metrics, respectively. Finally, the third evaluation method was pre-post experimental tests by only one study (Dinc et al., 2021), focusing on performance metrics such as experiment completion time and error rate metrics. However, the remaining 16 (44.4%) studies did not provide their evaluation metrics.

### 4.1.5 Virtual environment

The studies reviewed utilized a variety of learning environments in different areas of study. The Denver Assistance Lab (Watters et al., 2020) was developed for general science. Likewise, Amrita Lab (Achuthan et al., 2011) was designed for science and engineering. The

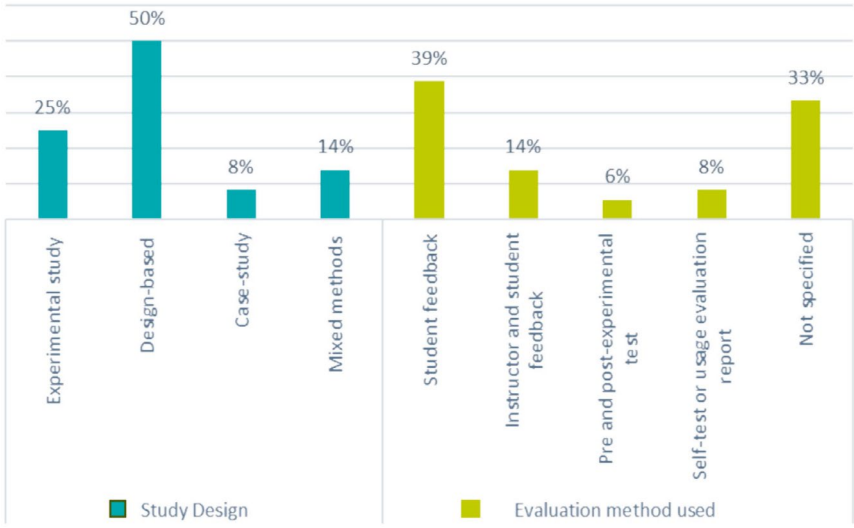



FIGURE 5
Study design and evaluation method used in studies.

3D Museum Lab (Ceccacci et al., 2022) and E-Culture Lab (Drigas et al., 2006) were developed for public users and academic purposes.

In the field of computer science, there are labs such as the Virtual Robotic Lab (Gómez-Chabla et al., 2017), the Emperica Lab (Almaatouq et al., 2021), and the Intrusion Detection System Lab (Li, 2022), Electronic Circuit Lab (Tejedor et al., 2008) was developed for computer science lessons. The AR-based Lab (Dinc et al., 2021), and Microsoft Form based lab (Galang et al., 2022) for the chemistry lab. The Medical Imaging Lab (Dikshit et al., 2005) was developed for the biomedical field. The remaining labs were developed for various engineering disciplines. For example, the Microprocessor Lab (Bähring et al., 2004) and the Photorealistic 3D Lab (Adamo-Villani et al., 2006) were designed for Electrical and Computer Engineering. The Satellite Image Processing and Analysis Lab (Buddhiraju et al., 2018) is used for Geomatics Engineering. The Control System Lab (Guerra et al., 2009) used Java-based simulation for Electrical and Electronic Engineering. Similarly, the RLC Circuit Simulation Lab (Colak et al., 2014) employed for Electrical Engineering. The Photon Quantum Lab (Ghazinejad et al., 2021) and Wireless Sensor Networks Lab (Picovici et al., 2008) were developed for Mechanical and Aerospace Engineering.

In addition, the Optical Spectrometry Lab (Thoms and Girwidz, 2017) catered to Physics, while the Go Lab (de Jong et al., 2014) focused on STEM education. CALSB Lab (Yuan and Teng, 2002) was dedicated to Civil and Structural Engineering. The Noise and Machine Vibrations Lab (Redel-Macias et al., 2015) was developed for Electrical and Electronic Engineering. Lastly, the Microelectronics Lab (VijayKumar et al., 2002) was designed using visual 3D form for Electrical Engineering, and the Controller Design Lab (Sánchez et al., 2012) utilized a Java simulator for Control Engineering. The V-lab environment setting and the associated field study for the lab are shown in Figure 6.

## 4.2 Aspects of accessibility assessment

In this study, we investigated how the reviewed studies assessed their labs. Out of 36 studies, 15 (42%) examined the effectiveness of their labs, while four studies did not specify their evaluation results [S3, S15, S20, S22]. The effectiveness of these V-labs was evaluated using three different forms: students' or instructors' engagement levels, usability and interaction of the system with students/instructors, and user satisfaction and experience with the V-labs.

### 4.2.1 Usability and interaction with the V-labs

The effectiveness of the developed labs in the reviewed study was evaluated using the usability of the V-Lab and students' interaction with it. Seven studies [S1, S5, S17, S27, S28, S34, S35] focused on usability and interaction with the system. Each study was evaluated with a different focus on usability and interaction. For example, [S1] evaluated usability in terms of ease of use, user-friendliness, and complexity and concluded that 50% of the system is usable by the users. In addition, the student stated the system's functionality feature of the interaction was more than 76% and very helpful. In another study [S5], students' usability was assessed based on the V-Labs resourcefulness in resource-limited environments. In another study [S17], the authors evaluated usability and contribution to education and found that 97% of students' grades are improved by using visual design animation and online V-Labs support. The study by [S27], showed that the V-Lab was usable with 3.6 out of 5 points in terms of user-friendness, helpfulness, and encouraged motivation and had an effective user interface and adequate content 3.4 out of 5 points. In the study [S28], the authors measured usability in terms of ease of use and access received students responded 45% of usability and 27 responses were in the understanding of the course. In [S34], the improvement in examination results after the introduction of web-based tutorials was

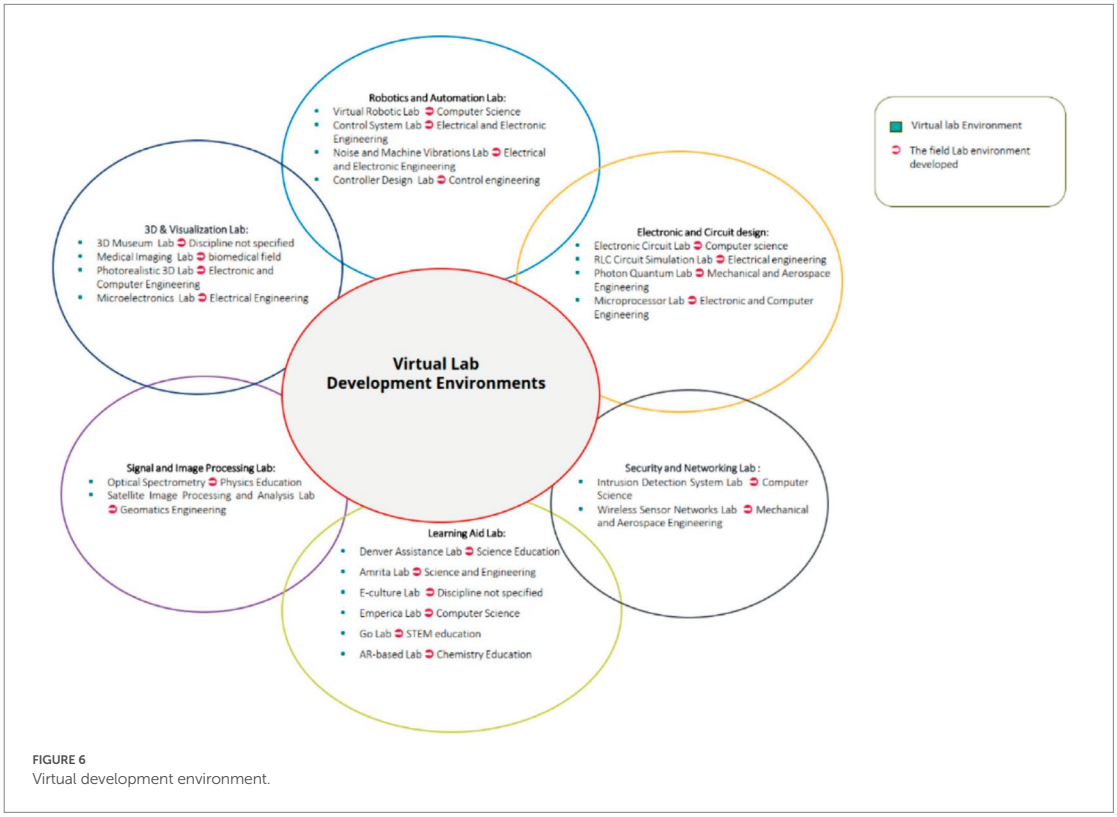



FIGURE 6
Virtual development environment.

observed. In [S35], the students found repetition opportunities, exercises, and feedback helpful. However, unclear instructions and the lack of a Back button created problems, highlighting the need for their inclusion to improve learning.

### 4.2.2 User experience and satisfaction

Three studies [S9, S13, S14] focused on satisfaction and measured user experience on the system. In [S9], the test results of the study experiments showed that the interactive performance on low-end devices is good, and 90% of students were satisfied with using gaming in the lab. In the study of [S13], students expressed high satisfaction (82%) with the innovative teaching method as well as the new way of teaching, understanding, and knowledge of concepts. In [S14], the students stated that they were satisfied with the acquired lab knowledge and practical skills.

### 4.2.3 Degree of engagement

In terms of interaction, two studies [S18, S19] assessed the degree of engagement of users' interaction with the system. The engagement level evaluation showed that 62% of the V-Lab was more engaging compared to other virtual exercises in [S18]. In [S19], students who used V-Labs for summative engagement found them helpful and user-friendly. More than 50% engaged through easy-to-use and accessibility and support the concept of learning materials. Table 4 offers a summary of the reviewed studies' assessment metrics, evaluation metrics, and study focus.

## 4.3 Accessibility categories

In this study, several accessibility features were identified and extracted from the reviewed literature. They were then organized into seven main categories based on similar themes (see Figure 7). The reviewed studies addressed accessibility either through the development of the lab environment or the design of prototypes for the V-Lab platform.

### 4.3.1 Instructional design support

In terms of instructional design, eight (22.2%) studies discussed how their lab environment supports 2D or 3D objects, simulations, or interactive animations [S2, S13, S16, S24, S27, S31, S32, S33]. Followed by seven (19.44%) studies [that] indicate[d] the lab they used supports video or provides audio content descriptions [S3, S12, S17, S21, S25, S28, S32]. In the context of zoom in and out, four (11.1%) studies mentioned improved zooming and viewpoint control [S3, S8, S16, S32].

### 4.3.2 Interaction support

Regarding the V-lab's technology interaction supporting platform, 13 (36.1%) studies reported that their labs support adjustable parameters and rich input options [S2, S5, S8, S10, S12, S14, S19, S20, S21, S22, S24, S26, S32], while seven (19.4%) studies were support submission and automatic grading for assessment of learner [S10, S14, S17, S20, S24, S28, S36]. Six (16.66%) studies emphasized the

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TABLE 4 Assessment metrics and aspects.

| Metrics of evaluation | Aspect of evaluations |
|---|---|
| Likert scale | Usability, interaction, and utility of architecture [S1]<br>Usability and contribution of the lab to education [S17]<br>Understanding of topics and ease of accessibility [S25]<br>Evaluated engagement and understanding of the lab [S18]<br>Usability and contribution of the lab to student learning [S27] |
| Open-ended questions | Satisfaction with the functionality of the lab [S5]<br>evaluated on the usability of the lab [S7] |
| Likert and open-ended question | Level of engagement, and performance in lab [S18]<br>On usability and motivation [S27]<br>Ease of accessibility of the lab [S20] |
| Attitudinal questions | Student satisfaction with the lab [S14, S19] |
| Experiment completion time, & error rate | Performance measures score on device type [S9] |
| Score rate | Social score rate on the outcome of individual performance [S22] |

Focuses

- User experience and satisfaction
- Usability and Interaction
- Degree of Engagement

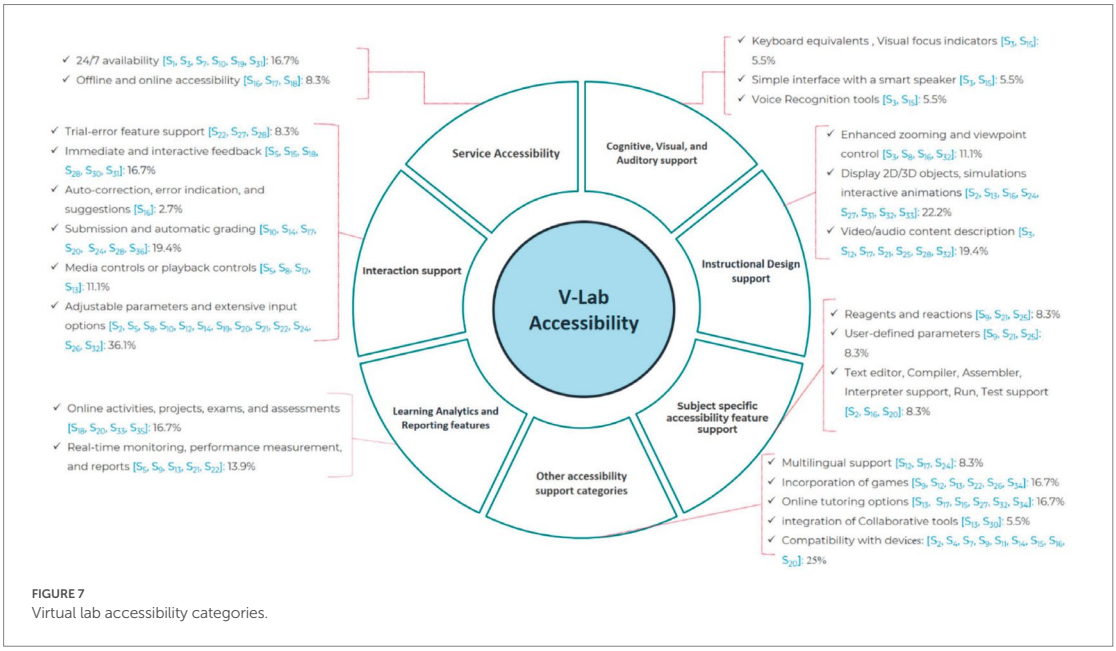



FIGURE 7
Virtual lab accessibility categories.

importance and incorporation of immediate and interactive feedback in the lab environment [S5, S15, S18, S28, S30, S31]. Additionally, four (11.1%) studies addressed media controls or playback controls [S5, S8, S12, S13]. Moreover, three (8.3%) studies indicated that their labs supported the trial-error feature [S22, S27, S28], and interestingly, only one study [S16] addressed the specifics of auto-correction, error indication, and suggestions.

### 4.3.3 Virtual service accessibility

While V-Labs service is expected to be accessible at all times, only a few studies have addressed the availability of virtual services. For example, six (16.7%) studies addressed the 24/7 availability of their V-Labs environment to provide various online activities, projects, exams, and assessments [S1, S3, S7, S10, S19, S31]. While, three studies (8.3%) addressed both offline and online accessibility [S16, S17, S18]. Some studies, for example [S7] reported the user needs to book the lab to use it for a specified time.

### 4.3.4 Cognitive, visual, and auditory support

Accessibility from the perspective of different forms of (dis)ability showed that only a limited number of lab environments have been developed. Notably, two (5.5%) studies addressed the integration of

assistive technologies into their environment, including keyboard equivalents, visual focus indicators, a simple interface with a smart speaker, and voice recognition tools [S3, S15].

### 4.3.5 Subject-specific accessibility support

In subject-based lab environments, three (8.3%) studies included specific features, including reagents and reactions, as well as user-defined parameters specified in chemistry-related domain lab environments [S9, S21, S25]. Additionally, three (8.3%) studies highlighted that their lab supported text editor, compiler, assembler, interpreter, execution, and test support for computer-related disciplines labs [S2, S16, S20].

### 4.3.6 Learning analytics and reporting features

Regarding reporting of lab environments, five (13.8%) studies highlighted that their labs support real-time monitoring, and performance measurement [S5, S9, S13, S21, S22]. Four (11.1%) studies reported the ability of lab environments to download and upload students' reports and statistics [S18, S20, S33, S35].

### 4.3.7 Other form of accessibility support

We categorized the remaining accessibility support into other categories. Among them, nine (25%) studies discussed the support and compatibility of their lab environment with all operating systems, Internet browsers, PCs, and smartphones [S2, S4, S7, S9, S11, S14, S15, S16, S20]. Additionally, six (16.66%) studies highlighted the integration of games for learning [S9, S12, S13, S22, S26, S34], and another six (16.66%) studies highlighted online tutoring options in the lab environment for diverse learners [S13, S17, S15, S27, S32, S34]. Three (8.3%) studies confirmed the presence of multilingual support [S12, S17, S24]. Only two studies mentioned collaborative tools [S13, S30].

## 5 Discussion

This section aims to discuss of our main findings guided by the research questions.

**(RQ1):** What essential accessibility features are integrated into V-Labs to cater to diverse user needs?

To address the research question, we examined the reported studies' accessibility features of the labs, framed them, and provided them with different perspectives.

*Perspective 1*: Accessibility of the lab features to users with cognitive, visual, and auditory limited abilities. While there have been studies that developed V-lab environments for students, limited attention has been given to how students with learning disabilities navigate these environments without restrictions on their abilities (Gavronskaya et al., 2021). Gavronskaya emphasized the importance of providing appropriate lab interfaces that focus on students' abilities, enabling them to gain a sense of mastery through experimentation. However, the review revealed a scarcity of studies addressing the integration of assistive technologies. Only two studies [S3, S15] demonstrated certain features, such as keyboard equivalents, visual focus indicators, simple interfaces with smart speakers, and voice recognition tools, which contribute to the inclusion of users with diverse abilities. For example, students with motor or visual limitations can navigate and interact with V-Labs without needing a mouse, while the integration of smart speakers may facilitate experiments for users with visual impairments.

Despite the inclusion of these features, the studies did not highlight the benefits of such assistive tools on learning outcomes, task completion, or deeper understanding. In this regard, Lin et al. (2017) emphasized the importance of designing teaching activities and the flexible application of technology tools in integrated education. Further exploration is needed to determine practical teaching strategies for enhancing effectiveness using V-labs.

On the other hand, Bong and Chen (2024) raised concerns about the challenges faced by students with disabilities due to the inaccessibility of digital tools and learning materials. Our study suggests that existing V-labs are inaccessible to students with various abilities, including those with cognitive, visual, and auditory limitations. Adhering to digital technology accessibility guidelines (Chisholm et al., 2001) is essential, necessitating a focus on labs designed for different abilities. For example, students with cognitive limitations may benefit from V-labs with simplified interfaces, clear instructions, step-by-step guidance, and interactive feedback. Considering these features and incorporating multiple learning modes and customization options ensures accessibility and accommodates the needs of cognitively limited students. Additionally, addressing the challenges faced by students with limited vision requires V-labs with screen reader compatibility, text-to-speech functionality, keyboard navigation, and compatibility with assistive listening devices (Butler et al., 2017; Bell and Foiret, 2020). Further research on virtual learning tools that address the diverse needs of students with different abilities and explore their impact on the educational process is warranted.

*Perspective 2*: Accessibility of the lab features for users with diverse cultural and linguistic backgrounds. Considering diversity and equity in education, and exploring multiple approaches to teaching learners while fostering motivation in both practical and laboratory work, are essential requirements for effectively preparing students and facilitating their mastery of practical skills. Hunter et al. (2010) stated that thoughtful design of laboratory activities can positively impact students with diverse educational, social, and cultural backgrounds. However, less attention has been given to how laboratory experiments can promote diversity and inclusion (Nonnemacher and Sokhey, 2022). To promote the equity and inclusion of V-lab to students from different backgrounds, limited studies highlight accessibility features that are beneficial to users from different backgrounds. For example, in the reviewed study, only three studies [S12, S17, S24] indicate the integration of multilingual support in V-Labs. While this feature addresses the language barrier during experiments, promoting linguistic diversity in instructional approaches by offering multilingual support needs further attention. To solve this issue, Hackl and Ermolina (2019) suggest establishing an inclusive cultural feature within laboratories.

Studies explored integrating online tutoring as another feature of V-Lab that engages students in learning through active participation rather than passive observation. While integrating online tutoring into V-Labs, the diverse student is assisted by several means, e.g., either error messages warning students about wrong actions or hints related to the next action of the practice (Paladines et al., 2023). In the context of tutoring, only 16.7% of studies were included, demonstrating the integration of the feature into developed lab environments. The included studies did not explore how tutoring contributes to real-time

support for a student who experiences challenges or difficulties during lab sessions.

In addition, the integration of collaborative tools as a feature into V-lab is another feature that supports diverse cultural students. The included studies [S13, S30] highlighted a collaborative tool that promotes teamwork. According to De La Torre et al. (2013), an extension of collaborative tools to V-labs fosters students' practical skills through working in groups of two or more, mutually searching for understanding and exchanging ideas on difficult concepts. Regarding collaborative tools, several open areas need to be further investigated from this perspective, e.g., Which modalities of online tutoring (e.g., chat, video, audio) are most effective in the context of V-Labs and what contribution does online tutoring make to peer-to-peer learning in the V-Lab environment? How can cultural nuances and language-specific pedagogical approaches be taken into account when implementing multilingual support in V-Labs? These need to be addressed to encourage strong linguistic backgrounds and the inclusion of different cultures to use the V-Labs in academia.

*Perspective 3*: Accessibility of the lab features in terms of instructional design and content availability. Augmenting instructional design with V-Labs has several effects on student engagement in critiquing experiments, enhancing students' critical judgment of inquiry processes and their scientific literacy (Liu et al., 2022). This shows various instructional design accessibility features improve educational outcomes using lab experiments. Interestingly, many function-related aspects are addressed in the reviewed studies. For example, educational games (16.7%), immediate and interactive feedback (16.7%), 2D or 3D simulations (22.2%), zooming and viewing angle control (11.1%), and Video or audio content descriptions (19.44%) in V-Labs. Our study extends the idea of instructional design features mentioned in Chan et al. (2021), by identifying further the elements of design features. This customized feature enables the achievement of educational goals by entertaining the educator by integrating the gameplay into the virtual learning environment. The inclusion of 2D or 3D simulations or animations promotes student engagement and adds to the educational experience. On the other hand, improved zooming and viewing angle control are used for different lab forms such as labs designed for students with different (dis) abilities. Adamo-Villani et al. (2006) demonstrated that the ability to zoom in or out feature allows students with limited vision to connect better with small devices.

The availability of V-lab content at any time, place, and on any device is crucial. Several reviewed studies have discussed this issue. For example, 6 out of 36 studies stated that their developed V-Labs are fully accessible to students in both offline and online forms, while one [S7] study required booking lab sessions to access V-Lab services, including exams, projects, and other online experiments. We believe that offering 24/7 services, both online and offline, to accommodate students with varying schedules, time constraints, or residing in different time zones, can significantly contribute to providing equitable access to educational resources for students worldwide. Boukil and Ibriz (2015) support this notion, asserting that cloud-based lab environments offer restricted access to either time or location. In such circumstances, instructors and learners can access lab documentation and resources at their convenience, facilitating individual or collaborative engagement from any location. However, it prompts us to rethink the vulnerabilities, privacy, and security of labs. These labs may present risks of degrading essential and optional educational characteristics and technical difficulties. Another aspect to explore is how the offline accessibility of virtual labs caters to the needs of students who may opt for offline study or face limitations in accessing a stable internet connection. Therefore, further investigation is required.

*Perspective 4*: Accessibility of the lab feature in terms of interaction design supporting features. Sahin and Gul (2014) raised interesting points regarding the design practices of digital technology, particularly how designers grapple with the complexity inherent in design activities. These insights emphasize the importance of integrating design-supporting features into digital educational tools, including V-Labs. Our study aimed to evaluate the alignment of interaction design support, revealing a thorough examination of numerous accessibility features. Building on this, Wray et al. (2019) emphasized the necessity for users to provide a model of their motivations, goals, and contexts to inform the development of user-centric interventions. However, the incorporation of such considerations was largely limited in the studies reviewed, with only a few addressing integration design accessibility.

Among the 36 studies analyzed, 13 identified adjustable parameters and extensive input options. This focus on customizable parameters is commendable, as it contributes to fostering an inclusive learning environment and encourages creative experimentation. Höök and Löwgren (2021) corroborated the benefits of utilizing digital tools with adjustable parameters and extensive input options to enhance learning experiences.

Furthermore, three studies delved into the utility of trial-and-error features, highlighting their role in facilitating experimentation within structured learning objectives. Horsman (2018) noted that students employing a trial-and-error approach exhibited statistically lower levels of test anxiety compared to their peers. However, the reviewed studies failed to elucidate how trial-and-error features could help destigmatize errors and achieve desired outcomes through error recognition and elimination. Another significant aspect, intelligent suggestions, as well as error display and correction, received limited attention in certain studies [S16]. Although discussed in a different context, Ai (2017) identified corrective feedback as effective in aiding learners to self-identify and rectify issues, albeit with the occasional necessity of intervention from an onsite tutor when the system falls short. Crucial questions pertaining to autosuggestion and error correction in lab experiments, such as the scope of errors autocorrection can address and how V-Labs can provide meaningful feedback on correcting and refining experimental techniques, remain unexplored by the studies included in our analysis.

A study, Ballantyne et al. (2018) examined the design accessibility of universal guidelines within selected mobile applications, thus contributing to a broader comprehension of accessibility evaluation within design contexts. Although their evaluation centered on mobile applications, our study expanded this assessment to the domain of V-Labs. Consequently, there remains untapped potential for further investigation into V-Labs, particularly in the development of virtual learning tools. The integration of features aimed at enhancing accessibility could not only improve the overall user experience but also advance the ongoing evolution of inclusive educational technologies.

*RQ.2:* How do existing studies assess the effectiveness of accessibility of V-Labs and which aspects do they assess?

The effectiveness of V-Labs has been examined in several ways. For example, Faias et al. (2007) stated it in two ways: either heuristic evaluation conducted by experts based on the usability guidelines or practical tests with users. While there are accessibility guidelines (e.g., WAG) to evaluate the effectiveness of digital tools such as websites, to our knowledge, there are no specified standard guidelines to evaluate the effectiveness of V-Lab. We found that the focus of the evaluation varied based on the study's aim. For example, Manyilizu (2023) evaluated the effectiveness of the lab on the impact on student learning outcomes.

In this review, while we found the assessment of the accessibility of the V-Labs in three focuses, only less than a third (11 out of 36 studies) of the reviewed studies have evaluated and stated assessment aspects of the V-Labs. Our result aligned with the findings of Lai et al. (2022) that the evaluation of the effectiveness of V-Labs has a gap and requires further investigation. The reviewed study primarily found three viewpoints of evaluation aspects including usability and interaction of the lab with users, user experience and satisfaction of the lab, and level of engagement with the system.

Several topics by different scholars were raised related to evaluating the assessment of usability and interaction of the system. While reviewed studies such as [S1], [S27], and [S28] evaluate the focus on technical usability checklists by users, other studies like [S17] pedagogical usability such as the contribution of the lab to the improvement of their examination in their result or grade. Interestingly, our result also aligned with Kumar et al. (2018) findings regarding the usability analysis of V-Labs, where they identified 15 technical usability checklist items and 21 pedagogical usability checklist items to evaluate the usability of Labs. However, there are still unsolved problems. For example, identifying types of interactions (e.g., simulations, virtual experiments, collaborative features) are most effective for promoting active learning in V-Labs.

Our review report also showed an evaluation of the lab through rating in terms of user experience and satisfaction. Estriegana et al. (2019) investigated students' perceived satisfaction as a critical issue in better understanding learners' behavioral intention to use online learning Environments (e.g., V-Lab). In our study, the reports [S13], and [S14] discussed the satisfaction of students with the integration of gamification in the V-Labs, satisfaction with the understanding of course topics, and the practical skills acquired. Further studies are required to examine key indicators of user satisfaction when using V-Labs. The report also showed that they looked at the level of engagement but did not specify the level of interactivity that contributes to student engagement and understanding of complex concepts.

On the other hand, we found that only 42% of studies evaluated V-Labs. In our review, the studies used different assessment methodologies, even for the same accessibility aspects, which may introduce bias in determining the true effectiveness of the developed labs. Although the objective of studies varies, either they need to develop and validate evaluation instruments or adopt developed assessment instruments such as Kumar et al. (2018) technical, and pedagogical instruments.

## 6 Limitations and future research

Our study acknowledges several limitations in terms of search accessibility and study inclusion. Primarily, we conduct searches and select studies exclusively from three databases, focusing on conference proceedings and journal articles. Our search criteria were limited to two terms, "Virtual Lab" and "Accessib*," potentially overlooking relevant studies on V-Labs during our search process. We acknowledge that different fields use various terms, such as remote lab, online lab, remote experiment, online experiment; simulation in medical science. Further research could be explored by incorporating these key terms for future studies.

In the context of the study's focus, we aimed to assess the accessibility of V-Labs. We found that achieving equitable and accessible V-Labs is still a distant goal, prompting us to consider accessibility from four different perspectives. This study also calls for investigations into the accessibility of other emerging virtual education technologies including Learning Management Systems, e-learning platforms, VR, and AR within an educational context for various learning ability students.

Several unanswered questions emerge, necessitating further research. For example, how can the increased accessibility of VR and AR technologies benefit across different education levels and various learner demographics (diverse learners)? How do adaptive learning technologies, such as AI applications, address accessibility issues and enhance the effectiveness of V-Labs or other learning environments in delivering person-centric learning experiences? Additionally, how do students understand accessibility concepts in classrooms during the development of learning environments or systems that are considered for all users? These guided questions serve as a foundation for future work stemming from our studies.

## 7 Conclusion

Addressing accessibility in V-Labs is pivotal to making students fully participate and promoting diversity and innovation in the field. To harness the potential of V-Labs in the education domain requires a multi-faceted approach that ensures proper functionality and accessibility while addressing existing gaps in accommodating diverse users' needs. In this work, we reviewed the accessibility of V-Lab technologies by examining how studies address accessibility within the V-Lab environment and specific aspects under assessment. While we emphasize the significance of accessibility features for enriching V-Labs, our study highlighted a predominant focus on higher education institutions in existing V-Labs. However, a noticeable gap has been identified in studies exploring various (dis) abilities in K-12 education, indicating a pressing need for more inclusive and accessible lab development.

The study revealed that understanding the aspects of accessibility such as usability, interaction, and satisfaction of students with the V-Lab is crucial for promoting inclusive learning environments and ensuring equal opportunities for all learners, regardless of their abilities. The study uncovered a diverse array of accessibility categories encompassing instructional design,

interaction support, virtual service accessibility, cognitive, visual, and auditory accessibility support, and subject-specific accessibility support. It showcases the multifaceted nature of accommodating diverse learners in V-Lab environments.

Augmenting instructional design with V-Labs positively impacts student engagement and academic literacy. Furthermore, the availability of V-lab content at any time, place, and on any device is fundamental for equitable access to educational resources. However, we found that existing V-labs limited attention has been given to how students with learning disabilities navigate these environments without restrictions on their abilities.

Our review highlights the necessity to evaluate the effectiveness of V-Labs, particularly concerning usability, user experience, satisfaction, and engagement. Further research is imperative to address the current gaps and inconsistencies in evaluating V-Labs, ultimately enhancing their impact on student learning outcomes, and ensuring widespread accessibility.

In conclusion, our review has shed light on issues related to the assessment of accessibility in V-Labs and other emerging virtual educational technologies, emphasizing the need to consider accessibility elements during the design and development of various systems. It has provided valuable insight into areas that still harbor significant gaps, paving the way for future studies to address these shortcomings and contribute to the advancement of accessible and inclusive virtual learning environments.

## Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

## Author contributions

FD: Conceptualization, Formal analysis, Methodology, Writing – original draft. MS: Conceptualization, Supervision, Validation, Writing – review & editing. MT: Funding acquisition, Supervision, Validation, Writing – review & editing.

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## Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

The author(s) declared that they were an editorial board member of Frontiers, at the time of submission. This had no impact on the peer review process and the final decision.

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## ARTICLE II

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## ARTICLE III

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## ARTICLE IV

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**FITSUM DERIBA**

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This dissertation examines accessibility in educational technologies, identifying pedagogical gaps in design education and exploring the role of artificial intelligence in advancing inclusive design. Using a sequential mixed-methods approach, it provides insights into how AI enhances accessibility education and its lasting impact, equipping students and stakeholders to design inclusive technologies in resource-constrained contexts..



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